World Health Organization

Onchocerciasis elimination mapping

A handbook for national elimination programmes





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Contents

Preface v

Acknowledgements viii

Glossary of key terms and abbreviations ix

Introduction 1

Scope of this handbook 2

Who should use this document? 5

Justification for OEM 5

Challenges in elimination mapping 6

Overview of the OEM process 8

Chapter 1.
Step 1: Desk review of existing data and exclusion mapping 10

1.1 Background 12

1.2 Defining the Transmission Zone and Assessment Unit 12

1.3 Desk review 14

Chapter 2.
Step 2: Breeding site assessment 18

2.1 Timing of the assessment 21

2.2 Community mobilization 21

2.3 Breeding site assessments and identification of *S. damnosum* and *S. neavei* 22

2.4 *Simulium neavei* group breeding site assessment 25

2.5 Obtaining information on vector status from communities 27

2.6 Challenges with breeding site assessments only 28

2.7 Conclusions 28

Chapter 3.
Step 3: Epidemiological surveys in OEM 30

3.1 Rationale for the sampling strategy 32

3.2 Community mobilization and timing of epidemiological surveys 33

3.3 Procedure to follow upon arrival in villages 33

3.4 Step 3A: Epidemiological survey in first-line villages 34

3.5 Step 3B: Risk-stratified secondary survey 36

Chapter 4.
Step 4: Laboratory analysis 38

4.1 Choice of diagnostic test 40

4.2 Laboratory supplies 41

4.3 Quality Control and Quality Assurance 41

Chapter 5.
Step 5: Managing and interpreting data 42

5.1 Critical cut-offs for various diagnostic tools 44

5.2 Interpretation of epidemiological survey results 45

5.3 Reporting of results 46

5.4 Generation of action maps 48

5.5 Reporting results to WHO 48

Chapter 6.
OEM planning considerations 50

6.1 Team composition 52

6.2 Budget and supplies for OEM implementation 53

Training of survey teams 55

Training considerations 55

Selection of trainees 58

Selecting and preparing trainers 60

Survey coordination and supervision 60

Training evaluation 61

Piloting of Step 3 epidemiological surveys 61

References 62

Annex 1.
Additional mapping and geospatial modelling resources 65

Annex 2.
Overview of steps in the OEM process 72

Annex 3.
Standard operating procedure for collecting dried blood spots 73

Annex 4.
Tools for generating maps for desk review and action maps based on results of onchocerciasis eliminating mapping 77

Annex 5.
River prospection form 82

Annex 6.
Entomology community questionnaire 83

Annex 7.
First-line village recording form 84

Annex 8.
Crab collection form for *Simulium neavei* immature stages 85

Annex 9.
Standard operating procedure for packaging and shipping dried blood spots 86

Annex 10.
Onchocerciasis elimination mapping survey sample builder 89

Annex 11.
Ov16 RDT with dried blood spots standard operating procedures 90

Annex 12.
Onchocerciasis Elimination Program for the Americas: standard operating procedures for enzyme-linked immunosorbent assays 95

Annex 13.
Reporting onchocerciasis elimination mapping results using the epidemiological reporting form 99

Annex 14.
Evaluation forms for trainers 101

Annex 15.
Evaluation forms for trainees 103

Preface

Human onchocerciasis, also known as river blindness, is a neglected tropical disease (NTD) that results from infection with *Onchocerca volvulus*. It causes severe itching, disfiguring skin conditions and visual impairment, including permanent blindness, for millions of people.

This ancient disease has a huge economic impact in endemic areas, including substantially reducing productivity and household welfare and increasing health-related expenditures. The World Health Organization (WHO) has set a target for elimination of onchocerciasis by 2030 in its road map for NTDs 2021–2030. Efforts to combat the disease through mass distribution of ivermectin since the early 1990s have registered notable successes, leading to a programmatic shift in focus from control to elimination.

The WHO African Region has made tremendous progress towards eliminating onchocerciasis. In 2022, 159.1 million people received treatment for onchocerciasis, and 14 of 26 countries achieved 100% geographical coverage. Regional coverage of the 244.5 million population who required preventive chemotherapy for onchocerciasis was 65.1%. Preventive chemotherapy was administered in 23 of the 26 African countries where evidence indicates that MDA for onchocerciasis is required. Despite the progress made, treatment with ivermectin has historically targeted only areas where the prevalence of onchocerciasis was estimated to be high, leaving untreated areas of lower transmission.

The existence of ivermectin-naive areas where onchocerciasis transmission is ongoing remains a critical challenge to elimination efforts, and all remaining areas with active transmission must be identified to achieve elimination. Additionally, as countries progress towards submission of a dossier for elimination, they will need to have data or a strong rationale for why an area was declared non-endemic. According to WHO's guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis, dossiers should include the justification, rationale and criteria for determining the geographical extent of foci and ascertaining that no transmission occurred in areas outside the known foci.

To help countries meet these challenges, a process called onchocerciasis elimination mapping, or OEM, was developed to (i) determine exactly where interventions are required and (ii) provide rigorous evidence supporting the classification of non-endemic areas. Pilot surveys conducted by a number of countries in the African Region have generated evidence to inform the drafting of a simplified method presented in this document. This handbook captures a collation of discussions, documents, advice and evidence submitted by numerous countries and partners. At its heart is the aim to make OEM possible, practical and standardized. The WHO Onchocerciasis Technical Subgroup (OTS) has reviewed the available evidence for OEM and endorsed most aspects of the strategy for programmatic use while provisionally recommending others, namely the Step 3B risk-stratified secondary epidemiological survey described in 3.5.

The purpose of this document is, therefore, to guide countries on how best to conduct OEM with the tools available to programmes now. This will help to achieve the aims set out in each national NTD masterplan. The handbook provides step-by-step instructions on how to conduct OEM, including desk review of available data, exclusion mapping, river prospections, identification of first-line villages, population sampling for assessment, choice of diagnostic test and interpretation of results. It is inevitable that as new tools and evidence become available, it will need to be updated. However, the evidence generated by following the guidance herein will be robust, and countries should feel confident in embarking upon their elimination mapping using this document as a guide.

For the purposes of OEM, an assessment unit (AU) is defined as a convenient local government administrative unit that is used for exclusion mapping and for the two phases of epidemiological mapping. An AU will be classified as *endemic* (and therefore requiring mass treatment) when the population living in at least one surveyed community has a prevalence of *O. volvulus* serology above the threshold defined in this handbook. In areas that have not undergone mass treatment for onchocerciasis, the AU is considered a proxy for ongoing active transmission that warrants mass treatment. Based on the results of mapping, the country will need to determine the appropriate implementation unit (IU) for mass treatment. It may be determined that the IU should be the same administrative area as the AU, but this is not a requirement.

The five steps described in Chapters 1–5 of this handbook provide a systematic approach that may help national programmes to correctly classify IUs as *endemic* or *non-endemic*; however, not every step may be required in all settings, and the health ministry, with its advisory National Onchocerciasis Elimination Committee and/or Technical Advisory Committee for NTDs may decide how the programme may elect to proceed through Steps 1–5.

Summary of OEM steps[a]

Step 1:	Action	Explanation
Desk review and exclusion mapping	Identify suitable transmission areas	Conduct a thorough review of clinical, entomological, epidemiological, geographical and geospatial modelling data to define the AU, identify potentially suitable transmission areas and exclude unsuitable areas.
Step 2	**Action**	**Explanation**
Entomological assessments	Identify breeding sites and first-line villages	Visit river locations identified during desk review to search for black fly larvae and spot-catch biting adult flies.
Step 3:	**Action**	**Explanation**
Epidemiological assesments	Step 3a: conduct epidemiological survey in first-line villages	Conduct epidemiological survey collecting finger-prick blood samples from adults in 5 or more first-line villages for *O. volvulus* analysis.
	Step 3b: conduct risk-stratified secondary survey	Conduct epidemiological survey collecting finger-prick blood samples from adults in 20 secondary villages for *O. volvulus* analysis.
Step 4:	**Action**	**Explanation**
Laboratory analysis	Process specimens collected during epidemiological assessments	Analyse specimens from Step 3 (epidemiological assessments) using best available diagnostic tool (Ov16 RDT, OEPA ELISA, etc.).
Step 5:	**Action**	**Explanation**
Analysis and interpretation of data	Develop summary tables	Develop table of summary results for all AUs including number tested, number positive and diagnostic used.
	Generate action maps	Generate maps indicating formation of IUs that require treatment and AUs that can be excluded. Share maps and tables.

AU: assessment unit; ELISA: enzyme-linked immunosorbent assay; IU: implementation unit; OEPA: Onchocerciasis Elimination Program for the Americas; RDT: rapid diagnostic test.

[a] OEM is an iterative process. Some steps may not be required, depending on the particular context and/or the results of previous steps. The decision to follow the steps as listed is at the discretion of each national programme. Endemicity is defined as the detection of infected humans above a threshold that warrants intervention. Identification of infected humans should be the outcome of the process independent of the steps followed in each particular context.

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Its contents are based on reviews from relevant published papers, reports and information from leading scientists in onchocerciasis elimination mapping (OEM), partners and NTD programme managers from some countries that pilot tested OEM. The technical content is based on the third (2019), fourth (2020) and seventh (2023) reports of the Onchocerciasis Technical Subgroup, in which the OEM strategy was discussed and finalized or provisional recommendations were provided. WHO/NTD assisted in reaching out to countries and partners to obtain important information on OEM activities. Information was also extracted from databases managed by the Expanded Special Project for Elimination of Neglected Tropical Diseases (ESPEN) portal for diseases amenable to preventive chemotherapy at the WHO Regional Office for Africa.

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Preparation of this publication was facilitated by a grant from the Mectizan Donation Program. The consultancy to develop this handbook was supported by WHO.

Glossary of key terms and abbreviations

The terms defined below are used in this handbook; they may have different meanings in other contexts.

African Programme for Onchocerciasis Control (APOC)

A global programme launched in 1995 with the objective of controlling onchocerciasis as a problem of public health importance in the remaining endemic countries in Africa not covered by the Onchocerciasis Control Programme in West Africa.

antibody

A protein component of the immune system that circulates in the blood, recognizes foreign substances such as bacteria and viruses, and neutralizes them.

assessment unit (AU)

A convenient local government administrative unit used for exclusion mapping and for the two phases of epidemiological mapping.

biological threshold

The prevalence above which onchocercal transmission becomes self-sustainable.

black fly

Any member of the family Simuliidae. Common names of members of the Simuliidae include blackfly or black fly (two words). The latter is used throughout this publication according to standard entomological notation, indicating members of the order Diptera (true flies), and to the Oxford Concise Medical Dictionary (10th edition). The family consists of over 2300 species, the genus *Simulium* being the largest (more than 1800 species). These insects are usually small, black or grey, with short legs and antennae

buffer zone

An area without transmission of onchocerciasis that lies between and separates transmission zones. It must be wider than the flight range of appetitive blood-hungry vectors and is usually unsuitable for vector breeding (although transmission may be absent because the area is unpopulated by humans – if it is a reserved area or swamp, for example).

clusters

Geographically defined collections of households used to construct a sampling frame in a cluster-sampling strategy.

critical cut-off value

The maximum number of positive tests that can be observed for the survey area to be considered below the biological threshold.

DBS

dried blood spot or sample

district

For onchocerciasis control and elimination purposes, a district is defined as the normal administrative unit for health care management. It corresponds to the administrative unit classified as "Admin 2"; however, in some other countries other names may be used.

ELISA

enzyme-linked immunosorbent assay

EPI quasi-randomised method

A sampling method of the Expanded Programme on Immunization that approximates random sampling through selection in a random direction, starting with a household number from a central location within the cluster, and then continuing to enrol households from the starting household using a nearest neighbour approach, until the required sample size and gender quotas are reached.

EPIRF

Epidemiological Data Reporting Form

Expanded Special Project for Elimination of Neglected Tropical Diseases (ESPEN)

A project of the WHO Regional Office for Africa the aim of which is to accelerate elimination of neglected tropical diseases, including onchocerciasis.

first-line village

The closest community to an active or suspected breeding site, or a river that can support breeding, with no other village or dwellings in between. Most commonly, a first-line village is defined as a community located within 5–10 km of a breeding site or a river thought to be suitable for breeding.

GIS

geographical information system

GPS

global positioning system

hyper-endemic area

An area where nodule prevalence was 40% or more among those individuals examined. These areas were targeted for treatment with ivermectin (Mectizan) following APOC REMO activities.

hypo-endemic area

An area where nodule prevalence was less than 20% during REMO surveys, and not treated with Mectizan through community-directed treatment with ivermectin. OEM focuses on these areas.

implementation unit (IU)

The administrative unit in a country used for mass drug administration.

mass drug administration (MDA)

Distribution of medicines to the entire population of a given administrative setting (for instance, state, region, province, district, subdistrict or village), irrespective of the presence of symptoms or infection; however, exclusion criteria may apply.

meso-endemic area

An area where nodule prevalence was 20% or more but less than 40% during REMO surveys. These areas were targeted for treatment with Mectizan following APOC REMO activities.

NOEC

National Onchocerciasis Elimination Committee

neglected tropical disease (NTD)

A diverse set of 21 diseases and disease groups that disproportionately affect populations living in poverty, predominantly in tropical and subtropical areas. They impose a devastating human, social and economic burden on more than one billion people worldwide.

NTD TAC

Technical Advisory Committee for Neglected Tropical Diseases

Onchocerciasis Control Programme in West Africa (OCP)

A global programme launched in 1974 that operated until 2002 with the objective of controlling onchocerciasis through mainly aerial spraying insecticides against black fly larvae (vector control) by helicopters and airplanes. This activity was supplemented by large-scale distribution of ivermectin from 1989.

onchocerciasis elimination mapping (OEM)

The process of identifying all those areas previously excluded from onchocerciasis control programmes because they had been defined as hypo-endemic or assumed to be non-endemic and must now be reassessed to determine whether or not onchocerciasis is endemic at a level above the threshold at which ongoing transmission is possible.

Onchocerca volvulus

A nematode that causes onchocerciasis, the second leading infectious cause of blindness worldwide after trachoma.

Onchocerciasis Technical Advisory Subgroup (OTS)

A subgroup established in 2017 to advise WHO on OEM methodology (and other relevant topics) in accordance with the terms of reference developed for the subgroup.

PCR

polymerase chain reaction

programme manager

The individual with overall responsibility for planning and executing activities related to onchocerciasis mapping, control and elimination.

prospection

The process of searching for aquatic stages of the onchocerciasis vectors and identifying their breeding locations.

rapid epidemiological mapping of onchocerciasis (REMO)

A tool that rapidly assessed the onchocerciasis situation using palpable nodules and provided geographical information to identify communities likely to be at high risk of infection. It was developed by APOC to identify hyper- and meso-endemic settings focusing on suspected first-line villages.

RAPLOA

rapid assessment procedure for *Loa loa*

registrar/enroller

The individual who records information with the ESPEN collect mobile phone App about each cluster, and enrols individual participants, including gaining informed consent.

RDT

rapid diagnostic test

RSS

risk-stratified secondary survey

standard precautions

A set of infection control practices used to prevent transmission of diseases that can be acquired by contact with blood, body fluids, non-intact skin (including rashes) and mucous membranes. They are a basic level of infection control precautions, which are to be used as a minimum.

survey coordinator

The individual responsible for deploying technicians and recorders to undertake the onchocerciasis survey, and ensuring that all the necessary logistical arrangements are in place, in order that surveys can be conducted efficiently and effectively.

technician

The individual who extracts blood from each survey participant and applies it to the dried blood spot.

trainer/supervisor

The individual responsible for training and overseeing the work of the technicians and recorders, and providing guidance and assistance as needed.

training coordinator

The individual with overall responsibility for identifying, inviting and preparing trainers; choosing and booking the training venue; choosing and preparing sites for community-based training sessions; and making other logistical arrangements necessary for the training sessions to be a success.

transmission zone

A geographical area where transmission of *O. volvulus* occurs by locally breeding vectors, and which can be regarded as a natural ecological and epidemiological unit for intervention.

WHO

World Health Organization

Human onchocerciasis remains endemic in 34 countries around the world.



Introduction

Human onchocerciasis remains endemic in 31 African countries[1], one Asian country (Yemen) and two Latin American countries (Bolivarian Republic of Venezuela and Brazil; WHO, 2019a). This neglected tropical disease (NTD) results from infection with *Onchocerca volvulus*, a parasitic worm transmitted to humans through repeated bites of infected female black flies (genus *Simulium*) that breed in fast flowing rivers and streams, lending onchocerciasis its common name: of "river blindness" (WHO, 2019). In addition to blindness, onchocerciasis causes skin disease, including unsightly skin lesions, debilitating itching and excess mortality among highly infected people. It is also a major risk factor for epilepsy and nodding syndrome (Tekle et al., 2016). In 2022, more than 246 million people were living in areas known to be endemic for onchocerciasis (WHO, 2023a). An estimated 21 million people are actively infected, of whom 1.1 million suffer from onchocerciasis-mediated visual impairment or blindness (GBD, 2017).

In 2009, the World Health Organization (WHO) began shifting programmatic focus from control of disease to elimination of transmission. Treatment with ivermectin, donated under the brand name Mectizan®, is the core strategy for achieving onchocerciasis elimination. *Ending the neglect to sustain the Sustainble Development Goals: a road map for 2021–2030* ("the road map") sets an ambitious target to eliminate onchocerciasis through interruption of transmission, with a milestone for all 34 endemic countries to have stopped mass drug administration (MDA) in one or more foci by 2030 (WHO, 2020b).

Notwithstanding this ambitious aim, there are several challenges to eliminating onchocerciasis, most notably the need to identify all areas where transmission is present that have yet to receive mass treatment.

Previous transmission delineation surveys focused on identifying areas of moderate to high endemicity for priority treatment. Moderate to highly endemic areas were accurately defined and treatment prioritized for those at highest risk of blindness (Annex 1). In the context of elimination, however, areas previously classified as either low-level endemic (hypo-endemic), or areas previously assumed to be non-endemic, must be revisited and assessed. A strategy known as onchocerciasis elimination mapping (OEM) has been developed to identify all remaining areas of active transmission by conducting entomological and epidemiological assessments, in addition to desk review and exclusion mapping, in areas that are naive to ivermectin and previously considered hypo-endemic or of unknown endemicity. In an elimination programme, it is essential to carry out treatment in all areas where transmission is taking place to arrest transmission in hypo-endemic or unknown areas and to prevent spill-over from these areas from reseeding transmission in meso- and hyper-endemic areas where treatment has been delivered. Without completing OEM and accurately determining the extent and intensity of current transmission in ivermectin-naive areas, the onchocerciasis elimination goals will not be reached. Based on the results of OEM, countries and programmes may need to expand MDA programmes into new areas.

[1] Angola, Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Chad, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Equatorial Guinea, Ethiopia, Gabon, Ghana, Guinea, Guinea-Bissau, Kenya, Liberia, Malawi, Mali, Mozambique, Niger, Nigeria, Rwanda, Senegal, Sierra Leone, South Sudan, Sudan, Togo, Uganda and the United Republic of Tanzania.

Scope of this handbook

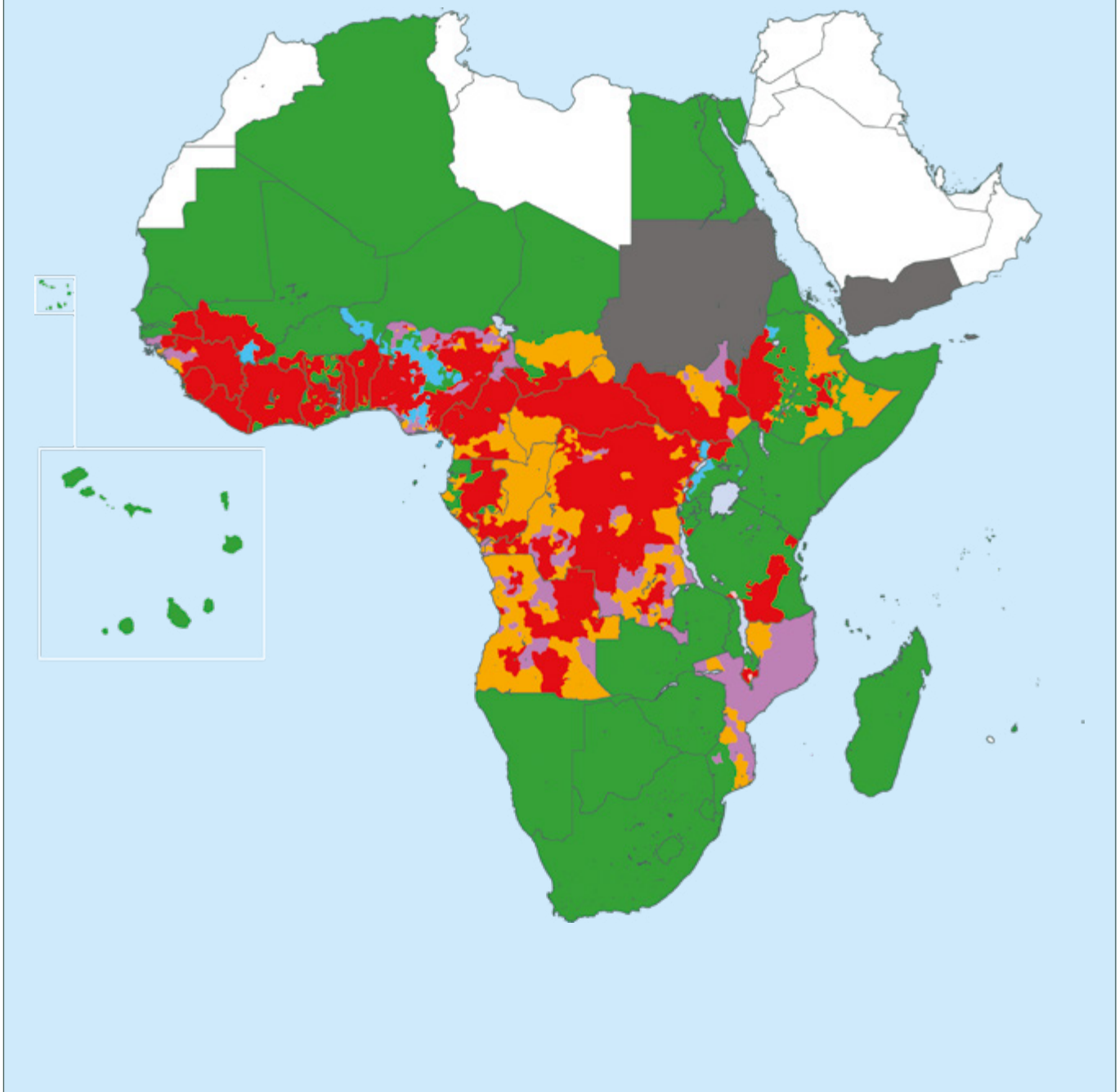
The WHO guidelines for verification of onchocerciasis elimination focus on stopping treatment safely and completing post-treatment surveillance requirements (WHO, 2016). This handbook provides guidance for the section of the Ministry of Health or equivalent ("health ministry") responsible for elimination of onchocerciasis to determine the current transmission status of previously untreated areas (hypo-endemic or unknown/unclassified endemicity) where ongoing transmission is a possibility (labelled in Fig. 1 as "unknown"). It is also relevant for any health ministry onchocerciasis advisory bodies such as the National Onchocerciasis Elimination Committee (NOEC) and/or the NTD Technical Advisory Committee (NTD TAC).

Some hypo-endemic areas have received ivermectin for years as part of efforts to eliminate lymphatic filariasis in areas where both diseases are co-endemic and may need to be assessed using a different strategy to determine if transmission of onchocerciasis has been interrupted or remains active. Formal guidance on this integrated assessment strategy is forthcoming. Other areas are co-endemic for *Loa loa* at moderate or high levels and have not received ivermectin for onchocerciasis due to the risk of serious adverse events. Areas of meso- and hyper-endemic loiasis may require a different strategy. Areas that are estimated as being hypo-endemic for loiasis (prevalence < 20%; Fig. 2) have never reported serious adverse events despite millions of treatments with ivermectin being delivered every year. These areas may be considered appropriate for conducting OEM since MDA could be safely administered if onchocerciasis transmission was discovered (Zouré et al., 2011). However, the decision to proceed in these areas is ultimately at the discretion of the national programme and its advisory bodies and should be made only after careful consideration of all available evidence as to the prevalence of loiasis in a given area.

This handbook covers countries that have not been fully mapped; its scope therefore excludes currently endemic countries (e.g. Bolivarian Republic of Venezuela and Brazil) and previously endemic countries (i.e. countries already free of onchocerciasis) in Latin America. Where countries have already generated evidence for declaring areas as non-endemic, the guidance provided herein should not be applied retroactively. However, sufficient justification for determining where to provide onchocerciasis treatment and which areas to exclude will need to be provided when submitting elimination dossiers.

Its scope is limited to areas previously classified as onchocerciasis hypo-endemic (or now considered unknown) that are (i) ivermectin-naive and (ii) estimated as being loiasis non-endemic. Strategies for areas that are co-endemic for lymphatic filariasis or at risk for loiasis will be addressed in subsequent documents. The endemicity status maps for onchocerciasis (Fig. 1) and loiasis (Fig. 2) are available through the Expanded Special Project for Elimination of Neglected Tropical Diseases (ESPEN) data portal (https://espen.afro.who.int/). The maps do not include countries endemic for onchocerciasis outside the WHO African Region.

Fig. 1. Endemicity status of onchocerciasis in the WHO African Region, 2020



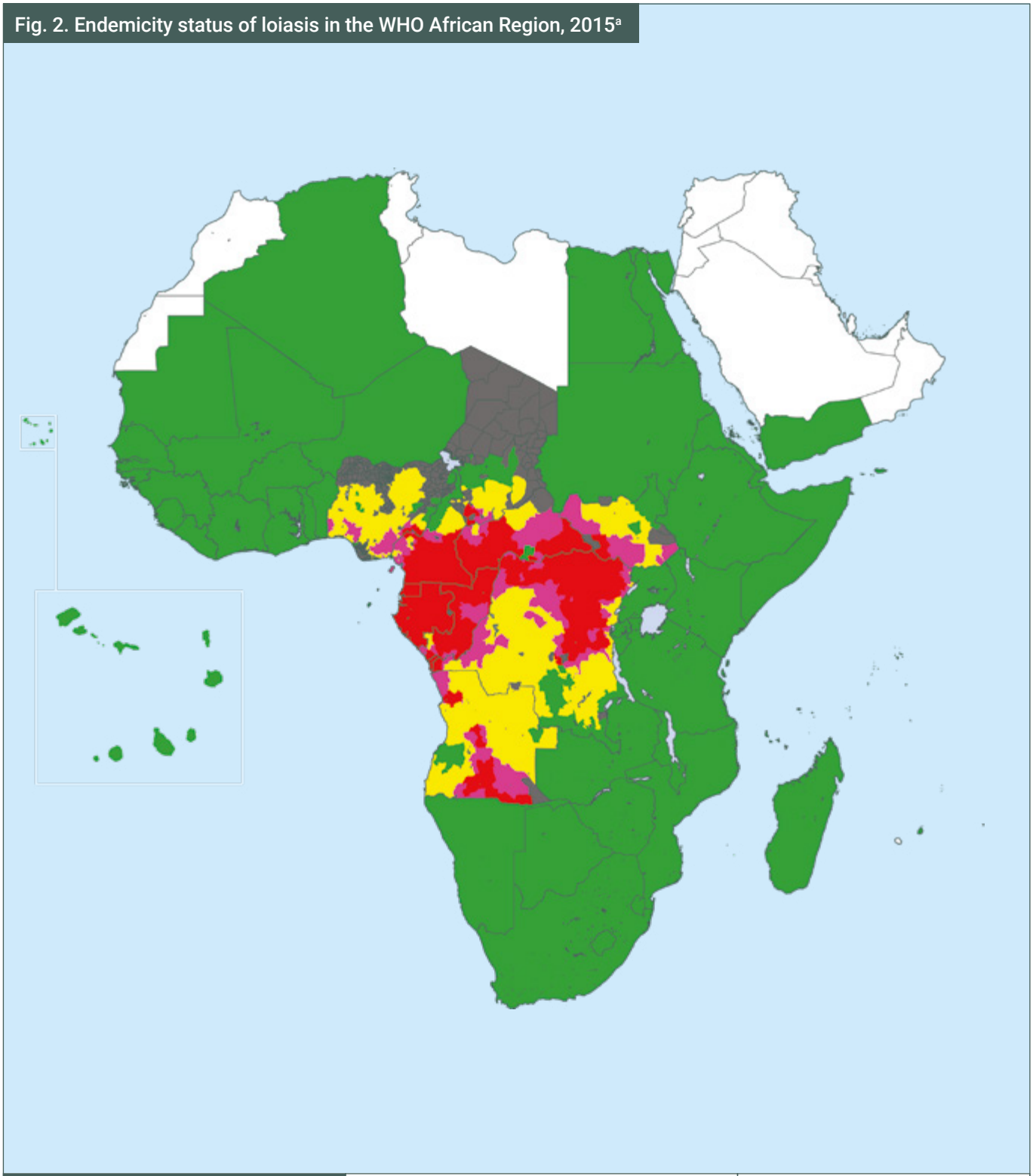
Oncho endemicity/MDA status		0 1,000 2,000 Km				© WHO-AFRO 2023. All rights reserved
Non-endemic	Endemic (requiring MDA)	Endemic (pending IA)	Unknown (under LF MDA)	Endemic (under post-intervention surveillance)	Unknown (consider Oncho Elimination Mapping)	Not reported

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Data source: Data provided by health ministries to ESPEN through WHO reporting process. All reasonable precautions have been taken to verify this information

Map production: ESPEN/AFRO/WHO NTD Mapping project funded by the Bill & Melinda Gates Foundation.

Fig. 2. Endemicity status of loiasis in the WHO African Region, 2015[a]



Loiasis endemnicity					
0 - 5% No risk	>5 - 19.9% Low risk	20 - 39.9% Moderate risk	>=40% High risk	Endemic	No date available

0 1,000 2,000 Km

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Data source: Data provided by health ministries to ESPEN through WHO reporting process. All reasonable precautions have been taken to verify this information

Map production: ESPEN/AFRO/WHO NTD Mapping project funded by the Bill & Melinda Gates Foundation.

The handbook was created to provide a standardized strategy for countries wishing to conduct OEM. It is not meant to be prescriptive but rather to guide countries in the implementation of OEM activities. Thus, countries have the autonomy to apply methods and tools as most relevant to their settings, as long as they conclusively generate the needed data and satisfy the requirements of initiating MDA or otherwise. However, for any country yet to submit an elimination dossier, all the steps in this handbook are recommended. It is designed to provide NTD programme managers and trainers with the skills needed to train staff responsible for implementing OEM surveys. A step-by-step process for conducting OEM is provided, including desk review, exclusion mapping, identification of breeding sites and first-line villages, population sampling for assessment, choice of diagnostic test, interpretation of results, and generation of action maps for the health ministry, the NOEC and the NTD TAC.

Its objectives are:

- to assist onchocerciasis technical focal points in health ministries and community-based health workers in planning and implementing OEM; and
- to provide step-by-step procedures for conducting OEM.

Who should use this document?

This handbook is meant to be used by national onchocerciasis country programmes, and their respective onchocerciasis elimination committees (NOEC or NTD TAC), to follow a standardized process of assessment to determine if there is a need for onchocerciasis treatment or not in previously untreated areas. NTD programme managers and health workers at national and district levels will find it a valuable tool for training lower-level technical staff.

Justification for OEM

In transitioning from control to elimination of onchocerciasis, most endemic countries have intensified efforts to eliminate transmission of *O. volvulus*. A number of countries have demonstrated that it is technically feasible to eliminate onchocerciasis using ivermectin alone (Traore et al., 2012; Miri et al., 2023) or in combination with vector control measures (Katabarwa et al., 2014).

Rapid epidemiological mapping of onchocerciasis (REMO), the main mapping tool used during the era of the African Programme for Onchocerciasis Control (APOC), identified hyper, meso- and hypo-endemic zones in most countries; however, priority for treatment was given only to hyper- and meso-endemic zones. Additionally, REMO surveys took place from 1989 until 2012 and, while epidemiological and entomological conditions are not often subject to dramatic change over short periods, meaningful changes impacting transmission dynamics have likely taken place over the past decade in some areas, particularly given the impact of climate change in sub-Saharan Africa. Regions where no nodules were detected during REMO could still exhibit a microfilaraemia prevalence exceeding 20%. This observation was substantiated when areas initially categorized as hypo-endemic through nodule palpation in Gabon were subsequently identified as meso- or hyper-endemic when assessed using skin snips (Atsame et al., 2016). Thus, the initial REMO baseline data cannot be wholly relied upon to rule out transmission, as it was not specifically designed to identify low transmission, and circumstances may have evolved since its collection (Cantey et al., 2018). The omission of areas of ongoing transmission (previously considered hypo-endemic and left untreated, or where endemicity is unknown/unclassified) will compromise the elimination effort and have been proven to support onchocerciasis transmission in some countries (Katabarwa et al., 2010). Since elimination targets were set for 2020, 2025 and 2030 for some countries in the road map (WHO, 2020), none of these goals are likely to be met if the extent of onchocerciasis infection remains uncertain.

"Elimination mapping" has to be adopted to determine exactly where additional interventions are required and provide rigorous evidence supporting any classification of areas as "non-endemic". Pilot surveys conducted by a number of countries have demonstrated that this strategy works and have been used to identify areas of active transmission; however, this was done in the context of operational research. There is now a critical need to translate this approach into a simple, standardized method for implementing OEM to scale up this strategy. The results of OEM will help national programmes to make appropriate decisions in extending treatment where required.

Challenges in elimination mapping

There are several challenges to conducting OEM. These include ecological changes impacting black fly breeding sites, lack of context-specific data, conducting OEM in cross-border areas, agreeing on the tools and strategies, and adapting strategies in areas co-endemic for other filarial infections (Rebollo et al., 2018).

During REMO surveys, knowledge of the biology of the vectors and their breeding habitats was an important factor (Ngoumou and Walsh, 1993), and so too for OEM. However, the ever-changing ecology due to land use changes, climate change, and population changes and movement could create breeding sites or make them disappear. In most *S. damnosum sensu lato (s.l.)* biting areas, breeding is seasonal and is often absent during the dry season, depending on local geography and ecology. Transmission will therefore be higher typically during the rainy seasons when density of flies is high and bites are frequent. Therefore, the timing of OEM has to consider transmission seasonality in the area to be mapped so that breeding site assessments are conducted at times most likely to identify evidence of black flies.

Additionally, previous knowledge and data on black fly breeding habitats and ecological preferences have been largely based on hyper- and, to a lesser extent, meso-endemic areas. OEM focuses on known or suspected hypo-endemic areas, in addition to areas of unclassified endemicity, where transmission dynamics, vector ecology and host–vector interactions are likely to differ from those observed in areas of higher endemicity. Therefore, careful and thorough investigations are required during OEM to ensure no areas are missed that are capable of sustaining transmission. Hypo-endemicity is a distinct, under-studied ecological and epidemiological setting. Hypo-endemic areas may not always have active breeding of black flies and infected individuals may be infected by flies from neighbouring meso- or hyper-endemic settings.

Onchocerciasis is widely observed to traverse borders. This has mainly been associated with river systems forming boundaries of two countries or having their source in one country and traversing across another country. Cross-border collaboration is necessary if any mapping is to be successfully implemented. Recent cross-border efforts in Africa have yielded good returns and have accelerated the interruption of transmission of onchocerciasis (Lakwo et al., 2018). This experience can be used in OEM so that both sides of the border can be mapped at the same time. Most borders in Africa are porous, and there is usually free population movement, but this complicates elimination efforts if the intervention is not coordinated across the borders. Furthermore, this is not only confined to population migration but also to black fly migration along river basins since in some countries river systems are shared.

In some countries, there are areas previously considered not endemic where there is little or no evidence to disregard the potential for transmission because they have never been mapped or assessed properly. Furthermore, REMO assessments are now out of date and they may need to be reviewed and updated; while they should be used to guide OEM prioritization, it is not advised they be the only criterion.

By following the steps listed in the simplified strategy laid out in this document it is possible to overcome most, if not all, of these challenges. While the tools available currently are not perfect, when used properly they can generate valid, useful programmatic data for decision-making and accelerate progress towards onchocerciasis elimination. Entomological surveys are resource-consuming and can take a long time. They also require expertise that is not always present. Even in hyper- and meso-endemic settings, identification of breeding sites by experts could take days in one single setting. In addition, some hypo-endemic areas may not have breeding sites and be dependent on breeding sites located in neighbouring hyper- or meso-endemic settings. Therefore, given that endemicity status is based on the presence of human infections, health ministries/ NOECs may decide when the entomological studies described under Step 2 are not required and the epidemiological surveys can be conducted directly.

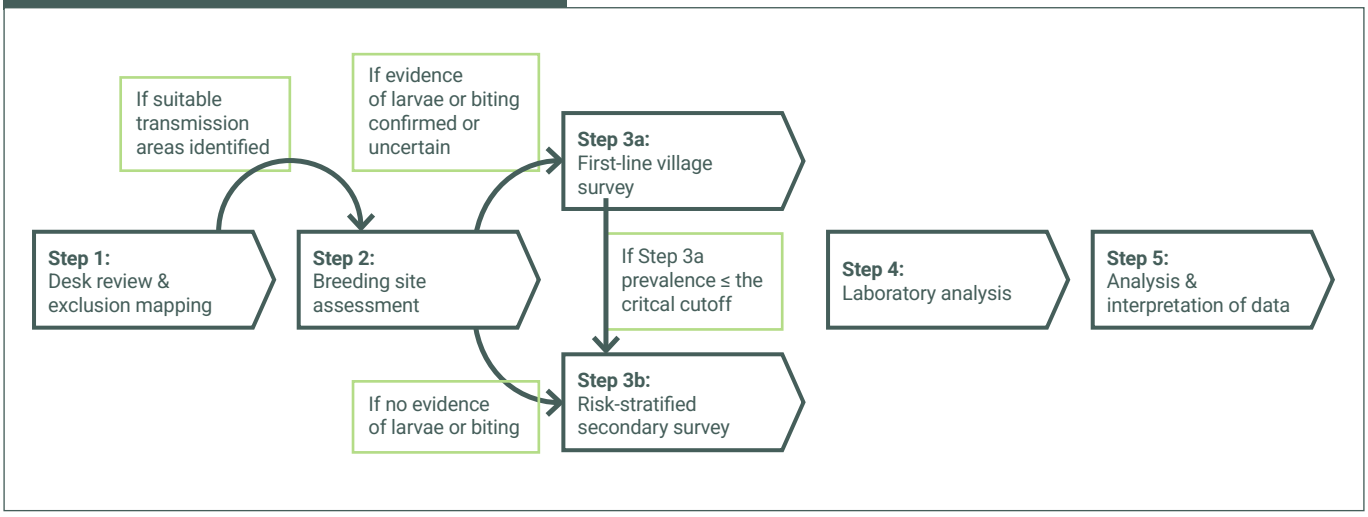
Overview of the OEM process

Onchocerciasis elimination mapping refers to identifying all those treatment-naive areas previously excluded from onchocerciasis control programmes because they had been classified as hypo-endemic or assumed to be non-endemic. These areas must now be reassessed to determine whether or not onchocerciasis is endemic. Endemicity is defined as the presence of anti- *O. volvulus* antibodies at a level above a threshold that has been agreed as a proxy for active ongoing transmission. OEM consists of five distinct steps, which can be further broken down into smaller parts. An overview is given in Fig. 3. A more thorough diagram outlining the methods and options can be found in Annex 2.

The steps are explored in greater detail in Chapters 1–5. Briefly, the steps are:

1. **Desk review and exclusion mapping (Chapter 1).** The desk review collects entomological, epidemiological and case report data on onchocerciasis from the areas under consideration, in addition to recent hydrological maps and geostatistical modelling results, where available. The local knowledge of onchocerciasis and black flies by community members should be collected and considered when available. All data should ideally be geo-referenced and can be used to determine if and where physical surveys are needed. River systems and their tributaries that are potential breeding sites for black flies should be identified and marked. The desk review is also critical for countries to define the assessment unit (AU). Broadly, this should be the administrative level at which MDA would be conducted, if required. Given the focal nature of onchocerciasis, and depending on the context, countries may need to delineate treatment areas in units smaller than the administrative units or districts.

Fig. 3. Overview of steps in the OEM process



NOTE: If desk review and exclusion mapping indicates that there are possible suitable areas for the presence of blackflies but no evidence of larvae or biting is found during breeding site assessments, and thus no confirmed first-line villages are identified, programmes can decide to proceed directly to Step 3B: risk-stratified secondary survey or Step 3A: surveys on suspected first-line villages.

2. **Breeding site assessment (Chapter 2).** During the appropriate season, a small team visits rivers in targeted AUs to conduct prospection for black fly larvae and spot checks for biting adult flies. Other evidence to collect through discussion with residents around the potential breeding sites includes their experience of adult black fly biting in the community, and seasonality and daily timing of bites. If larvae are found/adult flies are caught, or if the river is suspected to be suitable for black fly breeding, then nearby village names and coordinates are noted by the survey team. These villages are then classified as *first-line villages* for the next step in OEM.

3. **Epidemiological assessments (Chapter 3).** When Step 2 is able to identify first-line villages (due to proximity to breeding sites or to adult black fly presence), a two-step epidemiological survey is conducted as follows:

   3A: First-line village survey: In each AU, at least five first-line villages will be identified. Sampling will include a minimum of 100 adults aged 20 years or older at each location. From each selected individual, a dried blood spot or dried blood sample (DBS) will be collected. Small (10 uL) blood samples (from a finger prick) are blotted on an absorbent paper. The sample is then allowed to dry for storage and later analysis.

   3B: Risk-stratified secondary survey (RSS): The purpose of RSS is to reduce the chance that any unknown areas of transmission are missed. It is a spot check to be conducted only if the first-line village survey has not yielded conclusive results or if the breeding site assessment fails to identify first-line villages due to a lack of black fly larvae or biting in suitable areas. Full details are given in Annex 2 where Step 3B is a provisional recommendation pending endorsement from the WHO Onchocerciasis Technical Subgroup (OTS) for programmatic use. Programmes opting for other methods to conclusively rule out transmission should compare it to the provisional strategy.

4. **Laboratory analysis (Chapter 4).** Given the current available and recommended diagnostics, DBS collected from participants in Steps 3A and/or 3B are properly packaged and transported to a field laboratory or central laboratory for analysis by SD bioline Ov16 rapid diagnostic test (RDT). Point-of-care testing by SD Bioline Ov16 RDT directly from finger-prick blood is not recommended due to inferior performance relative to laboratory-based testing with DBS. Standard operating procedures for the collection of DBS are included in Annex 3.

5. **Managing and interpreting data (Chapter 5).** The results will be analysed at the village and AU level and interpreted relative to the appropriate *O. volvulus* critical cut-off for the diagnostic test used. Results should be reported in a comprehensive table and shared with the health ministry/NOEC. Action maps will also be generated to indicate which IUs require treatment and which can be excluded.

NOTE: Steps 4 and 5 will occur immediately following Step 3A. If Step 3B is warranted, Steps 4 and 5 will occur a second time after Step 3B is complete. However, programmes may decide to conduct Steps 3A and 3B concurrently (see note in chapter 3.5).

Chapter 1.

Step 1: Desk review of existing data and exclusion mapping

Step 1 describes the initial steps in the OEM process, which involve a thorough desk review of existing data and exclusion mapping to identify suitable areas for further investigation.

Importance of desk review



The desk review process is essential for cost–effective and efficient onchocerciasis elimination efforts. By using existing data and environmental models, the review helps narrow down areas that need ground mapping, thereby saving resources and focusing efforts on the most critical areas. The data collected support the planning and implementation of further steps in the OEM process, ensuring a systematic approach to onchocerciasis elimination.

The chapter underscores the importance of thorough planning and data analysis in the fight against onchocerciasis, emphasizing collaboration among experts and the use of advanced geospatial tools to inform decision-making.

Objectives of the desk review



The desk review aims to identify eligible areas for entomological teams to visit and search for breeding sites of *Simulium* vectors or evidence of biting by adult flies. This step is crucial to minimize costs and resources by leveraging already available information to decide whether ground mapping is necessary. The desk review process involves:

Team formation:



A team with expertise in onchocerciasis, including an epidemiologist, an entomologist and a geographer is convened. Members of the NOEC are often consulted.

Data sources:



The review involves gathering data from various sources such as national databases, research institutes, and published literature. Key data include epidemiological information, entomological data on black flies, maps, and local health reports.

Assessment units:



The AU is typically defined at the district or subdistrict level. This choice is influenced by local disease distribution, district size and ecological complexity. The team uses this unit to conduct exclusion mapping and epidemiological surveys.

Desk review process



The review process involves several steps:

Identification of river systems:



Rivers are crucial as *Simulium* larvae breed in fast-flowing, well-oxygenated rivers. Maps and satellite images help identify these rivers and potential breeding sites.

Environmental suitability models:



These models use environmental data (e.g. climate, topography) to predict the suitability for onchocerciasis transmission. The data help exclude areas predicted to be unsuitable and prioritize those that need further assessment.

Data collation:



Information from epidemiological and entomological data, ivermectin treatment records, hydrological and geological maps, and local reports is collated to define AUs requiring further investigation.

Exclusion mapping:



The team marks uninhabited or unsuitable areas (e.g. game parks, deserts, lowlands) on maps. Suitable areas are highlighted, especially those near rapid changes in elevation or known breeding sites.

1.1 Background

Simulium flies choose their larval breeding sites in fast-running rivers. A major determinant of onchocerciasis risk is the location of a village in relation to the nearest breeding site. A first-line village is defined as the closest community to an active breeding site, with no other village or dwellings between that village and the breeding site. In most settings this means villages within 5 km of a breeding site, but it can be further away in some settings (OTS3, 2019). These villages are in maximum contact with black flies, while second and third-line villages benefit from a certain protective effect from the buffer provided by the first-line villages. Onchocerciasis is therefore most severe in first-line villages, even if these are located at a fair distance from the river (Katabarwa et al., 2018). The OEM process begins with the identification, through review of existing resources and data, of areas that may be suitable for onchocerciasis transmission.

1.2 Defining the Transmission Zone and Assessment Unit

The distribution of onchocerciasis and the endemicity levels attained are determined by the ecology and behaviour of the black fly vector. In addition to the fact that the flies breed in fast-flowing, well-oxygenated rivers/streams with nutrients, one other behavioural trait is that they have an effective appetitive (blood-hungry) flight range, unlikely to exceed 20 km. This means that the highest onchocerciasis transmission is mainly in villages that are within a 20 km radius of breeding sites. By this, the villages severely affected by onchocerciasis are almost invariably located within a 20 km radius of the vector-breeding sites.

During efforts to control onchocerciasis, identification of endemic zones where mass ivermectin distribution should be implemented was based on the results of REMO (Ngoumou and Walsh, 1993; WHO, 1995). REMO relied on clinical examination for onchocercal nodules from a sample of 30–50 resident adults aged > 20 years in selected villages. Given the ecology and behaviour of the vector flies, a random selection of villages for REMO would be ineffective and epidemiologically inappropriate. Therefore, the selection of REMO villages was optimally biased towards those black fly breeding sites (situated in the likely worst areas) with due regard to adequate geographical coverage. Hence two types of villages were selected on both the main river and the tributaries for the exercise: high-risk or first-line villages and second-line villages. First-line villages are villages closest to the breeding sites (without other human settlements between them and the river). For every high-risk village, a second-line village is selected 15–20 km radius to the right and to the left of the first-line or high-risk village. The objective of the second-line village is to obtain some indication of the distribution and overall severity of the disease. The overriding consideration in selecting the village/site is its location in relation to potential breeding sites of black flies.

In line with WHO guidelines (WHO, 1992; WHO, 1998), the results of REMO exercises were grouped into four categories: hyper-endemic zones where the community nodule rate is ≥ 40%; meso-endemic areas where the community nodule rate is ≥ 20–39%; hypo-endemic areas where the community nodule rate is 1% < 20%; and non-endemic area where the community nodule rate is 0%. With the introduction of the community-directed treatment with ivermectin (CDTI) strategy by APOC, MDA was limited to areas where the community onchocercal nodule rate was ≥ 20%. The CDTI zones are classified as areas whose members are at risk of developing severe and disabling ocular, dermal or lymphatic complications. REMO was used by APOC to identify the priority areas for CDTI and was also used to estimate the numbers of individuals requiring treatment in 20 African countries (Noma et al., 2002; 2014). This principle of selecting REMO villages has continued to guide the selection of villages for epidemiological and entomological evaluation of onchocerciasis elimination programmes (Nwoke et al., 2023).

Defining transmission zones is important in demarcating the epidemiological units for intervention. A transmission zone (TZ) is defined as a geographical area where transmission of *O. volvulus* occurs by locally breeding vectors, and which can be regarded as a natural ecological and epidemiological unit for intervention (WHO, 2016). The TZ is often defined based on the geography of onchocerciasis prevalence and vectors, taking into account the biology of transmission. However, in cases where onchocerciasis seems to be distributed over large areas, it may be necessary to divide a large TZ into suitably sized areas which are convenient as implementation units (while taking into account the biology of transmission as far as possible). These areas can be defined as operational transmission zones (OTZs) and will be defined on a case-by-case basis by the NTD programme with expert advice from the NOEC, taking into account important local factors (such as hydrology). As TZs are programmatically difficult to define, countries can decide to use their administration units as a proxy, noting that several units can be part of a large TZ. Further, what is required following OEM is the number of administrative units that require treatment.

The AU will be used for exclusion mapping and for the two phases of epidemiological mapping. It is usually defined as a convenient local government administrative unit. The AU should be identified before the assessments and mapping of designated AUs take place. In most situations, either the second or third administrative level (usually the district or subdistrict) is suitable as the AU. Normally, the choice of which administrative level will constitute the AU is made at the national level by the national NTD programme of the health ministry with advice from the NOEC, and the district (or health district) is often considered to be the most suitable. However, the choice should also take into account feedback received from lower administrative units on the distribution of the disease within those units, and also the size of the districts and the ecological complexity. If onchocerciasis transmission is believed to be highly focal, a lower administrative level (for instance, subdistrict) might be chosen as the AU, whereas if the infection is more widespread, a higher administrative level (for instance, district) might be chosen. Efforts should be made to estimate the population in the AU using census data or other sources. The average population size of an AU may vary within and among countries. In many cases, the AUs and IUs may be the same. However, guidance from the health ministry/NOEC in defining the AU/IU may be required.

Existing administrative boundaries can be used for AU boundaries for convenience when conducting OEM. However, caution must be exercised where rivers demarcate administrative boundaries, as is often the case. Since it is most convenient to use a district as the AU, when considering the number of districts to be included during planning, all districts on either side of a suitable river border should be included. Where the border is international, then cross-border collaboration locally and nationally is required.

1.3 Desk review

The desk review aims to define eligible areas for entomological teams to visit and search for breeding sites of *Simulium* vectors or evidence of biting by adult flies. Generally, ground mapping of diseases is expensive and requires a significant amount of resources. To minimize costs, every effort should be made to access already available information, assess its quality and decide whether ground mapping is necessary. In many cases, appropriate data might already exist and be able to provide enough information for making decisions on whether physical OEM is necessary.

NOTE: All villages with a known history of onchocerciasis in the area of interest should be carefully considered for inclusion as potential first-line villages during desk review (Step 1), breeding site assessment (Step 2) and/or epidemiological surveys (Step 3A).

1.3.1 Desk review process

The programme should convene a small team with a strong background in onchocerciasis and other NTDs to undertake desk review. The team should consist of at least one epidemiologist with a good quantitative understanding of onchocerciasis, an entomologist with the relevant understanding of the *Simulium* vectors and a geographer with a good knowledge of the study area (Ngoumou and Walsh, 1993). If possible, members of the NOEC and/or NTD TAC should be on the review team. This team should undertake a review of historical data, paying particular attention to coordinates or references to specific locations within each AU. It would also be useful to include in the team staff with good knowledge in geographical information systems (GIS) in order to leverage all the spatial data that can support the desk review process, and ultimately restrict the areas to be targeted for OEM. The following sources of information (in addition to others known to the programme) should be considered.

1. The first place to consult should be the NTD TAC, the NOEC and the national onchocerciasis programme management in the health ministry, to access available existing and historical data on disease burden and information on vectors from the national databases. Medical schools and researchers in various universities and research institutes as well as national libraries can also be visited for some archived records.
2. Epidemiological data from any source, for example programme reports, scientific literature, published literatures from partners (OCP, APOC, WHO, ESPEN, nongovernmental organizations, universities, research institutes, etc.).
3. Review of entomological data focusing on the presence of black flies, but also collating information on cytospecies, biting rates, infection/infective rates, etc., where available.
4. Review existing REMO maps and any other onchocerciasis assessments that have been conducted in the area.
5. Contact local health authorities and community health volunteers (by phone or email) to gather information on clinical cases of onchocerciasis presenting at health centres, or complaints of black fly biting in the local area.
6. Review data on ivermectin treatment in neighbouring areas.
7. Review recent and available hydrological and geological maps and geospatial modelling estimates.
8. Collate the data from all of the above steps to define AUs that require further investigation, and exclude empty/unsuitable areas. Such areas can be exlcuded if they are without human population, without rivers or are totally unsuitable for black fly vectors to breed.

1.3.2 Geospatial modelling

The transmission and distribution of onchocerciasis are linked to the distribution of its vector, black fly species of the genus *Simulium*, and also to the presence of human hosts and circulating *O. volvulus*. Therefore, for the occurrence of transmission, there must be suitable conditions for the vector populations to breed and thrive, and for the parasite to complete its life cycle in both human and vector hosts. Most of the factors that make onchocerciasis transmission possible are environmental (climate, topography and vegetation) and human-related, such as settlement distribution.

In recent years, ecological niche modelling methods, generally applied to species distribution modelling, have been used to model the distribution of diseases (Cano et al., 2014; Mylne et al., 2015; Messina et al., 2016; Pigott et al., 2016; Messina et al., 2019; Simpson et al., 2021; Deribe et al., 2020). The purpose in all cases is to identify the environmental factors that enable the disease to be present in specific areas and to delineate these areas geographically. In 2021, the Local Burden of Disease project led by the Institute of Health Metrics and Evaluation developed a predictive model of the environmental suitability for onchocerciasis in Africa (Cromwell et al., 2021). They combined survey data available in the ESPEN NTD Data repository, mostly through REMO surveys, gathered from the scientific literature, with a suite of environmental datasets related to factors known to be associated with the presence of the onchocerciasis vector: climatic factors (aridity, precipitation and day-time temperature), vegetation, topography (slope and elevation) and hydrography (distance to streamways) factors, and urbanicity, assuming rural areas to be more exposed to onchocerciasis transmission. They used a machine learning algorithm called *boosted regression trees* to calculate the probability of having onchocerciasis transmission across the African continent in regular areas of 5 km x 5 km based on the selected environmental factors (Fig. 4). More details on the methodology and results are given in the publication presenting this work (Cromwell et al., 2021). A secondary product of this environmental suitability map was a binary map identifying the areas where the occurrence of onchocerciasis is expected with high certainty (see Annex 1).

NOTE: The information obtained from these geostatistical models is intended to supplement programme data; it should not replace or override any observed indicators of transmission or environmental suitability.

This map of predicted environmental suitability for onchocerciasis can be used to support decisions on OEM at different steps of the process.

1.3.2.1 Environmental Suitability Models

An environmental model can be used to exclude areas that are predicted to be unsuitable or which show low suitability for the occurrence of onchocerciasis, and to identify priority areas for OEM based on the level of predicted suitability. **This information should be used in combination with all existing empirical evidence that has been collected by the desk review team and is not to be used as the sole source of evidence for defining areas as suitable for breeding sites.** To facilitate the management and interpretation of environmental suitability data, ESPEN has generated some key indicators around these datasets at the implementation unit level, namely:

- ✓ the minimum, median and maximum values of the predicted environmental suitability across each implementation unit;
- ✓ the percentage of the implementation unit area where onchocerciasis transmission is expected based on environmental conditions;
- ✓ based on the median of the predicted environmental suitability, a categorical classification of each implementation unit based on the following intervals:

- median suitability up to 0.15: *unsuitable*
- median suitability ≥ 0.15 to < 0.25: *very low suitability*
- median suitability ≥ 0.25 to < 0.50: *low suitability*
- median suitability ≥ 0.50 to < 0.65: *moderate suitability*
- median suitability ≥ 0.65 to < 0.85: *high suitability*
- median suitability ≥ 0.85: *very high suitability*

Some examples of maps generated using this classification are provided in Annex 1.

Country teams should contact ESPEN for support with up-to-date environmental suitability maps and scores.

1.3.3 Exclusion mapping and identification of suitable areas

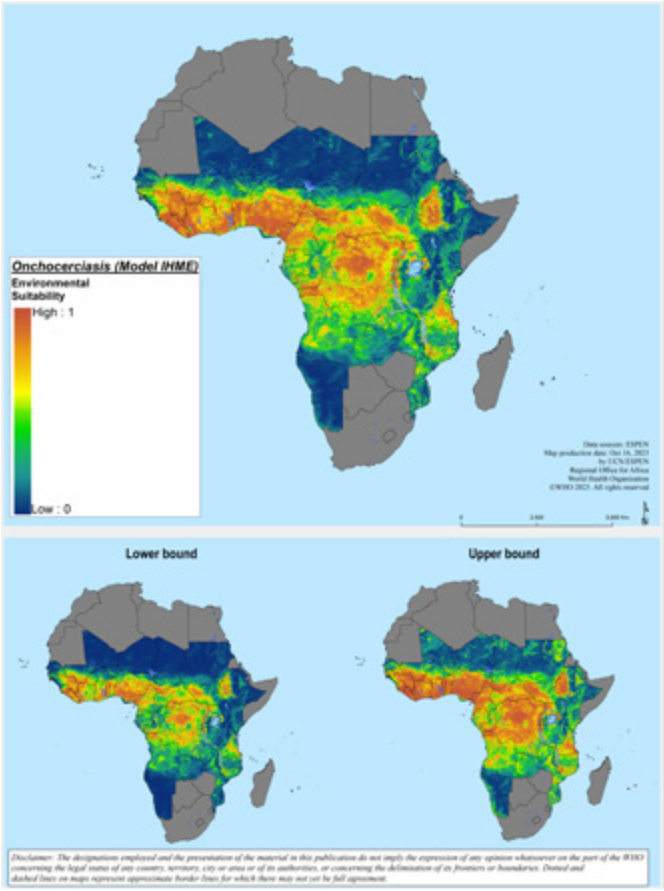
Once the above information is compiled, the team should combine it with maps of the area of interest to select areas for breeding site assessment and eventual epidemiological surveys. The team's geographer should use high-quality digital maps if possible, or else paper maps of scales 1:50 000–1:250 000. Examination of a geological map (typically 1:1 000 000 or 1:2 000 000) gives a good indication of the likelihood of rapids on rivers. Google Maps provide excellent detail in many countries and can be freely accessed. On paper or digital maps, elevation, rivers and other major geographical features should be visible. Overall, any topographical map that has a clear portrayal of the river systems, the relief of the region and the names of villages/communities can be used for the exercise. Results from the desk review process (including review of geospatial suitability modelling results) should serve as a starting point for the following steps in each area to be mapped.

1. Identify the river systems.
2. Create the AU around the river system.
3. Exclude empty and unsuitable areas: mark off uninhabited game parks and forest reserves, but if there are populations living in or very close to such areas, they should not be automatically excluded unless it is certain there is no potential for transmission. Also mark off any other uninhabited areas, lowland areas, flatland, deserts, deltas and swamps.
4. Mark potentially suitable areas on the maps: highlight any rivers that flow through rapid changes in elevation, and search Google Maps or other satellite images for white water rapids, which are often easy to see. If there is information on black fly breeding sites from this area or surrounding districts, this should be marked on the map. This should allow an overview of potentially suitable or uncertain areas within the targeted AU. Suspected suitable and uncertain areas should be physically mapped. All suspected suitable or uncertain areas should be included in the AU.

Breeding site selection. Areas predicted to be environmentally suitable for the transmission of onchocerciasis can be included in the list of those to be prospected or searched for *Simulium* breeding sites. As explained in 5.4 and Annex 4, GIS analysis can be used to obtain spatial data on river networks and estimate the distance from different locations to the closest streamways, even with basic knowledge of GIS analysis. For this, the desk review teams would need to have a list of georeferenced locations, namely communities with available geographical coordinates, from the areas identified as potentially requiring OEM, which should be combined with the river network layer within a GIS environment. Then, it is possible to run calculations within the GIS framework to estimate the straight-line distance from every location to the nearest stretch of the river network. All the GIS applications such as QGIS or ArcMap are provided with tools to run this operation. In the absence of GIS capabilities, country teams should use paper maps to identify the communities and estimate the distance to the river. Useful resources of spatial data on river networks and georeferenced settlements are given in Annex 4.

The team should compile the desk review, geospatial modelling review and exclusion mapping report. Once the NOEC or NTD TAC has decided which areas need to conduct OEM, the team can then proceed to Step 2 breeding site assessment or Step 3 epidemiological assessments (if the health ministry/NOEC determines that Step 2 is not required).

Fig. 4. Onchocerciasis environmental suitability modelling results



Chapter 2.

Step 2: Breeding site assessment

Step 2 explains the method for identifying potential black fly breeding sites to assess the risk of onchocerciasis transmission. The primary focus is on thorough field assessments conducted by field teams to detect evidence of black fly larvae or adults in pre-selected river locations.

Objectives



The breeding site assessment aims to:

- confirm the presence of black fly breeding;
- identify suitable conditions for breeding, especially in areas with limited historical data; and
- inform decisions on proceeding with epidemiological surveys.

Challenges and considerations



- **Seasonality:** Breeding is often seasonal, with peak activity typically during or after the rainy season. Dry season assessments may give false-negatives if breeding sites are not active.
- **Environmental factors:** Changes in river conditions and accessibility can impact the ability to detect breeding sites.
- **Community involvement:** Mobilizing community support and ensuring their safety is crucial. Community drug distributors play a key role in this process.

Process

1. Team formation:

A team of 2–3 individuals with expertise in entomology and local ecology is formed to conduct site visits.

2. Site selection:

Pre-selected sites are chosen based on historical data, environmental models and community reports. The desk review and exclusion mapping phase (Chapter 1) heavily inform these selections.

3. Timing

The timing of the assessment is crucial. The breeding of *S. damnosum*, the primary vector, may be seasonal. Assessments are ideally conducted at the end of the rainy season or the beginning of the dry season to ensure adequate water flow in rivers and easier field movements.

Field procedures

1. Visual inspection

Teams inspect rivers for fast-flowing, well-oxygenated water, which is ideal for black fly larvae.

2. Larvae collection:

Teams search for and collect larvae or pupae from submerged vegetation, rocks and other surfaces. Collected specimens are preserved for laboratory analysis to confirm the species.

3. Community engagement

Teams gather information from local residents about biting flies, including seasonality and biting patterns. Community members can provide valuable insights, especially in identifying breeding sites.

4. Spot checks

In cases where breeding sites are inaccessible, spot checks of adult flies in nearby communities are conducted. This involves capturing flies that land on exposed skin, which helps verify the presence of black flies.

5. Laboratory confirmation

Collected larvae are examined in a laboratory setting using a dissecting microscope. *S. damnosum* larvae are identified by their distinctive dorsal tubercles and attachment organs. For *S. neavei*, larvae attach to freshwater crabs, requiring a different assessment method using traps baited with meat or fruit.

Decision-making

Based on the findings, decisions are made on whether to proceed with further epidemiological surveys. If larvae or adult flies are found, nearby villages are classified as first-line villages for subsequent epidemiological assessments. If no evidence is found, the team may proceed to more detailed surveys or confirmatory steps.

The breeding site assessment is a visit to pre-selected locations on rivers to look for evidence of and/or suitable conditions for black fly breeding. This more rigorous approach is useful, especially when there is no historical programmatic knowledge of breeding sites. However, it has serious limitations such as being resource and time-consuming, and it is not always possible to identify the evidence of existing breeding sites (seasonality) or they may not even exist (some hypo-endemic settings may be dependent on hyper- and meso-endemic sites, where flies can reach the hypo-endemic areas and infect individuals being a reservoir for infection). Countries may decide whether to conduct breeding site assessment and to what extent before making the decision to continue with epidemiological surveys (Steps 3A or 3B).

A team of 2–3 people should visit the rivers or streams and search for suitable sites where they can collect larvae or pupae of black flies. Depending on whether *S. damnosum* or *S. neavei* is suspected, the prospection should be done differently. Spot check collection of the biting black flies could also be conducted during the visit. If breeding sites are inaccessible due to environmental conditions or other factors, teams should still proceed to communities near suspected breeding sites if possible, to obtain information on vectors from residents and to conduct spot-check collections of black flies (see 2.4).

NOTE: Countries are encouraged to collect data during Steps 2, 3A and 3B using mobile phones or tablets. This practice lends itself to more accurate data and easier management of results. Programmes in the WHO African Region can take advantage of ESPEN Collect, a free mobile data collection tool designed to support disease-specific surveys. More information on ESPEN Collect, including standard forms for onchocerciasis surveys, can be found on the ESPEN website (https://espen.afro.who.int/tools-resources/espen-collect).

2.1 Timing of the assessment

It is important to consider the appropriate timing of breeding site assessments in order to maximize the possibility of identifying black flies. It has been noted that in most areas endemic for *S. damnosum*, breeding is seasonal, often stopping during the peak of the dry season when some rivers may completely dry. This may give the false impression that there is no breeding. Conversely, but unusually, the dry season can sometimes create better breeding conditions in some rivers, where a drop in water level exposes rocky substrates or reduces flow to a suitable level for black fly larvae. If OEM breeding site assessments are planned for the peak dry season, this must be supported by evidence that suitable river conditions exist during the dry season.

Where there is no evidence that the peak dry season could be suitable for black fly transmission, entomological activities should be carried out at the end of a rainy season or the beginning of a dry season. These are for two reasons: (i) most rivers would still have adequate water flow to support suitable larvae; and (ii) at this time there will be little or no rain, making field movements easier in difficult-to-traverse terrain, common in rural areas.

2.2 Community mobilization

Community mobilization and sensitization for breeding site assessments is particularly important for reasons of safety and facilitation of activities by the community (WHO, 2023b). Mobilization enables individuals and community groups to actively contribute to bringing about change. These actions involve identifying resources, disseminating information, garnering support, and promoting collaboration between the public and private sectors within the community. Mobilization is frequently characterized as the process of establishing community or collaborative partnerships, engaging with the community or building coalitions.

Specifically, for entomological assessments, community mobilization can lead to information pertaining to:

- identification of breeding sites through the awareness of community members on the presence, abundance and seasonality of biting black flies;
- safety of breeding sites, such as the presence of wild animals (e.g. crocodiles and hippopotamuses), dangerous sections of the river, or insecurity due to conflicts;
- prospection of breeding sites, by identifying suitable areas for prospection and assistance with activities.
- clearing and establishing capture points by community members;
- identification of locations for setting and protecting black fly traps; and
- collection of black flies through human landing collections by trained villagers.

Before and after all entomological activities, it is crucial to reach out to community leaders, providing them with information about the survey's objectives, its significance for public health outcomes, and the importance and roles of community involvement. It is imperative to incorporate community feedback and adjust the survey or prospecting activities accordingly. Community drug distributors, who are typically esteemed and trusted within the community, should play a role in community mobilization. This involvement should be discussed with village leaders during the mobilization planning phase. In areas where ivermectin has not been previously administered, the team should initiate a new round of community mobilization using appropriate onchocerciasis education materials, with a particular emphasis on the upcoming evaluations in the villages and their purpose.

2.3 Breeding site assessments and identification of *S. damnosum* and *S. neavei*

Detailed procedures for the identification of breeding sites, *S. damnosum* and *S. neavei* are contained in the *Entomological manual for onchocerciasis elimination programmes* (WHO, 2023). The breeding sites for *S. damnosum* are always in fast-flowing and well-oxygenated water, which often results in white water rapids. The larvae and pupae can also live in niches away from white water sections such as cascades, waterfalls, lake outlets, dam spillways, small tributaries and springs. Any fixed object submerged in water can be used as a larval attachment. Typical characteristics of black fly breeding sites are:

- the presence of fast-flowing water 0.5–2 m/s. On large rivers, one might not always see whitish or churning water but can observe that the flow is fast;
- the sound of rushing water and water breaking due to a rocky bottom creating rapids;
- the availability of supports for the vector to lay its eggs on (e.g. vegetation trailing in the water, submerged rocks, emerged supports kept wet by water spray, crabs for some *Simulium* species); and
- nearby community members complaining of biting flies.

2.3.1 Equipment

The following equipment is needed for *S. damnosum* breeding site assessments:

- maps marked with suitable and uncertain areas from desk review,
- rubber boots (Gumboots) or hip-high wading boots,
- life jackets,
- machetes,
- fine forceps,
- hand lens/magnifying glass,
- sample collection bags,
- portable global positioning system (GPS) or phone with GPS function,
- river prospection form, ideally electronic (see Annex 5),
- notebook and pen/pencil, and
- ethanol, glacial acetic acid (to make Carnoy's solution fresh if larvae are found).

2.3.2 *S. damnosum* breeding site assessment procedure

The following steps should be followed for *S. damnosum* breeding site assessments:

1. On arrival, inspect the river from the bank for potentially suitable black fly breeding habitats, and take photos. Look for any possible access to water. If there is no access, use a machete to cut an access route to the river, taking care not to cut through crops or farmland.
2. On entering the water, search for white waters, overhanging vegetation and trailing plants in the water. Remove submerged leaves, grasses, sticks and stones, then examine them for any attached larvae or pupae (see Fig. 5(B) and Fig. 6).
3. In some sites where there are rocks, inspect rock surfaces and any movable substrate in the water that is close to the surface or in a shallow area of the river.
4. At each site, 30–45 min should be spent examining submerged leaves, grasses, sticks and stones for larvae or pupae.
5. If you find larvae, do not remove them from the substrate they are attached to. Instead, remove the leaf, twig or stick from the river and bring it back to the riverbank, and, using a hand lens, try to identify the larvae.

   Continue carefully moving from the side of the river into the middle, searching for black fly larvae. In larger rivers this may require a boat to reach islands or rocky outcrops.
6. Collect some larvae or pupa and, while still in the field and with the help of a hand lens, pick a larva and try to check for the presence of large dark dorsal scales (tubercle), a morphological characteristic of *S. damnosum s.l.* larvae (Crosskey, 1995; Goodwin and Duggan, 1972). A dissecting microscope can also be used if available.
7. Take photos of larvae and, if possible, the location where they were found. Preserve a selection of larvae, (20–40 individuals from a couple of sites/substrates in the river, selecting larger larvae where possible) in Carnoy's solution for future analysis. Make up Carnoy's solution fresh at each site where larvae are found by mixing three parts of absolute alcohol (ethanol) with one-part glacial acetic acid. A laboratory identification to confirm if the larvae are actually *S. damnosum* is important. See 2.3.2 regarding laboratory confirmation of *S. damnosum* larvae.
8. Once a site on a river has been thoroughly searched, enter the results into the river prospection form (Annex 5). Do not forget to take the coordinates of the site using a portable GPS or a phone with a GPS application.
9. Once the river has been searched, go to the nearest 2–3 communities and conduct the entomological community questionnaire (Annex 6) and the first-line village recording form (Annex 7). Do this at all sections of the river searched regardless of whether or not larvae are found.
10. At least five sites should be visited in each IU to confirm vector presence in the river system. If the prospection yields no black flies/larvae, proceed with visiting more potential sites, conducting a spot check of adult flies in nearby villages (2.4.2), or elect to proceed to Step 3B.

Fig. 5. *Simulium damnosum* breeding sites. A typical *S. damnosum* breeding site in Cameroon (A); larvae and pupae of *S. damnosum* attached to a twig removed from water (B)

Source: Louise Hamill

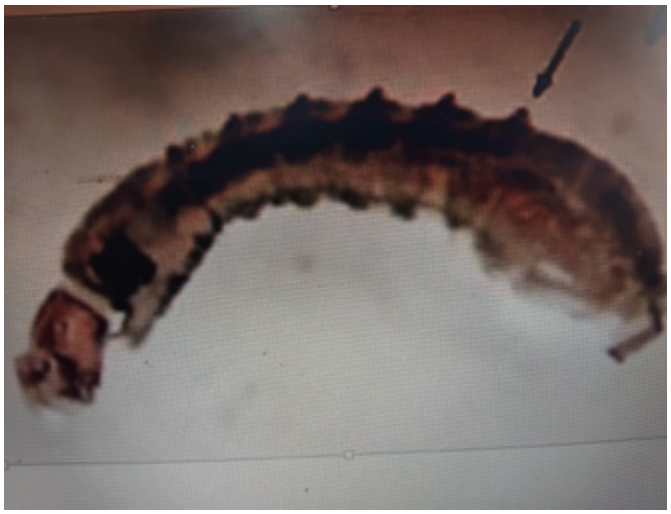


2.3.3 Laboratory confirmation of *S. damnosum* larvae

In the laboratory, the collected larvae are removed from the cool box or polythene bag. Using a fine forceps, remove one mature larva from the substrate and place it on a Petri dish with moist filter paper. Transfer the Petri dish to a dissecting microscope stage and adjust the magnification to x10 to allow you to examine the morphological characteristics. Magnifications can be adjusted higher to allow you to examine tiny features. *S. damnosum* larva is recognized by elongate characteristic shape. There are usually two attachment organs, a proleg crowned with hooks in the thorax and a posterior circlet formed by a crown of hooks. On the outline of the body, there are characteristic dorsal protuberances (dorsal tubercles) present on the first five or six abdominal segments (Fig. 6).

Fig. 6. Larva of *S. damnosum s.l.* with well-developed dorsal tubercles

Source: Thomson Luroni Lakwo



2.4 *Simulium neavei* group breeding site assessment

Prospections for the *S. neavei* group differ from those for *S. damnosum* because the larval stages of *S. neavei* attach themselves to freshwater crabs. To assess infestation status, freshwater crabs have to be caught and examined for immature stages of the vector. In preparation for the exercise you need to have:

2.4.1 Equipment

The following equipment is needed for *S. neavei* breeding site assessments:

- maps marked with suitable and uncertain areas from desk review,
- portable GPS machine or phone with GPS function,
- funnel basket traps,
- fresh meat/chicken or jackfruit,
- opes,
- machetes,
- bucket and
- a crab data collection form.

2.4.2 *S. neavei* breeding site assessment procedure

Walk to the river course and inspect the site which has a good flow. Then, along the river bank:

1. Prepare your basket trap by tying a small piece of fresh meat/chicken/jackfruit in a rope of about half a metre.
2. Insert the meat through the narrow side (posterior; Fig. 7 panel B) of the funnel basket trap and allow it halfway. Then block the narrow end with grass, leaving the funnel opening (anterior; Fig. 7 panel C) for ease of water flow. Additional interior views of a basket trap are provided in Fig. 8 panels A and B.
3. With your trap already with a bait, carefully enter the water and select a suitable site in a river with moderate flow but shaded.
4. Immerse the trap in water and cover it with grass or leaves to camouflage it since crabs are shy of bright objects. Use additional rope to anchor the trap to an object at the bank to avoid being swept by the water current. Leave it in water for 1 h.
5. In a river course, traps should be set at a distance of 10–15 metres apart depending on the number of traps the field team has. The more the traps the better the results.
6. After 1 h, check if freshwater crabs have been trapped inside. If no crabs are caught, move to the next site.
7. If freshwater crabs are caught, remove the grass used for blocking entry, empty the crabs into a bucket of clean water, and examine them one by one for the presence of immature stages (larvae/pupae) of *S. neavei* (McMahon, 1957).
8. Results should be recorded in designed crab forms indicating the sex, size of the crab, number of larvae, pupae, and pupal cases (Annex 8).
9. If no crabs are caught completely in a river course, you can leave one or two traps overnight to rule out the absence of freshwater crabs in the river/stream.
10. Before leaving the site, record the coordinates using a portable GPS or a phone.
11. Once the river has been searched, go to the nearest 2–3 communities and conduct the entomological community questionnaire (Annex 6) and the first-line village form (Annex 7). Do this at all sections of the river searched regardless of whether or not larvae are found.
12. Trapping in 3–5 sites will be adequate to confirm the presence or absence of *S. neavei*.

Fig. 7. Crabs and traps. Freshwater crab with larvae (A); posterior view of trap where bait (fresh meat) is inserted and also used for emptying crabs after trapping (B); anterior view of trap where the crabs enter looking for bait while in water (C)

Source: Thomson Luroni Lakwo

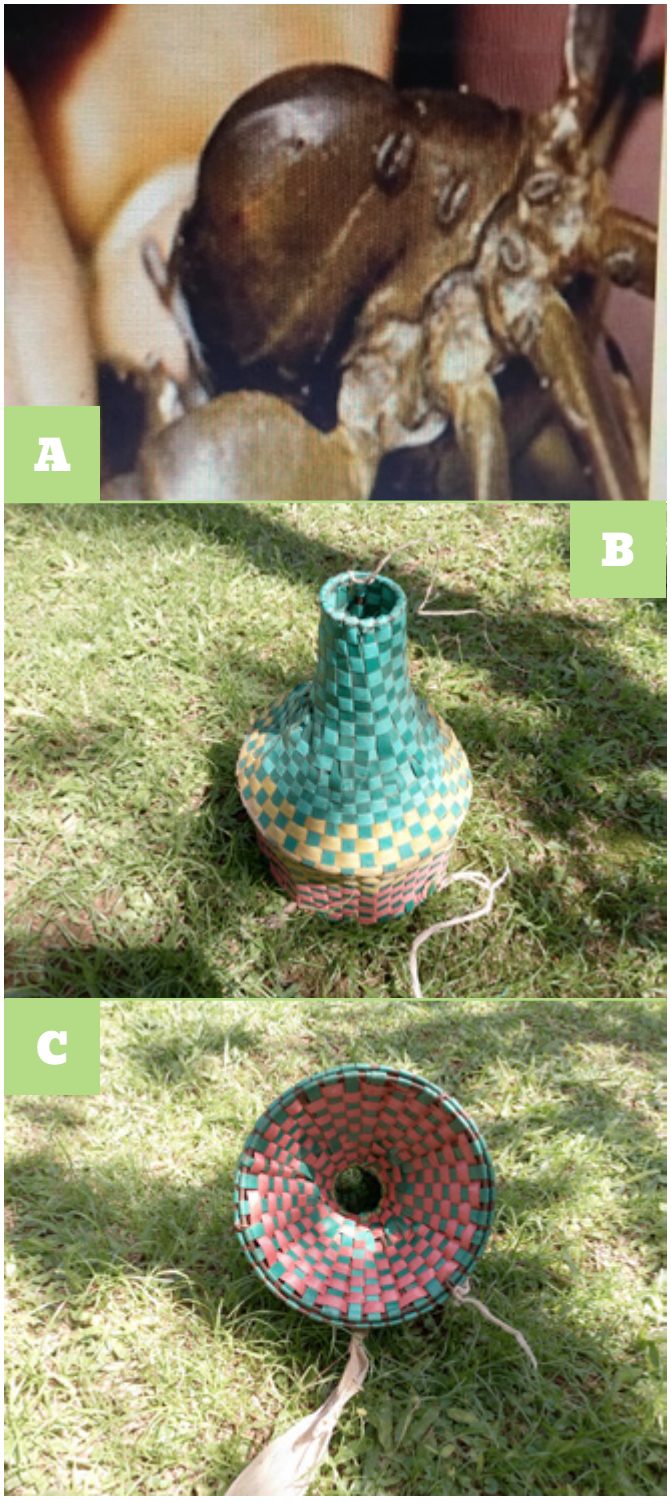
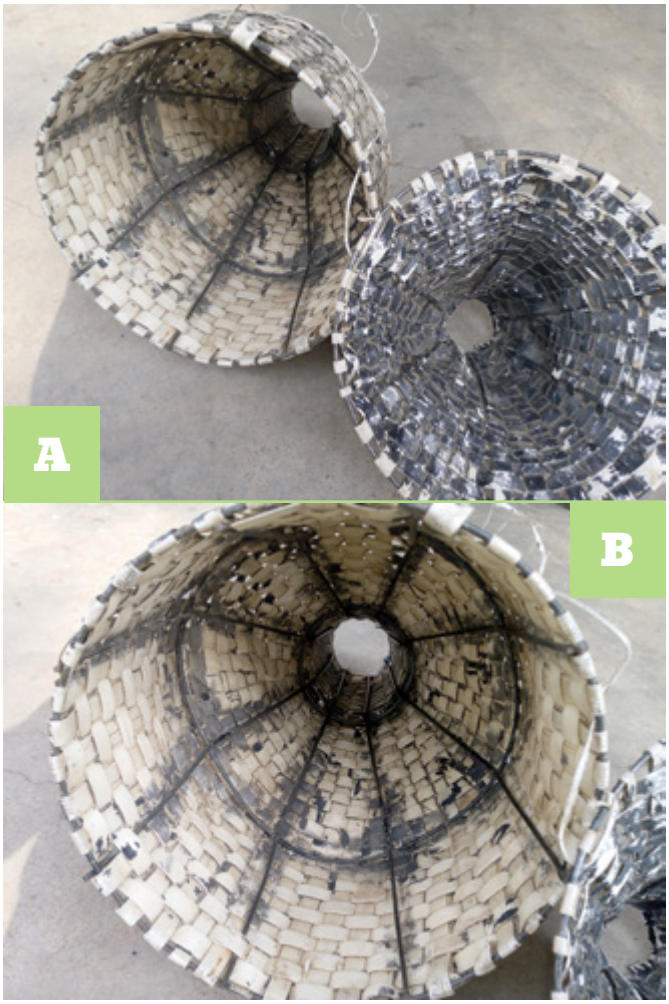


Fig. 8. Additional anterior views of a basket trap

Source: Thomson Luroni Lakwo



WARNING: Prospection activities can be risky. You may slide off a slippery rock and fall in fast-running water and drown! If rivers or streams are flowing too fast, such that a person entering the water could be swept away, do not enter the river. Team members participating in the exercise should at all times wear life jackets in or close to the river. This activity should not be done alone: there should be a minimum of three persons in a team, which helps in carrying survey tools and any rescue operations.

2.5 Obtaining information on vector status from communities

2.5.1 Community questionnaire

In addition to breeding site assessments, it is useful to collect information from communities residing along the rivers. In getting information about black flies, the team should be guided by key probing questions, as many flies exist along rivers in Africa that are not black flies. When completing the entomological community questionnaire (Annex 6), the team should ask:

- Are there problems with insects biting members of the community? If "yes", the team should probe further on the size since tsetse flies are also common along rivers; show a picture of a black fly or an actual one if available. (If 5–10 people answer "no", then discontinue the interview in the area);
- Do people report painful bites that at times leave some spot of blood on the body?
- What is the local name for "black fly" in the area, if available?
- How serious are black fly bites in the area? (probe for preventing people from normal gardening activities, etc.).
- At what times do black flies bite? (probe for early morning, afternoon or evening)
- During which period of the year are black flies abundant? (probe for season or months)

NOTE: People in the community who may have adequate information are fishermen, hunters, farmers or community chiefs or elders. In the absence of these categories, any member of the community can be interviewed.

2.5.2 Spot checks of adult flies

While visiting communities near suspected breeding sites to collect information from residents on the presence of flies, teams may also choose to conduct a spot check of adult flies. If a breeding site is inaccessible due to environmental constraints and cannot be confirmed, spot checks of adult flies can provide useful evidence of the presence or absence of flies in an area. The spot check may also be conducted at the river during the breeding site assessment if the team can access the site. The process is summarized as follows:

- Select 1 or 2 locations in the village as near as possible to the suspected breeding site (if unable to access the site).
- Choose 1 or 2 team members to perform the spot check.
- If possible, select a time of day when residents have indicated that fly biting is most common.
- At the designated time and location, each team member stands with their legs bare below the knee for a few hours.
- Any flies that land on their legs will be caught by inverting a small plastic tube over them.
- After capture the flies will be counted and kept in a cool place.
- Captured flies should be examined by an entomologist using a hand lens to verify the identification of black flies.

As with breeding site assessments, it is important to consider the appropriate timing of spot checks for adult flies to maximize the possibility of identifying black flies (2.1 provides further details). Fly collectors should be given ivermectin per the regular community mass drug treatment regimen.

2.6 Challenges with breeding site assessments only

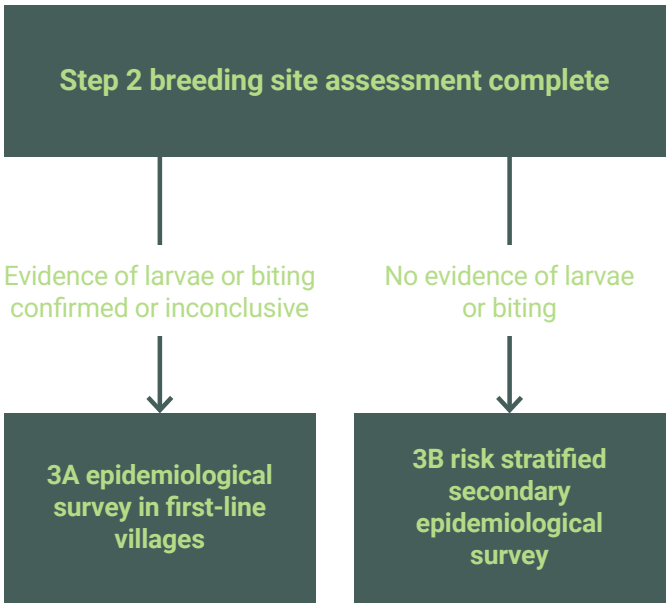
While breeding site assessment may reveal the presence of black flies, it should be noted that several species of black flies may exist in an area and not all of them may be human-biting or transmit *O. volvulus*. As such, it may become important to confirm the species and role of identified black flies in the transmission of *O. volvulus* in an AU. Entomological surveys comparing species of larvae and adults collected in an area, as well as dissection and PCR of adult flies to detect infection with *O. volvulus*, may be required. In the absence of this specific confirmation, the presence of black flies should be coupled with epidemiological surveys (refer to Chapter 3).

Likewise, breeding site assessment may also not reveal the presence of black flies, if not conducted properly or done at the wrong season and time of the year. For such a situation where no black flies are recorded from an AU, guidance from the NOEC may be required to confirm the findings or provide recommendations.

2.7 Conclusions

The recommended decision-making process regarding Step 2 breeding site assessment results is summarized in Fig. 9. To avoid ruling out suitable areas that may, in fact, have black flies that were not identified during breeding site assessments, health ministry/NOEC should consider taking the conservative approach of proceeding directly to Step 3B (RSS) to confirm whether or not an area is black fly-free if no evidence of larvae or biting is found during Step 2. The health ministry/NOEC may also consider proceeding first to Step 3A based on the permissive ecology near first-line villages if no evidence of active breeding was found in areas deemed to be suitable. If larvae or biting is confirmed or if results are inconclusive, teams should proceed to Step 3A (first-line village survey).

Fig. 9. Options for Step 3 epidemiological surveys based on Step 2 breeding site assessment results.



Breeding site assessment may also not reveal the presence of black flies, if not conducted properly or done at the wrong season and time of the year.



Chapter 3.

Step 3: Epidemiological surveys in OEM

Step 3 provides a comprehensive guide on conducting epidemiological surveys, which form a crucial step in the OEM process. The chapter is divided into several sections, detailing the rationale, methods and steps involved in conducting these surveys.

Rationale for the sampling strategy



The sampling strategy for epidemiological surveys in OEM aims to maximize efficiencies while minimizing the risk of missing areas with ongoing transmission. Traditional mapping often focuses on high-risk areas, but with the shift towards elimination, a more systematic and thorough approach is needed. The strategy is divided into two stages to ensure comprehensive coverage and accurate classification of areas.

Stage 3A: Epidemiological surveys in first-line villages

- This initial stage involves sampling 100 adults aged 20 years or older from at least five first-line villages. First-line villages are those closest to suspected or confirmed breeding sites. The purpose is to detect signs of transmission in the most likely places quickly and efficiently. Sampling adults, rather than children, is preferred as adults are more likely to have been exposed to the parasite over a longer period.

Stage 3B: Risk-stratified secondary survey (RSS)

- This stage is conducted only if the first-line village survey has not yielded conclusive results or if the breeding site assessment fails to identify first-line villages due to a lack of black fly larvae or biting in suitable areas. It involves sampling 50 adults from 20 villages selected systematically from a risk-stratified list. This step aims to enhance the detection of possible transmission across areas with low or uncertain transmission risk, especially in areas with incomplete understanding of the risk.

Community mobilization and timing

Effective community mobilization is critical for the success of epidemiological surveys. Field workers must engage local health facilities, authorities, religious and cultural leaders, and influential community members to ensure maximum participation. Timing of the surveys should avoid clashes with local events, holidays or busy farming periods to achieve a fair representation of the population.

Diagnostic testing

Testing samples collected during the surveys using the Ov16 RDT with DBS in the laboratory is recommended to ensure accurate detection of *O. volvulus* antibodies. The OEPA ELISA and AP ELISA may also be used in the assessment of the samples.

Interpretation of results

The interpretation of survey results involves comparing the observed prevalence of antibodies to predefined critical cut-off values. If the number of positive cases exceeds the critical cut-off, MDA is initiated in the AU. Otherwise, the area is classified as non-endemic, and no MDA is needed. The results are then presented to the health ministry and the NOEC for validation.

Reporting and action maps

The chapter also highlights the importance of reporting results to WHO through the EPIRF. Action maps are generated to indicate which areas require treatment and which can be excluded. These maps and tables are shared with the health ministry and NOEC to guide decision-making and resource allocation.

This section describes the epidemiological surveys to be conducted for OEM. The desk review and breeding site assessments should already have taken place to inform the selection of first-line villages. Epidemiological surveys for OEM are conducted using a two-stage approach, as detailed below.

3.1 Rationale for the sampling strategy

Traditional mapping calls for purposeful sampling in the areas of highest risk, but when mapping in areas considered to be "hypo-endemic' or of unknown endemicity, programmes may not always know which areas are at greatest risk. The shift from control to elimination requires greater certainty to declare an AU non-endemic and not in need of MDA. The epidemiological sampling strategy for mapping onchocerciasis was divided into two stages to maximize programme efficiency while minimizing the risk of missing areas with ongoing transmission.

Step 3A: Epidemiological surveys in first-line villages. This survey involves sampling **100** adults aged ≥ 20 years from at least **five** first-line villages. The purpose of this stage is to leverage existing programme knowledge and the presence of known risk factors to look for signs of transmission in the most likely places. This approach is intended to be simple, quick and biased in favour of finding areas clearly in need of MDA. The recommendation to sample in five first-line villages is based on modelling results, which used empirical mapping data and considered sampling 1–10 first-line villages. The results indicated that there were diminishing returns in sampling beyond five or six first-line villages, with regards to the probability of identifying areas that are in need of MDA. Consequently, it was determined that sampling in five villages is the appropriate balance between correctly classifying districts in need of MDA versus expending extra resources to add additional villages (OTS3, 2019). Similarly, the recommendation of how many adults to sample per first-line site was based on the trade-off between greater statistical precision vs. conserving resources; a sample size of 100 was deemed to be a good balance.

Step 3B: Risk-stratified secondary survey (RSS). This survey involves sampling **50** adults aged ≥ 20 years from **20** villages, selected systematically from a risk-stratified list. The purpose of this second stage is to improve the ability to detect possible transmission in the survey area. To increase confidence that transmission has not been missed, a rigorous sample that represents the different risk areas is important, particularly when the understanding of the risk in the area is incomplete. The 20-village requirement was based on modelling results, which used empirical data to simulate the impact of sampling 5–50 villages. The results indicated that there are diminishing returns after sampling beyond 20 villages (unpublished). Given the additional level of effort and costs required to visit 20 villages, careful consideration was given to the within-village sampling requirements. The experience of country teams engaged in operational research suggested that a sample size of 50 adults was feasible for a team to complete in a single day. Additionally, a sample size of 50 per village is the minimum required to observe a prevalence of 2%. The recommendation was therefore made (in the context of operational research) for a sample size of 50 adults per village across 20 villages (OTS3, 2019). However, guidance from the health ministry/ NOEC may be required on the use of Step 3B.

3.2 Community mobilization and timing of epidemiological surveys

There is a need for field workers to mobilize selected communities to achieve successful OEM activities and to time the activities to achieve maximum participation. Ensuring a good turnout and fair population representation is key for the serology surveys. It is important that these surveys do not clash with local events, cultural holidays, religious festivals or busy farming periods. An effective mobilization strategy is required in order to win community support and cooperation and to ensure their participation. To achieve this, the following should be done:

- Community mobilization meetings should be planned and arranged with the participation of staff in health facilities at AUs, local authorities, religious and cultural leaders, politicians and any influential local group.
- During sensitization meetings, information about the planned OEM activities and the benefits to the community should be provided.
- Additional information on onchocerciasis and the benefits of MDA should be provided.
- The dates when OEM activities will begin and how long they will take should be determined in consultation with administrative and local authorities, and subsequently conveyed to the community.

For more details, see training considerations and recommendations.

3.3 Procedure to follow upon arrival in villages

Before conducting the OEM surveys, the village local authorities (chairpersons, chiefs, elders) should be consulted for permission to perform the mapping surveys in their village, even if they have been previously informed during mobilization. The OEM survey teams, with assistance from village local authorities, community health workers, etc., will take the GPS coordinates of the village centre, ask village residents to participate in the survey, determine the age of the participants and obtain informed consent of the eligible participants.

NOTE: Countries are encouraged to collect data during Steps 2, 3A and 3B using mobile phones or tablets. This lends itself to more accurate data and easier management of results. Programmes in the WHO African Region can take advantage of ESPEN Collect, a free mobile data collection tool designed to support disease-specific surveys. More information on ESPEN Collect, including standard forms for onchocerciasis surveys, can be found on the ESPEN website (https://espen.afro.who.int/tools-resources/espen-collect).

3.4 Step 3A: Epidemiological survey in first-line villages

If suspected or confirmed breeding sites are identified during Step 2, epidemiological surveys should commence in first-line villages. In each AU, a minimum of five first-line villages should be selected from one or more suspected or confirmed breeding sites. Where possible, efforts should be made to space out the selection of first-line villages within an AU so that different breeding sites are represented and so that there is good geographical coverage of the suitable areas within the AU. Additional first-line villages should be added in large AUs or AUs with very dense river systems and/or many potential or confirmed breeding sites. In each first-line village, a sample of 100 adults aged 20 years or older is taken using EPI random walk sampling (see 3.4.2). Of the 100 adults, at least 50% should be male, given evidence from several countries that onchocerciasis infection is often more common among men than women (Kifle, Kifle and Nigatu, 2019).

Given the goal of onchocerciasis elimination, rather than exclude adults based on their years of residency in the village, all adults should be included and residency information should be recorded during registration of participants. In the presence of suitable vectors, any infected persons can contribute to transmission and should be included in the surveys. If the presence of vectors has not been proven, non-residents may be excluded from the surveys. The presence of infected individuals who have not travelled outside the AU could show evidence of active transmission. If the only positive cases are from outside the AU (i.e. the individual does not spend the majority of the year in the community), the programme should consult with the district/regional health leaders in the place(s) from which the individuals migrated to ensure that those areas are under treatment or are being prioritized for OEM. This strategy should also be applied during Step 3B (RSS sampling).

Key question:

Why include adults instead of children?

The OTS carefully considered the age groups to be examined when conducting OEM to ensure that all areas of active transmission are accurately identified. In ivermectin-naive areas, never treated for onchocerciasis or lymphatic filariasis, it may be possible to detect anti-*O. volvulus* antibody responses in a range of age groups. OTS concluded that testing adults during OEM allows for greater sensitivity and, due to their longer potential exposure and greater attractiveness to black flies than children, adults are better markers of transmission in untreated areas than children.

In contrast, children are sampled in Stop MDA surveys because any antibody signal in a child aged < 10 years is indicative of active exposure to the parasite that took place after the initiation of MDA. Adults are not a suitable population for Stop MDA survey decisions because an antibody signal could represent current or historic transmission. The sampling procedure for determining transmission interruption in a given area by screening children is well documented in the Stop MDA survey recommendations (WHO, 2016). This would necessitate a sample of minimum 3000 children per transmission zone (for a threshold below 0.1% Ov16 serology) and 6000 black flies.

Determining active transmission in an area not exposed to treatment is easier; therefore, OEM surveys are specifically designed to be quicker and less costly than Stop MDA surveys (OTS3, 2019).

3.4.1 Sampling method for Step 3A first-line village surveys

The purpose of Step 3A sampling is to visit first-line villages – that is, those villages that are closest to a suspected or confirmed black fly breeding site.

- After desk review and breeding site assessment, select seven (five primary and two back-up) suspected first-line (or high-risk) villages for sampling. First-line villages are those closest to a confirmed or suspected black fly breeding site. No first-line village should be further than 10 km from the suspected breeding site. The two back-up first-line villages are selected in case any issues arise in conducting sampling or accessing the first five primary villages.
- In each selected village, sample 100 adults aged ≥ 20 years per village using EPI (random walk) sampling (see 3.4.2), and try to achieve an equal balance of men and women, but at least 50 males.
- If there are fewer than 100 adults present who agree to participate, move to the next closest village until 100 adults have been enrolled. Ensure that new mobilization and community permission to proceed is done in each new village the team moves to, including collection of new GPS coordinates for each location. For each participant, collect information on their community of origin during enrolment.
- From the adults selected by the EPI method, obtain DBS through finger prick. See Annex 3 for standard operating procedures for collecting DBS.
- Dry and package the samples and transport them to the processing laboratory. See Annex 9 for further instructions on packaging (and shipping, if necessary) DBS.
- Proceed to Step 4 (laboratory analysis) and Step 5 (analysing and interpreting data) to determine if MDA is warranted or if it is necessary to proceed to Step 3B (stage 2 RSS).

NOTE: This handbook was written to reflect the current recommendation that, for mapping purposes, DBS be collected for subsequent laboratory-based testing (see chapter 4.1 regarding choice of diagnostic test). It will be updated to reflect new recommendations if/ when reliable point- of-care diagnostics become available.

3.4.2 "Random walk" approach in sampling of adults aged > 20 years

The random walk approach is a simplified cluster sampling method developed by WHO that was originally implemented in the Expanded Programme on Immunization (WHO, 2008). The WHO OTS determined that this method was most feasible for OEM as it is a compromise between convenience sampling and systematic household sampling (to be done if preferred) and can enable programmes to achieve sex-balanced samples (OTS3, 2019). The method for randomly selecting households depends on the size, density and living arrangements of the population within the cluster. In small villages, full enumeration of households, followed by random sampling, is often possible and is the preferred approach. In scattered populations, randomly selecting a direction to walk (e.g. by spinning a bottle), selecting a random starting point and sampling contiguous households are common (Fig. 10). Random walk basic steps are:

1. Go to some central location within the community. This could be a market, a church, a health facility or the junction between two roads. If possible, find a local guide who can accompany you. He/she can introduce you to the families and help you find the households and the community boundaries.

2. Select a direction at random by spinning a bottle/ pen/pencil. Walk in the direction that the bottle/ pen/pencil is pointing and count every house in your path, starting from the centre of the village to the edge of the village. If possible, number each house with chalk as you go.

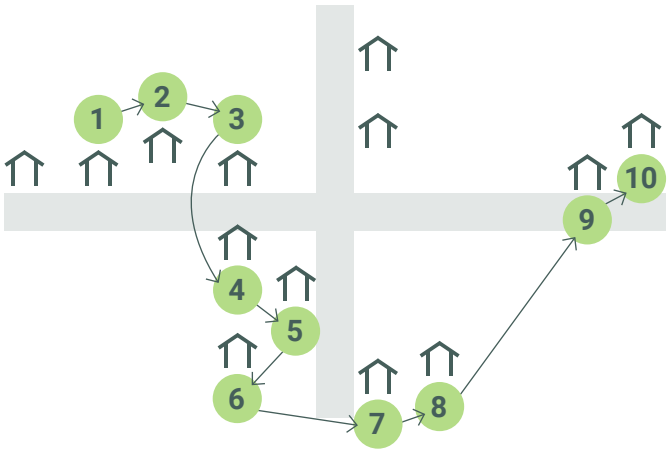
3. Draw a random number (e.g. using slips of paper from a hat) between 1 and the total number of households you counted in step 2. This number will represent your starting house. For example, if you count five households from the centre of the village to the edge of the village and you randomly pick the number 4, then the fourth household that you counted becomes your starting household.

4. Identify all adults aged ≥ 20 years and obtain their consent to take finger-prick blood for DBS. Register participants using a mobile data collection platform such as ESPEN Collect.
5. DBS collection can be done at each household, or the consenting adults can be invited to go to a central point in the village to meet the main team for DBS, while one team member and the community guide continue enrolling and directing people to the sampling point.
6. Proceed to the nearest neighbour household and obtain consent from all adults aged ≥ 20 years for finger-prick blood collection.
7. Continue using this nearest neighbour approach until DBS is taken from 100 people (Step 3A) or 50 people (Step 3B).

CAUTION: Households should be selected using the random walk approach so that those far from the village centre, or those at a distance from a main road, have the same chance of being sampled as more accessible households. Isolated families may have a different health status and it is important that they are not excluded from the survey.

Fig. 10. Sequence of the nearest next households beginning with the randomly selected household

Source: Training for mid-level managers: the EPI coverage survey (WHO, 2008).



3.5 Step 3B: Risk-stratified secondary survey

If the number of individuals with anti-*O. volvulus* antibodies in the selected first-line villages sampled during Step 3A does not exceed the critical cut-off (see Step 5, Chapter 5) then it is provisionally recommended to follow up with additional serosurveys in case there was a failure to identify the correct first-line villages and transmission occurs elsewhere in the AU. Careful selection of first-line villages during Step 2 would make RSS not required in many AUs where transmission is ongoing.

NOTE: Conducting RSS will be based on the results of the Step 3A first-line village sampling. If after sampling five selected villages under Step 3A, results do not indicate that MDA is warranted, countries should proceed directly to Step 3B RSS sampling. If this leads to the identification of a village that exceeds the critical cut-off (Chapter 5), then MDA should be initiated in the IU. There may be significant cost savings to be realized through economies of scale by conducting first-line and RSS sampling together, rather than carrying out only first-line village sampling and completing RSS sampling at a later date (Hamill et al., 2022). If cost and logistics make it more practical to conduct first-line and RSS assessments at the same time, countries could take the approach of first analysing first-line village samples before deciding whether or not to analyse RSS samples, to conserve laboratory resources.

The basis of the proposed RSS approach is to list all eligible villages within the AU in order of assumed risk. For example, those close to the river are assumed to be at higher risk than those very distant. Villages close to swampy areas with only slow-flowing waterways are assumed to be at lower risk. Villages close to fast-flowing rivers, breeding sites or first-line villages, as well as villages with reports of clinical cases, are assumed to be at high risk.

To ascertain the assumed risk of each village, ideally, a geo-referenced list of all communities should be shown on a map alongside rivers and elevation. If coordinates are not available, then, depending on the size of the AU, they could be collected during breeding site assessments, or risk can be assigned by local officials with good knowledge of the location of the villages within their domain. Once villages have been listed in order of risk, use the OEM survey sample builder (Annex 10) to determine the sampling interval and generate a list of villages to be included in the RSS. The RSS sampling will target 20 villages in total, with a sample size of 50 adults per village. The inclusion criteria for participants are the same as in Step 3A.

3.5.1 Sampling method for Step 3B risk-stratified secondary survey

The RSS survey involves the following steps:

- Using the OEM survey sample builder (Annex 10), list villages by risk and run the algorithm to systematically select 20 villages plus five reserves.
- Sample 50 adults per village using EPI (random walk) sampling, and try to achieve an equal balance of men and women, but at least 25 males. Follow the steps outlined in 3.4.2.
- From the adults sampled, obtain DBS through a finger prick. See Annex 3 for standard operating procedures for collecting DBS.
- Dry and package the samples and transport them to the processing laboratory. See Annex 9 for further instructions on packaging (and shipping, if necessary) DBS.
- Proceed to Steps 4 (laboratory analysis) and Step 5 (analysing and interpreting data).

Chapter 4.
Step 4: Laboratory analysis

Step 4 outlines the critical steps and considerations for laboratory analysis in the context of OEM. The chapter emphasizes the choice of diagnostic tests, the necessary laboratory supplies, quality control, and the proper setting and handling of samples.

Choice of diagnostic test



The primary diagnostic test recommended for OEM is laboratory-based DBS analysis using the SD Bioline Ov16 RDT. This test detects antibodies specific to *O. volvulus*. While ELISA tests are reliable, they are less user-friendly and require more extensive quality control and procurement of various reagents, making them less practical for many countries.

Laboratory supplies



The supplies needed for the laboratory analysis include:

- Ov16 RDT kits or ELISA kits;
- standard operating procedures and bench aids (detailed in the annexes to this handbook);
- positive and negative controls for quality assurance (QA) and quality control (QC); and
- necessary reagents and equipment for the chosen diagnostic method.

Countries should plan for long lead times in supply orders to ensure the timely availability of all necessary materials.

Quality control and quality assurance



QA and QC are essential to ensure reliable data generation. The chapter outlines measures for both:

- Quality control (QC): Involves using standardized positive and negative controls each time an assay is performed. This helps verify that the test is working correctly and maintains consistency across different technicians and laboratories.
- Quality assurance (QA): Aims to prevent errors throughout the entire process, from specimen collection to result reporting. Key areas of QA include:
  - ensuring reagent quality and equipment maintenance;
  - developing standardized operational procedures;
  - training staff adequately;
  - maintaining accurate records and good documentation practices; and
  - monitoring laboratory output and performance.

For the Ov16 RDT, adherence to the manufacturer's QC processes is crucial, and each assay should have established standards and controls.

Laboratory setting and sample processing



DBS collected during epidemiological surveys are tested in a laboratory setting, either in a field or central laboratory. The central laboratory is recommended due to its capacity to handle larger volumes of samples and the stability it provides for lengthy analyses. Proper drying, packaging, and transportation of DBS are vital to maintain sample integrity. If both Stage 3A and 3B samples are collected, Stage 3A samples should be processed first to conserve laboratory resources.

Managing and interpreting data



After the laboratory analysis, data from epidemiological assessments are reviewed and interpreted to classify areas as either endemic or non-endemic. Critical cut-off values for various diagnostic tools are used to determine if the observed prevalence of antibodies exceeds the biological threshold, indicating active transmission and the need for MDA. Results are compiled into summary tables and action maps, which are then shared with the health ministry and the NOEC.

4.1 Choice of diagnostic test

The first choice of a diagnostic test for OEM is the laboratory-based DBS on SD Ov16 RDT.

At its third (OTS3, 2019) and fourth (OTS4, 2020) meetings, the OTS made recommendations on the best use of currently available onchocerciasis diagnostics. All available methods rely on the detection of an antibody produced by the human body in response to *O. volvulus* infection. At the time of publication of this handbook, all available tests detect antibody responses to the *O. volvulus* specific antigen Ov16. The three commonly used Ov16 detection methods are:

- the OEPA ELISA, first developed by the onchocerciasis elimination programme for the Region of the Americas;
- a modified version of the OEPA ELISA, using alkaline phosphatase as the detection enzyme, AP ELISA; and
- the fully commercial Ov16 RDT, currently manufactured by Abbott.

While these diagnostics are currently in use by programmes, each has limitations, including both performance and ease of use. The OEPA ELISA has been used widely in programme settings with high specificity but low sensitivity (Sauerbrey, Rakers and Richards, 2018); however, it was not included in evaluations reported to OTS3. Additional analysis of the OEPA ELISA should be provided so that its performance can be compared with that of the other two ELISAs. In studies reported to OTS3, false– positive results were obtained in samples from areas non-endemic for onchocerciasis on ELISA-based tests (OTS3, 2019) ranging from 14.1% to 43.6%. The cause of these false–positive results is unclear, but they are reason for caution and concern. False–positive results were especially high among the ELISA-based tests, with the SD ELISA exhibiting particularly low specificity. In addition, the AP ELISA is subject to variability since its cutoff is tied to an optical density (OD) value. Ov16 RDTs used in the field on whole blood missed a large proportion of positives detected by other tests. However, the use of the Ov16 RDT in the laboratory, on eluted DBS rather than whole blood, was observed to improve the performance of the test comparable to the performance of ELISA (OTS4, 2020).

There are also ease-of-use limitations to the ELISA-based tests. The AP and OEPA ELISAs are not commercially available as kits and therefore require the procurement of various reagents and supplies, which makes them less user-friendly and challenging to implement QA and QC of testing. When cost and ease-of use is considered, the most sensible choice for most countries will be to use the laboratory-based DBS on Ov16 RDT methodology. Countries with existing laboratory capacity may wish to analyse a random subset of samples via ELISA to quality assure the laboratory DBS on Ov16 RDT results. Appropriate positive and negative controlsfor Ov16 RDTs and ELISAs, and appropriate QA and QC elements and processes, should be in place in the laboratory before beginning testing. In settings where there is a substantial backlog of DBS and timely analysis with ELISA cannot be assured, countries may consider the laboratory-based DBS on Ov16 RDT methodology.

OTS recommendations on onchocerciasis diagnostics
For OEM, DBS should be collected. These samples should be analysed using the Abbott Ov16 RDTs in the laboratory as the first choice. If countries have the capacity for ELISA analysis, they may wish to use this instead, but the advice is that the laboratory RDT is the preferred approach.

4.2 Laboratory supplies

Laboratory supplies will be dependent on the diagnostic test that is used. Standard operating procedures and bench aids outlining necessary supplies and procedures are provided for the laboratory-based DBS on Ov16 RDT (in Annex 11) and the OEPA ELISA (Annex 12). The standard operating procedure and bench aids for the AP ELISA are forthcoming. These annexes will be updated as new diagnostics become available, and programmes should ensure they are using the most up-to-date procedures. During the planning process, programmes should account for the possibility of long lag times in receiving supply orders and ensure that orders for laboratory supplies and diagnostics are placed well in advance of when they will be needed.

4.3 Quality Control and Quality Assurance

Proper QC and QA must be implemented for any diagnostic test used to ensure that the data generated are reliable. Quality control refers to measures that must be included each time an assay is performed to verify the test is working properly. Standardized positive and negative controls should be utilized during the analytical phase of testing to verify and monitor test performance across time and between different technicians and laboratories performing testing. A system that tracks the performance of controls (including positive and negative) based on the Westgard rules would be preferred (Westgard et al., 2002); data related to the performance of controls should be made available to the health ministry/NOEC.

The goal of QA is to prevent errors from happening by creating systems and processes for addressing potential sources of error that may occur throughout the entire process from specimen collection, transport/storage, testing (including quality controls), interpretation of test results and reporting.

The main areas of a QA programme include ensuring the quality of reagents and maintenance of equipment; developing standardized operating procedures; managing human resources so that staff are trained; implementing processes to maintain records and good document practices; and implementing systems for monitoring laboratory output and performance. For the assays that are laboratory-developed tests and not commercially available (the OEPA and the AP ELISAs), the QC and QA systems for antigen production and other critical reagents must be established before the beginning of testing. All QA/QC systems that laboratories implement should be well-documented and publicly available. For the Ov16 RDT, the manufacturer's quality control processes must be followed. Additionally, each assay should have standards and controls in place along with the quality assurance systems mentioned above.

4.4 Laboratory setting and sample processing

Dried blood spots collected during Steps 3A and 3B (if conducted) should be tested in a laboratory setting rather than in the field. Care should be taken to ensure that samples are properly dried and packaged in the field and transported or shipped securely to the testing location. DBS testing may take place in a field laboratory (RDT only; see 4.1) or in a "central" laboratory (RDT or ELISA), which may refer to a national/high-level laboratory or to any other permanent laboratory space available to programmes with the capacity to conduct the analysis. A central laboratory is recommended for feasibility purposes; however, if programmes decide to use a field laboratory the laboratory must be in a protected space, as the process of analysing the DBS on RDT can take up to 3 days (see Annex 11).

If Step 3A and Step 3B sampling were conducted simultaneously, it is recommended that Step 3A samples be processed first to conserve laboratory resources, as Step 3B samples may not be needed if Step 3A sampling indicates that MDA is required.

Chapter 5.

Step 5: Managing and interpreting data

Step 5 focuses on the critical steps of managing and interpreting data collected during the OEM process. The chapter emphasizes the importance of accurate data interpretation to determine which areas require MDA and which do not.

Objectives

The primary objective of managing and interpreting data is to analyse the results from the previous steps (epidemiological surveys and laboratory analyses) to decide on the necessity of MDA. This involves a detailed review of data to classify areas as endemic or non-endemic for onchocerciasis.

Data analysis process

1. Data compilation

- Village-level results: Data are collected at the village level, focusing on the prevalence of serological positivity for Ov16 antibodies, which indicate onchocerciasis infection.
- Operational transmission zones (OTZs): Areas are categorized into OTZs based on the collected data.
- Action Maps: Maps are generated to visually represent areas requiring MDA and those that can be excluded.

2. Data Interpretation

- Results from epidemiological surveys (Steps 3A and 3B) are interpreted using specific critical cut-offs (see Table 1).
- Step 3A (first-line villages): If any village exceeds the critical cut-off, the entire assessment unit (AU) is recommended for MDA.
- Step 3B (risk-stratified secondary survey): If multiple villages exceed the lower cut-off or any village exceeds the higher cut-off, MDA is recommended for the AU.

3. Handling conflicting results

- If there are discrepancies between epidemiological and entomological assessments, additional data may be required. Consultation with the health ministry and the NOEC is necessary to resolve such conflicts.

4. Reporting results

- Results are compiled into comprehensive tables and action maps.
- Tables: Summarize the prevalence data, the number of positive cases, the diagnostic tool used and the decision on whether MDA is required.
- Action maps: Visually indicate the endemic and non-endemic areas, assisting in strategic planning and resource allocation.

5. Submission to WHO

- Programmes are encouraged to report results to WHO using the EPIRF. This enhances transparency and aids in advocacy and resource mobilization.

Generation of action maps

Action maps are critical tools for visualizing the distribution of onchocerciasis and planning intervention strategies. These maps:

- label IUs based on their endemic status;
- include details on surveyed villages, prevalence rates, and sampling stages (Step 3A or 3B); and
- are used to guide decisions on MDA implementation and other control measures.

The objective of Step 5 is to interpret the data from Steps 2, 3 and 4 to determine which areas are free of onchocerciasis transmission and will not be subject to MDA, and which areas are subject to onchocerciasis transmission and should be subject to MDA.

Upon completion of Step 4 laboratory analysis, a thorough review and analysis of data from Steps 3 and 4 should be conducted, to develop a report and generate maps showing:

- village-level results (prevalence of serological positivity);
- operational transmission zones; and
- action maps.

As stated in 4.1, all of the currently recommended diagnostic tools for assessing the presence of *O. volvulus* in humans rely on the detection of Ov16 antibodies. Thus, the following sections on critical cut-off values and results interpretation will refer specifically to results collected using diagnostics that measure antibodies to Ov16. This handbook will be updated as new diagnostics become available.

5.1 Critical cut-offs for various diagnostic tools

The goal of an OEM survey is to correctly classify areas that require MDA and those that do not, while minimizing the number of areas for which an incorrect treatment decision is made. The agreed-upon tolerance for committing an over-treatment error (that is, providing MDA to an area that does not require it) was 10%, whereas the tolerance for making an "under-treatment" error (that is, failing to provide MDA to an area that requires it) was 5%. These values equate to an OEM strategy with 90% power and 5% Type 1 error.

MDA decisions for onchocerciasis are based on the results of population-based surveys whereby the observed survey prevalence is compared with the target threshold of 2% antibody in adults. This 2% threshold is referred to as the biological threshold, defined as the prevalence above which onchocerciasis transmission becomes self-sustainable. For any survey involving diagnostic tests, the observed prevalence will always be a combination of infected individuals who test positive ("true positives") and uninfected individuals who test positive ("false positives"). The probability that an observed positive result is indicative of a *true* positive (referred to as the "positive predictive value") is dependent on the underlying prevalence in the population and the performance of the diagnostic test.

For onchocerciasis, there are several different diagnostic tests available to national programmes with very different reported values for sensitivity and specificity. In the low endemic settings where OEM will take place, specificity ends up being a primary driver of the observed prevalence (see Fig. 11). Consequently, the critical cut-off value (that is, the maximum number of positive tests that can be observed for the survey area to be considered below the biological threshold) must take into account each diagnostic test's performance (Gass, 2020).

Fig. 11. Theoretical explanation of the critical cutoff and biological threshold for OEM

Source: Adapted from K. Gass, Task Force for Global Health, 2020 (unpublished)

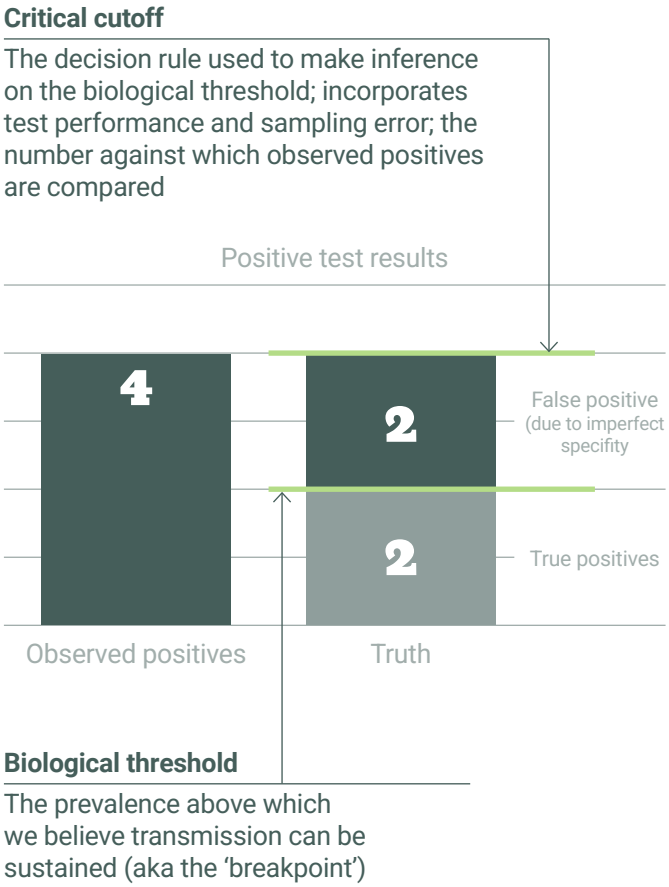


Table 1. Critical cut-off values for Step 3 epidemiological surveys by diagnostic tools

If the number of positive tests in a given village(s) exceeds the value shown in this table, then the entire AU/IU is determined to require MDA.

Diagnostic	Ov16 RDT with DBS	OEPA ELISA[b]	AP ELISA[c]
Sensitivity	80%	43%	88.2%
Specificity	99%	99.98%	99.7%
Step 3A critical cut- off[a] (≥1 village)	4	1	2
Step 3B lower critical cut-off[a] (≥2 villages)	2	0	1
Step 3B higher critical cut-off[a] (≥ 1 village)	4	1	2

AP: alkaline phosphatase; DBS: dried blood spot; ELISA: enzyme-linked immunosorbent assay; OEPA: Onchocerciasis Elimination Program for the Americas; RDT: rapid diagnostic test.

[a] Cut-off values represent the number of positive individuals found in a village.
[b] Richards et al., 2018
[c] Cama et al., 2018

5.2 Interpretation of epidemiological survey results

Results from Steps 3A (first-line village sampling) and 3B (RSS) should be interpreted as follows. The appropriate critical cut-off (Table 1) based on the relevant diagnostic must be used to interpret the results.

5.2.1 Step 3A First-line village sampling

- If the number of positive individuals exceeds the critical cut-off for Step 3A in **one or more villages**, then initiate MDA. Refer to Table 1 for the appropriate critical cut-offs based on the diagnostic used.
- If the number of positive individuals in all sites is **less than or equal to** the critical cut-off for Step 3A, then proceed to Step 3B as merited by context.
- If the results of the epidemiological assessments conflict with the entomological assessments, additional data may be required. The health ministry/NOEC should be consulted to provide guidance on the necessary steps.

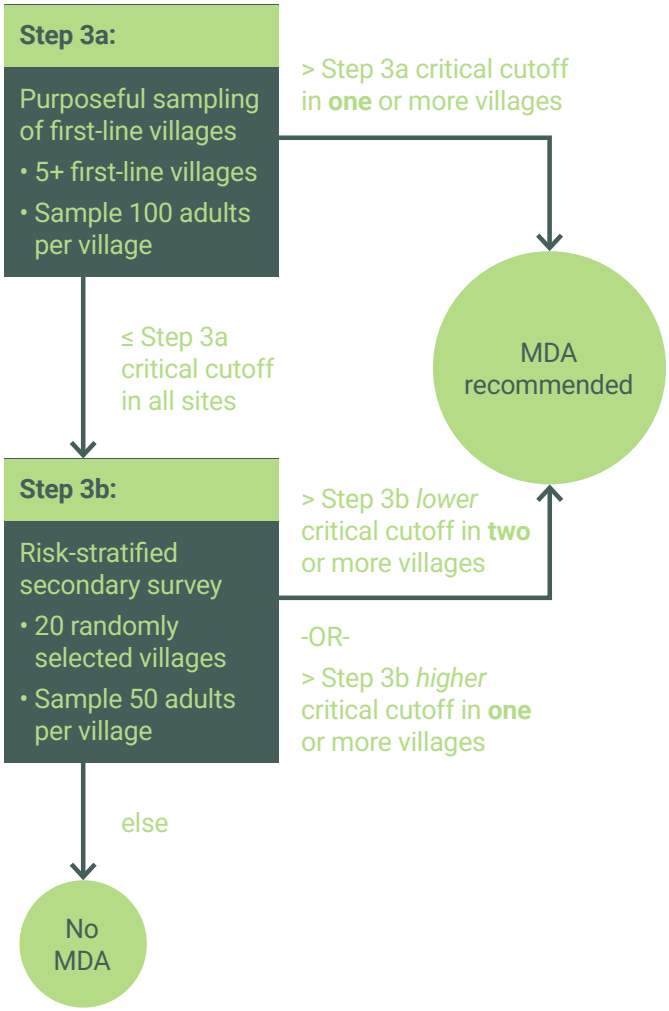
5.2.2 Step 3B risk-stratified secondary sampling

- If the number of positive individuals **exceeds** the Step 3B lower critical cut-off in **two or more villages**, or the Step 3B higher critical cut-off in **one or more villages**, then initiate MDA. Refer to Table 1 for the appropriate critical cut-offs based on the diagnostic used.
- If the above criteria are not met, and there are no additional concerns, then no MDA is indicated. Results should be presented to the health ministry/NOEC or NTD TAC for validation.
- OTS has recommended that initiation of MDA should be based on individual village results, rather than the average prevalence across all randomly selected villages.

NOTE: If onchocercal infection appears focal (positive villages are clustered in one area), subdistrict decisions can be made, and the survey area can be stratified. In each stratum that is not targeted for treatment, the total sample size should be completed by sampling additional villages to reach the full complement of 20 and the criteria for initiating MDA must be reapplied to that stratum using data from all 20 villages.

The interpretation of epidemiological survey results is summarized in Fig. 12. Refer to Table 1 for further detail on the diagnostic-dependent prevalence thresholds.

Fig. 12. Summary of results interpretation for OEM Step 3A and 3B sampling



5.3 Reporting of results

The reporting of results should follow the format prescribed in the EPIRF developed by WHO. For other presentations and analysis of the data, the outputs could be presented in tables summarizing the results at the village level and AU level (see sample Tables 2 and 3). Programmes may choose to add additional variables as they see fit. The tables may be accompanied by a narrative report that outlines key findings and recommendations. It is also essential to create a map for the AU, and any adjoining AUs, with the surveyed villages and the serological prevalence marked.

AUs should have been clearly defined during Step 1, and results presented using the agreed upon AU designations. Results for Steps 3A and 3B could be reported in one table or as separate tables.

Programmes are encouraged to consult with the NOEC to review the results and the topological map. This information will be needed to determine the appropriate IUs for treatment decisions. Care should be taken to ensure that the appropriate critical cut-offs, based on the chosen diagnostic, are used to make decisions at the AU level and that these thresholds are conveyed clearly in the reports. OTS has recommended that decisions to treat or exclude IUs be based on results at the individual village level and not on the average prevalence across all sampled villages within the AU.

Table 2. Sample table for summarizing epidemiological results at village level

Implementation unit	Village ID/ name	Village GPS coordinates	Population estimate	No. of individuals sampled	% male	No. of Ov16 positive

Table 3. Sample table for reporting results and treatment decisions at IU level

Implementation unit	Sampling stage	Diagnostic	No. of villages sampled	No. of villages exceeding Step 3A cut-off	No. of villages exceeding Step 3B lower cut-off	No. of villages exceeding Step 3B higher cut-off	MDA recommended?

5.4 Generation of action maps

The summary tables should be accompanied by maps that label the IUs of interest as either (i) onchocerciasis endemic or (ii) onchocerciasis non-endemic. Programmes may also wish to include the sampled villages on the maps with icons that indicate the observed prevalence as well as whether or not the villages were part of Step 3A or Step 3B sampling. Maps should be paired with the results tables in a summary report shared with the health ministry/NOEC.

There are currently different options for generating bespoke maps with information on onchocerciasis endemicity at IU level, depending on whether or not the programme has staff (data managers or data analyst) with experience in GIS software and conducting simple spatial analysis. Annex 4 provides greater detail on the available tools for generating action maps based on OEM results.

According to WHO recommendations, the implementation unit for MDA and other elimination activities is the TZ (WHO, 2016). These are areas that include hyper- and/or meso- and/or hypo-endemism for onchocerciasis as a result of transmission by local vectors. They are separated from other TZs by areas which are without onchocerciasis and are usually unsuitable for vectors, often called buffer zones. Buffer zones must be wider than the appetitive flight range of blood-hungry vectors (looking for a blood meal). This varies between vector species but is usually at least 20 km; otherwise, appetitive vectors will be able to cross them and carry parasites from one TZ to another (unless there is some sort of barrier to vector dispersal, such as a mountain range). In this way, a TZ is effectively independent from other TZs and acts as an independent epidemiological unit that is unaffected by activities in other TZs.

TZs are identified by examination of the maps showing village-level results (prevalence of serological positivity) which were produced in step 5 (see above). Maps are studied to identify clusters of villages with sero-positivity (above the threshold described in 5.1), which are separated from other clusters by areas without endemism.

Such clusters represent TZs and the gaps between them are buffer zones. The boundaries of TZs are set to follow district borders or subdistrict borders for administrative convenience. A district will not normally be divided into an endemic area for MDA and a non-endemic area without MDA unless the division is obvious and supported by relevant ecological differences. For example, it could be that positive villages are all clustered at one end of a district which is also adjacent to a positive neighbouring district, but the rest of the district is negative and neighbouring districts are also negative. These distinctions will usually be supported by differences in ecology, hydrology and/or geology (for example the positive areas may be uplands, but the negative areas could be part of a swampy plain). In such a situation, the district is likely to be divided into two parts: a part with positive subdistricts and a part without transmission. The positive subdistricts will join with adjacent positive districts as part of a TZ and be subject to MDA. This way of identifying TZs has worked well in places where onchocerciasis is focal, but care must be taken in defining the border. Although a few black fly species have longer appetitive flight ranges (Duke, 1975; Thompson, 1976; Renz and Wenk, 1987; Walsh et al., 2023), areas within 20km of known (or potential) breeding sites must be included within the TZ (not the buffer zone). Distances more than 20 km may be considered depending on the species in the area. A single negative village between two endemic areas is insufficient evidence for the existence of a buffer zone.

5.5 Reporting results to WHO

Programmes are encouraged to also report results of OEM to WHO through the EPIRF. This is an important step for transparency, advocacy and resource mobilization. For programmes that use ESPEN Collect for mobile data collection, data can be automatically exported to the EPIRF. Further detail on including OEM results in the EPIRF can be found in Annex 13.

Programmes are encouraged to also report results of OEM to WHO through the EPIRF. This is an important step for transparency, advocacy and resource mobilization.



Chapter 6.
OEM planning considerations

This chapter covers the essential planning considerations for successful implementation of OEM. The chapter emphasizes the importance of meticulous planning, resource allocation, team composition, training and coordination to ensure effective and efficient OEM activities.

Team composition for OEM

1. Core team:

The core team should consist of individuals with specific expertise:

- Epidemiologists: To lead and coordinate the mapping process.
- Entomologists: To handle the identification and assessment of black fly breeding sites.
- Laboratory technicians: To conduct and oversee diagnostic testing.
- Data managers: To manage data collection, entry, and analysis.
- Community mobilizers: To engage and educate local communities about the OEM activities.

2. Support staff:

Additional support staff, including drivers, administrative assistants and local guides are also crucial for logistical support and smooth operations.

Budget and supplies for OEM implementation

1. Budget planning:

Detailed budgeting is required to cover all aspects of the OEM process, including personnel costs, travel expenses, laboratory supplies and data management tools. Budget planning should also account for contingencies and unexpected expenses.

2. Supplies:

Essential supplies include:

- Field equipment: GPS devices, sample collection kits, protective gear.
- Laboratory supplies: Diagnostic tests, reagents, and consumables.
- Data management tools: Laptops, software for data entry and analysis, mobile data collection devices.

Training of survey teams

1. Training considerations:

Comprehensive training programs are essential to equip survey teams with the necessary skills and knowledge.

2. Selection of trainees:

Trainees should be selected based on their qualifications, experience, and ability to work in challenging field conditions. The training programme should include both theoretical and practical components.

3. Selecting and preparing train

Trainers should be experienced professionals with a deep understanding of onchocerciasis and the OEM process. They should be capable of delivering training effectively and addressing trainees' queries and concerns.

Survey coordination and supervision

1. Coordination:

Effective coordination among team members and stakeholders is vital for the smooth execution of OEM activities.

2. Supervision:

Supervisors play a critical role in ensuring the quality and consistency of survey activities.

Training evaluation

1. Evaluation forms:

Evaluation forms for trainers and trainees (included in the annexes) are used to assess the effectiveness of training programmes. These forms help identify areas for improvement and ensure that training objectives are met.

2. Continuous improvement:

Feedback from evaluations is used to refine training programs and enhance the overall effectiveness of OEM activities.

Piloting of Step 3 epidemiological surveys

Piloting Step 3 epidemiological surveys in selected areas helps identify potential challenges and refine methods before full-scale implementation. This pilot phase ensures that any issues are addressed, and processes are optimized for larger-scale surveys.

6.1 Team composition

To successfully conduct OEM, there is a need to constitute teams at two levels: a planning team at the national level, and a field team at the district level where OEM will be conducted.

6.1.1 National planning team

The team selected by the national onchocerciasis programme to support the planning of the OEM strategy should have some background in onchocerciasis and other NTDs. This team will be involved in conducting desk review, exclusion mapping and other relevant planning that the programme may require. The team should consist of at least 2–5 people with as much relevant knowledge of the area, the disease, vectors, other relevant environmental information and procurement processes as possible. Ideally, this should include:

- a representative of the national onchocerciasis programme and/or health ministry;
- a representative of the local health/district/AU/IU area or areas that are being considered for mapping;
- an epidemiologist (if possible with good understanding of *O. volvulus* transmission dynamics);
- an entomologist with some experience with *Simulium* vector species; and
- a logistics officer with understanding in procurement processes and national planning requirements at all levels.

If the area under consideration for OEM is along an international border, then the planning committee should include representatives from the national onchocerciasis programme and local health areas from the neighbouring country.

6.1.2 Field teams

The recommended composition of each of the entomology and epidemiology field teams is provided for guidance.

6.1.2.1 *Step 2 Entomological team*

- Entomologist, 1
- Entomological technician, 1
- Field guide, 1
- Driver, 1

This is the minimal number of an entomological team, however, this can be increased depending on the available budget and the size of river basin to survey.

6.1.2.2 *Step 3 Epidemiological team*

- Epidemiologist/parasitologist
- Laboratory technician, 2, with experience collecting finger-prick blood
- Registrar/enumerator, 1
- Driver, 1
- Field guide, 1, from within the community being visited

This team, when supported by community leaders, will be able to conduct the collection of DBS.

6.1.2.3 *Step 4 Laboratory team*

- Molecular scientist/laboratory scientist
- Laboratory technician, at least 2, with experience using the diagnostic and analysing DBS in the laboratory
- This can increase depending on the number of samples to be processed and timeline

6.1.2.4 *Step 5 Analysis and reporting team*

- Data analyst, 1, who may be an epidemiologist, biostatistician or other programme staff with data analysis experience
- GIS expert, 1, to generate action maps. External support may be requested.
- Other programme support staff, as needed

This team should include individuals with experience analysing basic epidemiological data, writing technical reports and generating basic action maps.

6.2 Budget and supplies for OEM implementation

A comprehensive budget should be developed by the supervisor and field team that accounts for all steps of the OEM process. Each step should be budgeted for separately, as the process is iterative and all steps may not ultimately be necessary. Suggested considerations for each step, and for key budget categories, are as follows.

6.2.1 Budgeting for Step 1:

The desk review team should have access to a computer and a telephone and dedicate at least 1 week to reviewing all available information.

6.2.2 Budgeting for Step 2:

For entomological activities, a budget should be prepared for conducting breeding site assessment for the identification of first-line villages. Some of the key budget line items to be considered are: personnel, equipment, transport, general supplies and cartography.

6.2.3 Budgeting for Step 3:

For epidemiological activities, the activities mainly involved are social mobilization, sampling individuals at household level, undertaking consents and collection/transport of specimens. A budget should be prepared based on the activity to be undertaken. The key budget line items are personnel, equipment, transport, general supplies and DBS collection materials. The importance of these budget line items is highlighted.

6.2.4 Budgeting for Step 4:

For laboratory analysis activities, budgets should include all supplies necessary to conduct diagnostic testing of DBS, including supplementary diagnostics and supplies for training, wastage and incidentals. Key budget line items include personnel, equipment (depending on the diagnostic used), diagnostics and ancillary supplies.

6.2.5 Budgeting for Step 5:

Activities in Step 5 include data analysis, report writing, action map generation and data dissemination. Key budget line items are personnel, equipment (including computers and software) and general supplies for preparing physical reports.

Personnel. Per diem and allowances will be required for those participating in OEM activities, and must be amicably agreed before any person is engaged in the activity. This should ideally be guided by the per diem/allowance rates of the health ministry provided by health personnel. Countries may choose to limit the number of field teams to help minimize costs.

Transport. The provision for transport is essential and will have to be carefully considered, whether to be hired or not. A 4WD vehicle will be required for easy movement of the survey team from village to village. In areas not readily accessible, provision should be made to use motorcycles. Running cost for vehicles and motorcycles should be budgeted.

Cartography. Existing maps of the survey area should be studied, or produced by the health ministry using GIS software. Google Earth and Google Maps provide a good level of detail on satellite imagery for many African countries, showing rivers with rapids often visible, and communities and dwellings marked. When national programmes have GIS expertise in their ranks, there are multiple quality resources of spatial data (vectorial and raster datasets) outlining river basins and providing a wealth of information about river network characteristics (i.e. flow direction, river channel widths, flow accumulation, etc.). All this information combined in a GIS framework can be used during the desk review step. A list of useful resources and examples of their use is provided in Annex 4. If digital resources are limited, maps of suitable scale should be purchased and will help the team during river prospections and identification of first-line villages. Maps of scales 1:50 000 can be obtained from surveying the department of a particular country or the geography department of any national university.

Diagnostics. Diagnostics should be budgeted for based on the number of samples to be analysed as well as a small additional amount to account for wastage and training. When RDT is used, orders for these kits should be made well in advance, preferably 3 months before the survey.

General supplies. There are other general supplies required for the surveys, all of which should be listed and budgeted for to ensure that nothing has been missed. The following is an example list of general supplies that may be needed for OEM:

- GPS (portable or use phone),
- REMO map for a country (if available, in colour to show zones),
- rubber boots,
- first aid kits,
- field notebook,
- blood lancets,
- sharps container,
- absorbent underpads,
- Whatman filter paper (Whatman 903 or TropBio “petal”),
- pencils (for drying DBS),
- styrofoam (to hold pencils while DBS dry),
- absolute alcohol,
- zip lock bags,
- desiccant,
- rubbish bags,
- Barcode labels or permanent marker,
- cool box,
- motorcycle,
- vehicle – preferably 4WD and
- laboratory supplies for preferred diagnostic (Ov16 RDT kit, e.g.).

NOTE: The above list may not be exhaustive, but the logistic officer in the planning team should ensure all the equipment and supplies are available in time for OEM activities. The quantities will be dictated by the number of districts/villages/AUs to be surveyed.

It should be noted that the cost for OEM activities will vary from country to country based on the number of AUs to be mapped and the number of days field teams take in conducting river prospections, identifying first-line villages and conducting first stage sampling of villages. It is important to remember that elimination mapping requires more rigour than required when the target was to control onchocerciasis. A more expensive strategy that correctly identifies all places in need of treatment should be preferred over a less expensive strategy that misses areas of transmission (Rebollo et al., 2018).

Training of survey teams

Training considerations

OEM requires considerable planning and preparation to ensure that the necessary official clearances have been obtained; that community teams have everything that they need on hand at the time they need it; and that the communities they intend to visit are prepared to welcome them.

Similar planning and preparation are required for the community-based component of training. These tasks are the responsibility of the training coordinator. Trainers should ensure that the necessary preparatory work has been completed; otherwise, community-based training sessions may be difficult or impossible. Classroom-based training sessions also require considerable practical preparation. It is therefore recommended that, before training starts, you ensure that:

The protocol has been finalized and approved

Each AU should have its own protocol, detailing exactly what activities will take place and in which specific locations. The protocol should describe steps 1–5, making clear the thresholds for proceeding with each part of the proposed protocol. The protocol should be approved by the health ministry/NOEC or NTD TAC, or, at a minimum, be in line with the national requirements for conducting surveys. The results of OEM may also be used to support a country's elimination dossier, given that it provides evidence and rationale for areas determined to be non-endemic. In this instance, it would be beneficial to include a detailed protocol for each area that will undergo OEM in the dossier, alongside the eventual results.

The survey procurement has been completed

Earlier chapters and annexes detail exactly what is needed for each stage of the survey. It is important, particularly for items that need international procurement, that everything is in country and has cleared customs before the survey training starts. These items will be needed for training, and it is best practice to begin the survey immediately when training is completed. Any unavailable items will delay the survey and cause additional costs.

The necessary official clearances have been obtained

Requirements for ethical clearance for OEM will vary from country to country. Obtaining formal review of the protocol by an ethics committee in advance of community-based work is best practice to ensure that the proposed methods are locally acceptable, are aligned with national standards and guidelines, and is **strongly recommended**. Without formal ethical approval or exemption from full review, determined by an ethical body in the country, it will not be possible to publish the results of OEM in a peer-reviewed journal.

Ideally, the national NTD programme will have taken the lead in communication and coordination with all relevant national, regional and district personnel; outlined the planned survey locations and schedules; and assisted in obtaining all the necessary ethical, administrative and political clearances and permissions. It is advisable to make a member of the OEM planning team directly responsible for obtaining ethical approval and associated permissions.

Training materials have been prepared specifically for the area to be surveyed

Some of the training materials will remain the same regardless of the area being mapped. However, training materials should include details such as the names of the villages being mapped, team composition, community mobilization, and sensitization materials and guidance in local languages. They should also include a summary of the protocol, to help the teams

appreciate why the work they are doing is so important. The training coordinator and planning team may wish to finalize the training materials through email or online meetings, or during a face-to-face meeting. The training materials should be closely aligned with the finalized protocol, so make sure the protocol has been finalized and approved before the training materials are finalized.

An appropriate training site has been selected

If possible, training should take place at a site where onchocerciasis is ivermectin-naive, to provide ample opportunity for practicing OEM in the community. Avoid conducting training elsewhere, such as in a capital city, just because it is convenient for the trainer or trainees. A good training site has the following characteristics:

- close to some rural communities;
- enough chairs and tables for trainees and trainers, and be arranged/have enough room to allow both working in small groups and working with all trainees at once;
- have electricity (or a generator/power pack) for a projector; and
- have facilities for tea and lunch so that trainers and trainees do not have to travel long distances at break times.

Local officials and residents are informed

Inform local health and district officials as well as residents of selected communities who will be involved in the community-based practical, of the training (and upcoming survey, if appropriate). They should be engaged in the process as soon, and as much as possible. Drivers and vehicles for community-based training sessions should be arranged ahead of time. The number of drivers and vehicles required will depend on the number of trainees and trainers that need to be taken to communities each training day.

All the materials required for training are available

Ensure that the following materials and equipment are ready for the training:

- digital projectors,
- laptop computers (for projecting PowerPoints, using the projector; with connecting cables, power adaptor and extension cords),
- spare USB memory sticks (for transferring files between trainers' computers in case of lack of Wifi),
- PowerPoint presentations,
- microphone and amplifier if the group (or the training space) is large,
- flip chart (or a whiteboard) and markers,
- filter-papers for DBS practice,
- sharps bin,
- supporting materials printed,
- soap and water,
- lancets,
- alcohol pads (for fingerpricks),
- absorbent underpads,
- alcohol gel hand disinfectant,
- gloves,
- cotton swabs (for DBS use),
- Android smartphones (1 for each technician plus 1 for the supervisor); with appropriate ESPEN collect surveys loaded, and SD and SIM cards with airtime,
- extra battery packs (1 for each Android),
- chargers (1 per Android) and surge protectors (1 per charger),
- 9.21 notebooks,
- rain-proof carry bags (1 each team member),
- if rain is likely, umbrellas or rain gear,
- pens,
- laminated sheet listing the onchocerciasis questions in the local language (if the forms on the Android are not in the local language),
- consent forms,
- QR code labels,
- thank-you gift for subjects participating in community-based training work (consider pencil, pen, soap, sweets; to be decided locally) and
- certificates of attendance for trainees.

Training of trainers has taken place

It is essential that a day or two in advance of the training, all trainers, supervisors and assistants meet to review and run through the finalized training materials, decide which trainer will lead on which sessions, how rooms will be set up, check that all materials and supplies for the training are available in the correct quantities, any manuals and feedback forms are printed, mobile phones for data collection are available and working correctly, how other trainers will observe the trainees, and how supervision and logistics for practical training will be done. Trainers can also ask questions about the protocol or anything regarding logistics that is uncertain. A representative of the planning team must be present for training of trainers, and ideally for the main training as well. From experience in several countries, training of trainers is most effective as a face-to-face session prior to the main training.

PowerPoint slide sets have been checked on the computer and projector

The trainer(s) should check that the PowerPoint slide sets display well on the projection system being used. Ideally, this would be done in the "training of trainers" or briefing session immediately before the training.

Selection of trainees

It is recommended that each survey team includes a minimum of four individuals: an assigned supervisor or team lead (which may be one of the trainers), one registrar/enroller, one technician (for collecting DBS) and one driver. A community guide should also be recruited to assist the team in the community. Local circumstances will dictate whether other people should be added to the team.

It is important to achieve gender balance within each survey team, and therefore in the selection of trainees. Community participants may feel more comfortable interacting with someone of the same gender. Evidence indicates that gender is a key issue influencing access to, and participation in, health-related activities across many settings. Therefore, the selection of women and men in equal numbers for OEM training is important to ensure that no additional barriers to the participation of men or women in the OEM surveys are created.

To ensure that high-quality standards are maintained, it is usually preferable for a country to train fewer teams and have them move around, rather than train many teams that all work in parallel. However, the more teams you have, the faster the work can be completed.

A decision on the optimal number of teams must be made at country level, but it is useful to have trained extra teams on a ratio of 4:1 should any adverse events arise. In terms of trainers, a ratio of 1 trainer to 12 team members can serve as a guide, with trainers expected to also act in a supervisory capacity. It is important that trainees understand the mapping process in its entirety, but also special attention is needed to ensure that individuals understand well their role within the surveying process.

Desirable trainee traits

Conducting OEM requires people with a number of generic skills. All team members should know how to interact well with residents of rural communities. This means some fluency in the local language, an understanding of the importance of greetings, and good inter-personal communication with village leaders and individuals being surveyed and their families.

Community residents volunteer their time to participate in training and surveys, and must be considered our partners in this work. As such, survey teams must treat them with respect. Additionally, team members should be able to walk long distances and work long hours in communities.

Requirements for Step 3 epidemiological survey technicians

Ideally, technician trainees will have already been trained in similar activities. Ideal technical trainees will be those who have experience with DBS and/or finger-prick blood samples (for example, those who have participated in pre-TAS or TAS (transmission assessment surveys) for lymphatic filariasis). They must be able to read and write, and have excellent attention to detail. Prior experience with smartphones is helpful. People selected for training as recorders need not be health care personnel. Hiring young people who are familiar with the use of smartphones may be a good option.

Requirements for Step 3 epidemiological survey registrars

Registrars or enrollers are the team members in charge of enrolling participants and collecting their personal information. They may also help with directing people towards the DBS collection area, and organizing queuing and movement of people towards and away from the DBS area. Registrars will also be in charge of conducting random sampling. It is important they are able to read and write very well, have good communication and organizational skills, and are happy to interact with community members. The registrar should ideally be able to communicate with community members in the local language.

Requirements for Step 3 epidemiological survey community guides

Within each survey team, a community guide is required to assist with team activities. The guide's role includes demonstrating where the village boundaries lie, introducing the team to survey households, ensuring that households are not inadvertently missed, providing crowd control, translating to and from the local language, and assisting the team in other ways, as needed. This person is not required to attend training, but is selected and briefed by the rest of the team upon arrival for the day's activities

Requirements for Step 1 and 2 entomologists

The role of an entomologist is for overall planning and coordination of entomological field activities. They are responsible for selection of proposed sites to be visited on physical maps or Google maps; confirmation of vector identification, entry of entomological data, analysis and preparation of reports. It is important that the entomologist has a strong understanding of the disease and its associated vector; or is an entomologist from a different discipline who could easily learn.

Requirements for Step 2 entomological technicians

In the breeding site assessment team there should an entomological technician. The role will be to assist an entomologist to carefully read the maps to locate sites where the team will have to visit. In the absence of the entomologist, he/she should be able to draw daily field programmes, prepare simple field budgets, and ensure that all the logistics/ supplies, data collection forms and field reagents are available. The person should be able to walk long distances in tough terrain.

Requirements for Step 2 entomological field guides

Each breeding site assessment team will require a guide who is recruited from the community based on its recommendations. The role will be to introduce the team to the village authorities and to lead them to some of the rivers that might have been identified on the map for assessment. The guide will also assist the team in carrying some of the field equipment/supplies. This person is not required to attend formal training, but is briefed by the other team members on the day.

Requirements for Steps 2 and 3 OEM drivers

If possible, as part of the rental agreement, drivers should be chosen for their willingness to assist the survey teams with the survey work in the community. If drivers can be involved, it is likely to be helpful to have them join part of the training, in order to understand the purpose of and overall plans for the survey. Where possible, the driver may help the survey teams within communities by:

- assisting with introductions in the community;
- assisting with crowd control; and
- encouraging families from selected households who are not at home but who are in the village at the time of the team's visit, to find the OEM team before the end of the day, with the help of local village residents.

Selecting and preparing trainers

Candidate trainers for epidemiological components should be very experienced in collecting DBS or conducting other serology diagnostic tests in the community and have experience as a supervisor. Candidate trainers for entomology components should be someone with knowledge and understanding of black flies, ideally a black fly entomologist or entomologist from a different specialty. In both cases, experienced health ministry or health care workers would be ideal. They should also be experienced in data collection and be thoroughly familiar with the operation of a smartphone. However, being good at performing a task is not enough to qualify an individual to be a good teacher for that task.

Choose candidate trainers who have both a strong skill set for the tasks at hand and an ability to impart that skill set to others. Ideally, trainers should be able to train in DBS collection, data entry on mobile phones and to have had previous experience in training others. Communication skills are key for trainers, and an ability to undertake public speaking and explain complex topics to trainees is essential. Even with previous experience in training teams for onchocerciasis assessments, trainers should ensure that they are completely familiar with all the details in this strategy document. Among the trainers, there should be at least one who has had experience in entomological field work with good knowledge in onchocerciasis disease.

Survey coordination and supervision

It is critical to consider the supervisory responsibilities of trainers and clarify what characteristics constitute the supervisory role. Supervision strategies can include:

- contacting teams by phone, WhatsApp or text messages;
- joining the teams in communities at the beginning of mapping;
- making periodic visit to communities to observe teams (scheduled or surprise);
- paying calls or visits to village leaders after sampling is completed;
- conducting routine follow up with persons providing logistic support;
- convening team meetings;
- reviewing data using the online dashboard and through daily email summaries; and
- as the first days of survey are often slow, encouraging teams to complete clusters properly.

It is recommended to choose several of the above approaches to have rigorous supervision, particularly if sample collection exceeds 5 days. The purpose of supervision is to provide extra support for inexperienced technicians to assist them with proper DBS procedures and make sure data are collected accurately and in full. There is a need to ensure that:

- the protocol is being followed correctly;
- use of QR code/barcode ID labels is well organized so as to avoid opportunities for introducing duplicate entries or missing data;
- daily communication with the OEM data manager occurs to identify any potential issues with submitted data quickly; and
- troubleshooting and support are provided for any challenges that arise in a particular location.

Ongoing supervisory considerations for during the survey and challenges require proactive supervision.

Training evaluation

After each day of training, an important process of reflection will be to evaluate the effectiveness of training. It is a vital element to any training in order to analyse, and to learn which elements successfully achieved their objectives, and which may have fallen short. The main purpose of evaluating the training will be to ascertain whether the objectives of the training have been met, and how lessons learnt can help inform subsequent adjustments and improvements to the training, as required. It is critical to gain perspectives from both the trainers and the trainees by using both focus group discussions and anonymous surveys. Evaluation forms are structured for trainers to have a daily debriefing and an end of training evaluation/ focus group discussion (Annex 14), and for trainees to have daily and end of training evaluation forms (Annex 15). It will be important for trainees to review and analyse these forms daily, and for further analyses to be undertaken at the end of the training.

Piloting of Step 3 epidemiological surveys

It is essential that the training for Step 3A and Step 3B include at least 2 full days of pilot testing the survey methodology in communities that were not selected for the study but have similar characteristics to the chosen sites. These should ideally be communities in the same AU as is being targeted for OEM. The purpose of field piloting is to allow the survey teams an opportunity to practice the methodology, including enrolment, finger-prick collection and DBS creation. It is an opportunity to identify the best workflow in the field and make any changes before the start of data collection. After each day of pilot testing in the field, the survey teams should meet with the trainer to debrief on the experience and make any changes before the next day. Learnings and best practices should then be incorporated into the workflow during the actual surveys.

References

Atsame J et al (2016). Abstract 51. Filariasis in Gabon: empiric assessments refine distribution and treatment strategies for onchocerciasis and loiasis. Am J Trop Med Hyg 2016; 95 (Suppl 5):17 (https://www.astmh.org/ASTMH/media/Documents/ASTMH-2016-Annual-Meeting-Abstract-Book.pdf, accessed 10 July 2024).

Cama VA et al (2018). Evaluation of an OV-16 IgG4 enzyme-linked immunosorbent assay in humans and its application to determine the dynamics of antibody responses in a non- human primate model of *Onchocerca volvulus* infection. Am J Trop Med Hyg;99(4):1041–1048. doi:10.4269/ajtmh.18-0132.

Cano J et al (2014). The global distribution and transmission limits of lymphatic filariasis: past and present. Parasit Vectors; 7:466. doi:10.1186/s13071-014-0466.

Cantey PT et al. Transitioning from river blindness control to elimination: steps toward stopping treatment. Int Health. 2018 Mar 1;10(suppl_1):i7–i13.

Cromwell EA et al (2021). Predicting the environmental suitability for onchocerciasis in Africa as an aid to elimination planning. PLoS Negl Trop Dis; 15(7):e0008824. doi:10.1371/journal.pntd.0008824.

Crosskey R (1995). The natural history of blackflies. West Sussex, England: John Wiley and Sons Ltd.

Deribe K et al (2020). Predicting the environmental suitability and population at risk of podoconiosis in Africa. PLoS Negl Trop Dis; 14(8):e0008616. doi:10.1371/journal.pntd.0008616.

Duke BOL (1975). The differential dispersal of nulliparous and parous *Simulium damnosum*. Tropenmed.Parasitol; 26:88–97.

Gass K (2020). Time for a diagnostic sea-change: rethinking neglected tropical disease diagnostics to achieve elimination. PLoS Negl Trop Dis; 14(12):e0008933. doi:10.1371/journal.pntd.0008933.

GBD (2017). GBD 2017 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease study 2017. Lancet; 392:1789–1858. doi:10.1016/S0140-6736(18)32279-7.

Goodwin L, Duggan A (1972). A new tropical hygiene and human biology. London, England: George Allen and Unwin.

Hamill L et al (2022). Navigating the way to onchocerciasis elimination: the feasibility and affordability of Onchocerciasis Elimination Mapping. Int Health; 14(Suppl 1)i17–i23. doi:10.1093/inthealth/ihab083.

Katabarwa MN et al (2010). Does onchocerciasis transmission take place in hypo-endemic areas? A study from the North Region of Cameroon. Trop Med Int Health; 15:645–52.

Katabarwa MN et al (2014). Transmission of *Onchocerca volvulus* by *Simulium neavei* in Mount Elgon focus of Eastern Uganda has been interrupted. Am J Trop Med Hyg; 90:1159–66.

Katabarwa MN et al (2018). After 70 years of fighting an age-old scourge, onchocerciasis in Uganda, the end is in sight. Int Health; 10:i79–i88. doi:10.1093/inthealth/ihx044.

Kifle B, Kifle W, Nigatu M (2019). Prevalence of onchocerciasis and associated factors among adults aged ≥ 15 years in Semen Bench District, Bench Maji Zone, Southwest Ethiopia: community based cross-sectional study, Adv Public Health; 7276230. doi:10.1155/2019/7276230.

Lakwo TL et al (2018). Cross border collaboration in onchocerciasis elimination in Uganda: progress, challenges and opportunities from 2008 to 2013. Glob Health; 14(1):16.

McMahon JP (1957). Notes on the *Simulium neavei* group of Simuliidae with particular reference to *S. nyasalandicum* and *S. woodi.* Bull Entomol Res; 48(3):607–17.

Messina JP et al (2016). Mapping global environmental suitability for Zika virus. Elife; 5:e15272. doi:10.7554/eLife.15272.

Messina JP et al (2019). The current and future global distribution and population at risk of dengue. Nat Microbiol; 4:1508–15. doi:10.1038/s41564-019-0476-8

Miri ES et al (2023). Two Nigerian States (Plateau and Nasarawa) have eliminated transmission of human onchocerciasis: a report of post-ivermectin mass drug administration surveillance. Am J Trop Med Hyg; 108(1):37–40.

Mylne AQ et al (2015). Mapping the zoonotic niche of Lassa fever in Africa. Trans R Soc Trop Med Hyg; 109(8):483–92. doi:10.1093/trstmh/trv047.

Ngoumou P, Walsh F (1993). A manual for rapid epidemiological mapping of onchocerciasis. Geneva: World Health Organization (https://apps.who.int/iris/handle/10665/59537).

Noma M et al (2002). Rapid epidemiological mapping of onchocerciasis (REMO): its application by the African Programme for Onchocerciasis Control (APOC). Ann Trop Med Parasitol; 96(1):S29–S39.

Noma M et al (2014). The geographic distribution of onchocerciasis in the participating countries of the African Programme for Onchocerciasis Control: (1) Priority areas for ivermectin treatment. Parasit Vectors, 7:325- 339.

Nwoke BEB et al (2023). Onchocerciasis in Nigeria 1: history of control efforts. Niger J Parasitol, Special Issue No. 1, May 2023 (https://njpar.com.ng/index.php/home/article/view/356/381).

OTS3 (2019). Report of the third meeting of the WHO Onchocerciasis Technical Advisory Subgroup, Geneva, 26–28 February 2019. Geneva: World Health Organization (https://apps.who.int/iris/handle/10665/332257).

OTS4 (2020). Report of the fourth meeting of the WHO Onchocerciasis Technical Advisory Subgroup: virtual meeting, 28–29 October 2020. Geneva: World Health Organization (https://apps.who.int/iris/handle/10665/348383).

OTS7 (2023). WHO Onchocerciasis Technical Advisory Subgroup: report of the seventh meeting, Saly, Senegal, 4 November 2023 (https://iris.who.int/handle/10665/380313).

Pigott DM et al (2016). Updates to the zoonotic niche map of Ebola virus disease in Africa. eLife:5. doi:10.7554/elife.16412.

Rebollo MP et al (2018). Onchocerciasis: shifting the target from control to elimination requires a new first-step—elimination mapping. Int Health; 10:i14–i19. doi:10.1093/inthealth/ihx052.

Renz A, Wenk P (1987). Studies on the dynamics of transmission of onchocerciasis in a Sudan-savanna area of North Cameroon I. Prevailing *Simulium* vectors, their biting rates and age-composition at different distances from their breeding sites. Ann Trop Med Parasitol; 81(3):215–228.

Richards FO et al (2018). Operational performance of the *Onchocerca volvulus* "OEPA" Ov16 ELISA serological assay in mapping, guiding decisions to stop mass drug administration, and posttreatment surveillance surveys. Am J Trop Med Hyg; 99(3):749–752. doi:10.4269/ajtmh.18-0341.

Sauerbrey M, Rakers LJ, Richards FO (2018). Progress toward elimination of onchocerciasis in the Americas. Int Health. 10(suppl_1):i71-i78. doi:10.1093/inthealth/ihx039.

Simpson H et al (2021). Mapping suitability for Buruli ulcer at fine spatial scales across Africa: a modelling study. PLoS Negl Trop Dis. 15(3):e0009157. doi:10.1371/journal.pntd.0009157.

Tekle AH et al (2016). Progress towards onchocerciasis elimination in the participating countries of the African Programme for Onchocerciasis Control: epidemiological evaluation results. Infect Dis Poverty. 5:66. doi:10.1186/s40249-016-0160-7.

Thompson BH (1976). Studies on the flight range and dispersal of *Simulium damnosum* (Diptera: Simuliidae) in the rain-forest of Cameroon. Ann Trop Med Parasitol; 70(3):343–54.

Traore MO et al (2012). Proof-of principle of onchocerciasis elimination with ivermectin treatment in endemic foci in Africa: final results of a study in Mali and Senegal. PLoS Negl Trop Dis; 6(9):e1825.

Walsh JF et al (2023). A field investigation of short-range dispersal by female *Simulium damnosum s.l.* Acta Trop; 240:106863. Erratum in: Acta Trop; 242:106898.

Westgard JO et al (2002). Chapter 6. In: Basic QC practices, training in statistical quality control for healthcare laboratories, second edition. Madison, WI: Westgard QC, Inc;77–88.

WHO (1992), Methods for community diagnosis of onchocerciasis to guide ivermectin based control in Africa: report of an informal consultation held in Ouagadougou from 19–21 November 1991. Geneva: World Health Organization (https://iris.who.int/handle/10665/204268).

WHO (1995). Supplemental guidelines for rapid epidemiological mapping of onchocerciasis. Geneva: World Health Organization (TDR/TDF/ONCHO/95.1 https://tdr.who.int/docs/librariesprovider10/meeting-reports/oncho-guidelines-1995-pdf.pdf).

WHO (1998). Guidelines for analysis of REMO data using GIS. UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases (https://iris.who.int/handle/10665/64124).

WHO (2008). Training for mid-level managers: The EPI coverage survey. Geneva: World Health Organization Expanded Programme on Immunization (https://apps.who.int/iris/handle/10665/337065).

WHO/APOC (2009). Informal consultation of elimination of onchocerciasis transmission with current tools in Africa – "Shrinking the map". Ouagadougou, Burkina Faso 25 – 27 February 2009. Ouagadoudou: African Programme for Onchocerciasis Control (https://iris.who.int/handle/10665/275793).

WHO (2016). Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis. Criteria and procedures. Geneva: World Health Organization (https://apps.who.int/iris/handle/10665/204180).

WHO (2019). Onchocerciasis elimination mapping of endemic countries is key to defeating river blindness. Geneva: World Health Organization (In: WHO/News [website]. Geneva: World Health Organization (https://www.who.int/news/item/29-11-2018-onchocerciasis-elimination-mapping-of-endemic-countries-is-key-to-defeating-river-blindness, accessed 10 July 2024).

WHO (2020). Ending the neglect to attain the Sustainable Development Goals: a road map for neglected tropical diseases 2021–2030. Geneva: World Health Organization (https://apps.who.int/iris/handle/10665/338565).

WHO (2022). Onchocerciasis. In: WHO/Newsroom/fact sheets [website]. Geneva: World Health Organization] (https://www.who.int/news-room/fact-sheets/detail/onchocerciasis, accessed 10 July 2024).

WHO (2023a). Elimination of human onchocerciasis: progress report, 2022–2023. Wkly Epidemiol Rec. 98 (45):572–582. Geneva: World Health Organization (https://iris.who.int/handle/10665/373991).

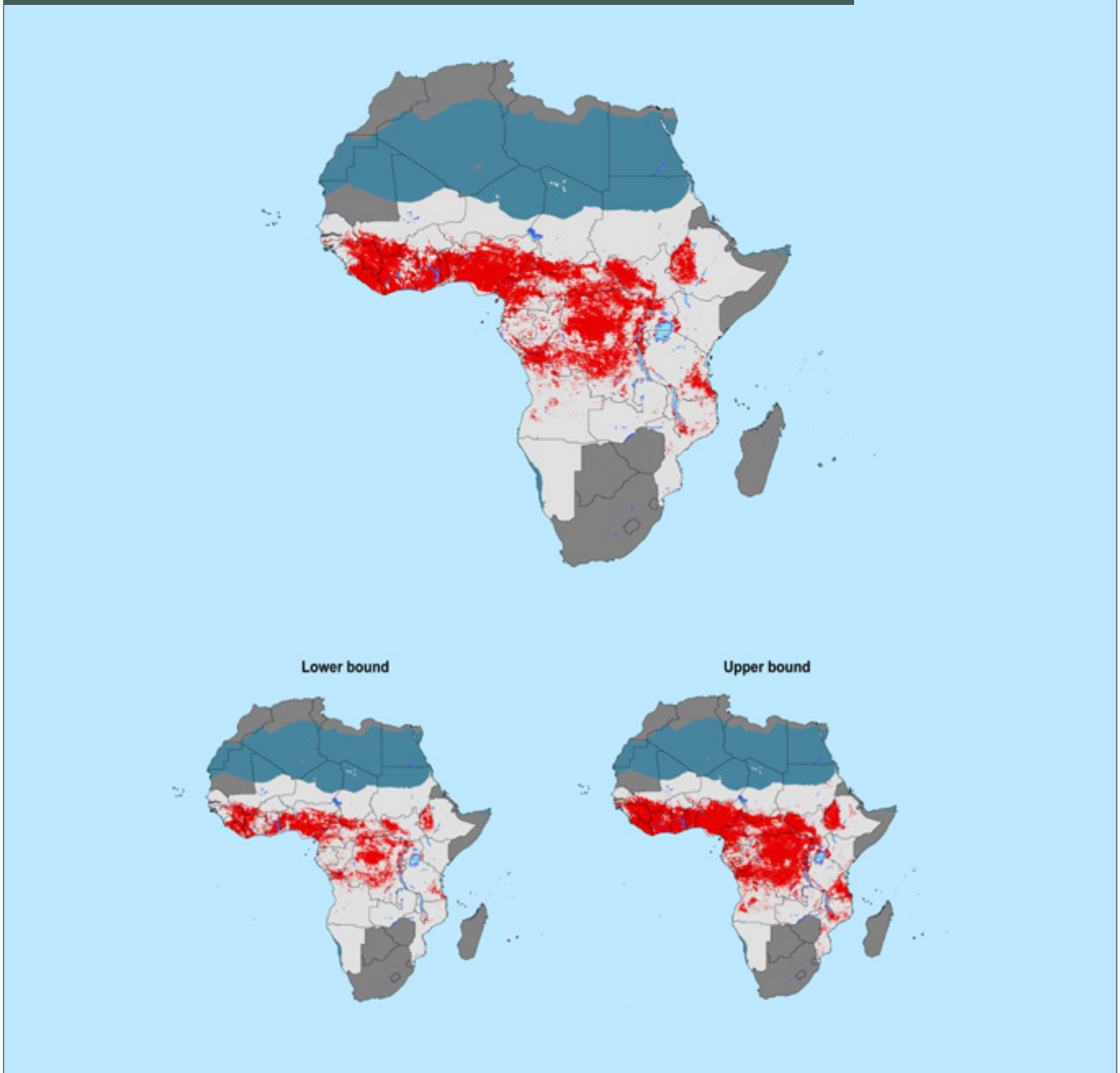
WHO (2023b). Entomological manual for onchocerciasis elimination programmes. Geneva: World Health Organization (https://iris.who.int/handle/10665/371688).

Zouré HGM et al (2011). The geographic distribution of *Loa loa* in Africa: results of large-scale implementation of the Rapid Assessment Procedure for Loiasis (RAPLOA). PLoS Negl Trop Dis; 5(6):e1210. doi:10.1371/journal.pntd.0001210.

Annex 1.

Additional mapping and geospatial modelling resources

Fig. A1.1. Predicted occurrence of onchocerciasis based on environmental modelling



Predicted occurrence Onchocerciasis (Model IHME)

0 2,500 5,000 Km

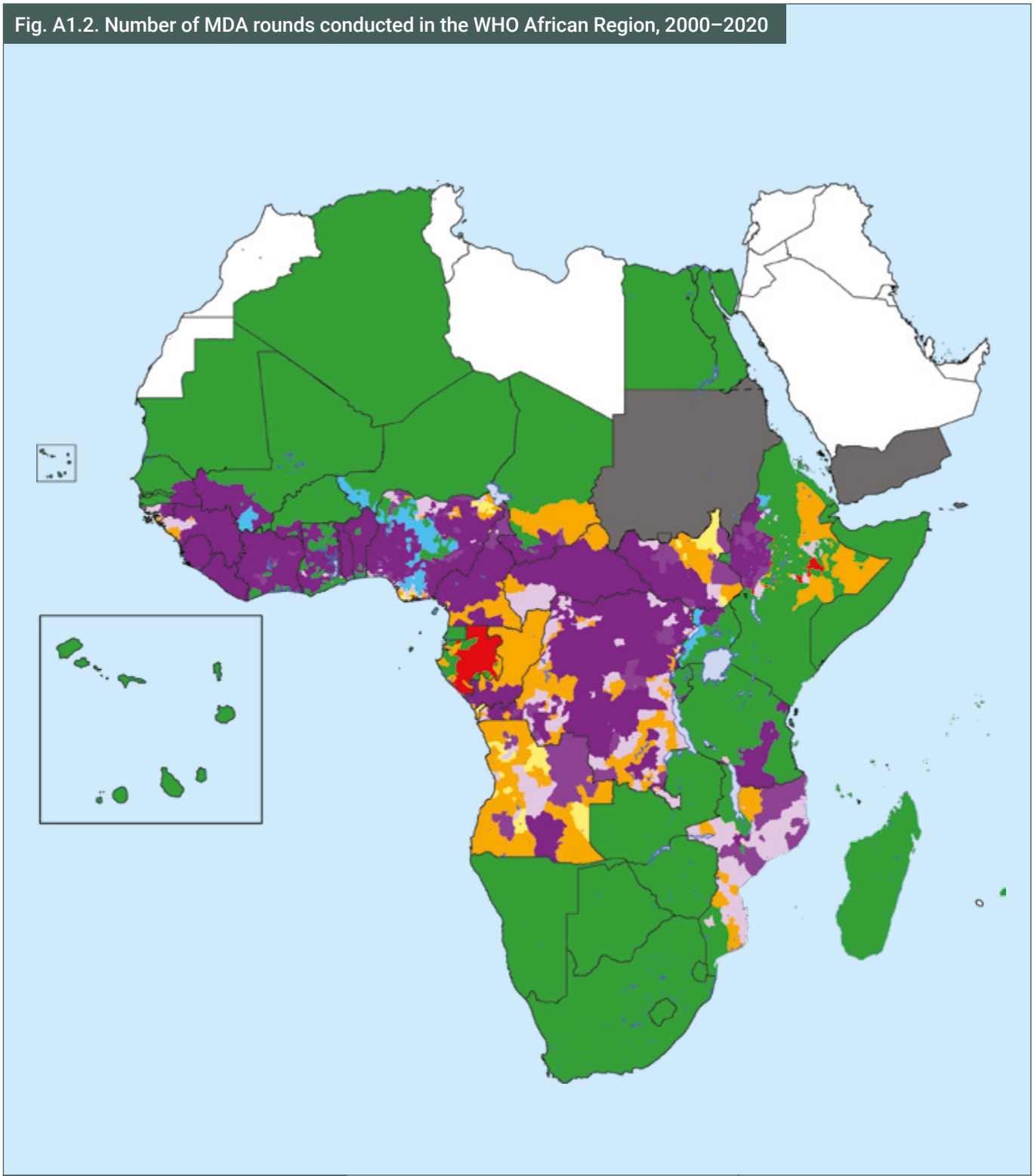
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Non-endemic	Endemic (requiring MDA)	Endemic (pending IA)	Unknown (under LF MDA)

Disclaimer: The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN Regional Office for Africa World Health Organization

Fig. A1.2. Number of MDA rounds conducted in the WHO African Region, 2000–2020



Onchoceriasis No. MDA rounds	0 1,000 2,000 Km	© WHO-AFRO 2023. All rights reserved

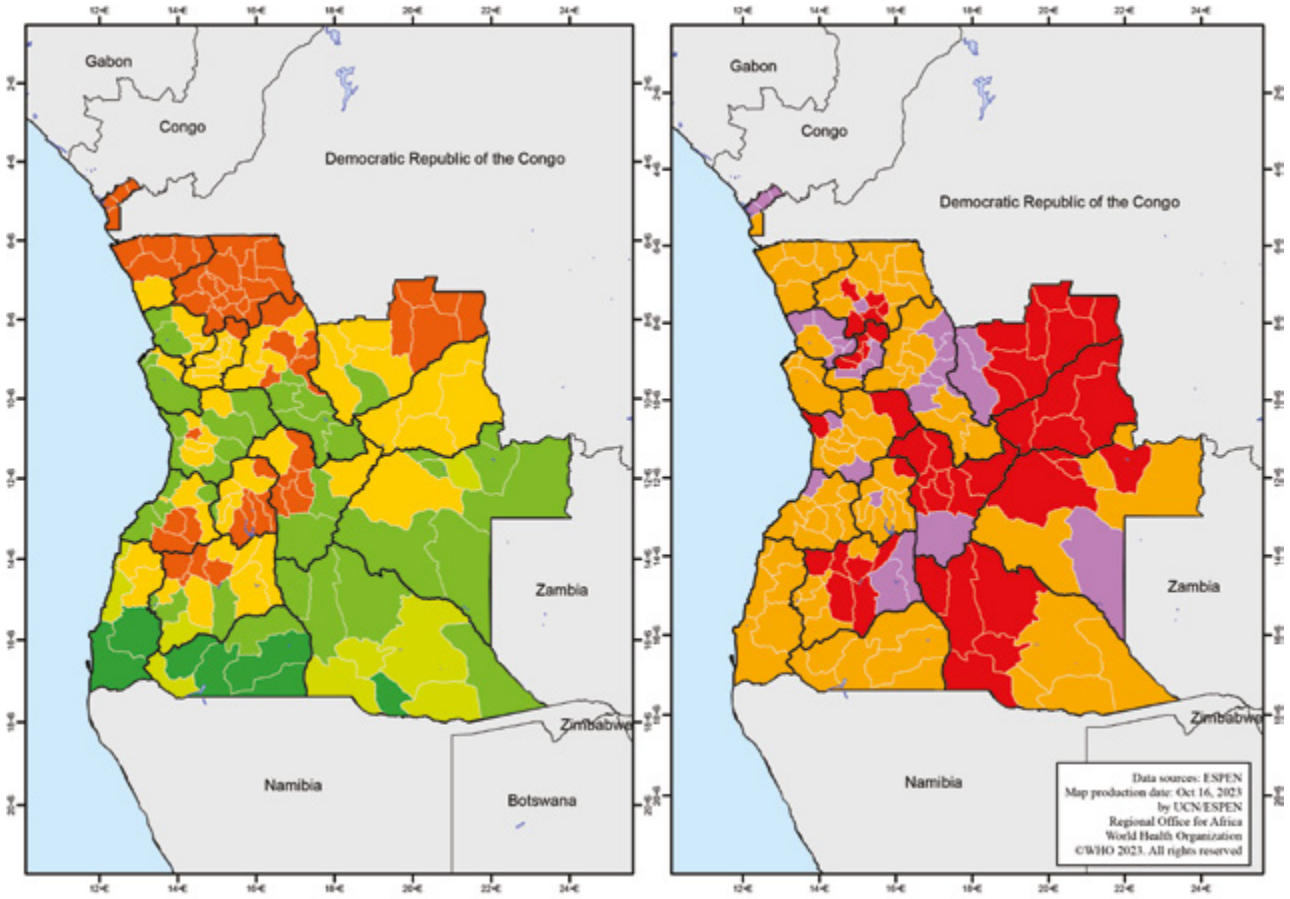
<5 rounds	= 5 - 10 rounds	= 10 rounds	Endemic (MDA not delivered)	Non-endemic	Endemic (pending IA)	Endemic (under post MDA surveillance	Inknown (under LF MDA)	Unknown (consider OEM)	Not reported

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Data source: Data provided by health ministries to ESPEN through WHO reporting process. All reasonable precautions have been taken to verify this information

Map production: ESPEN/AFRO/WHO NTD Mapping project funded by the Bill & Melinda Gates Foundation.

Fig. A1.3. Predicted suitability for onchocerciasis and reported endemicity (2020) at implementation unit level in Angola



Predicted Suitability Categorized	
0.85 - 1: Very high suitability	0.15 - 0.23: Very low suitability
0.65 - 0.85: High suitability	0 - 0.15: Unsuitable
0.5 - 0.65: Moderate suitability	N/A
0.25 - 0.5: Low suitability	

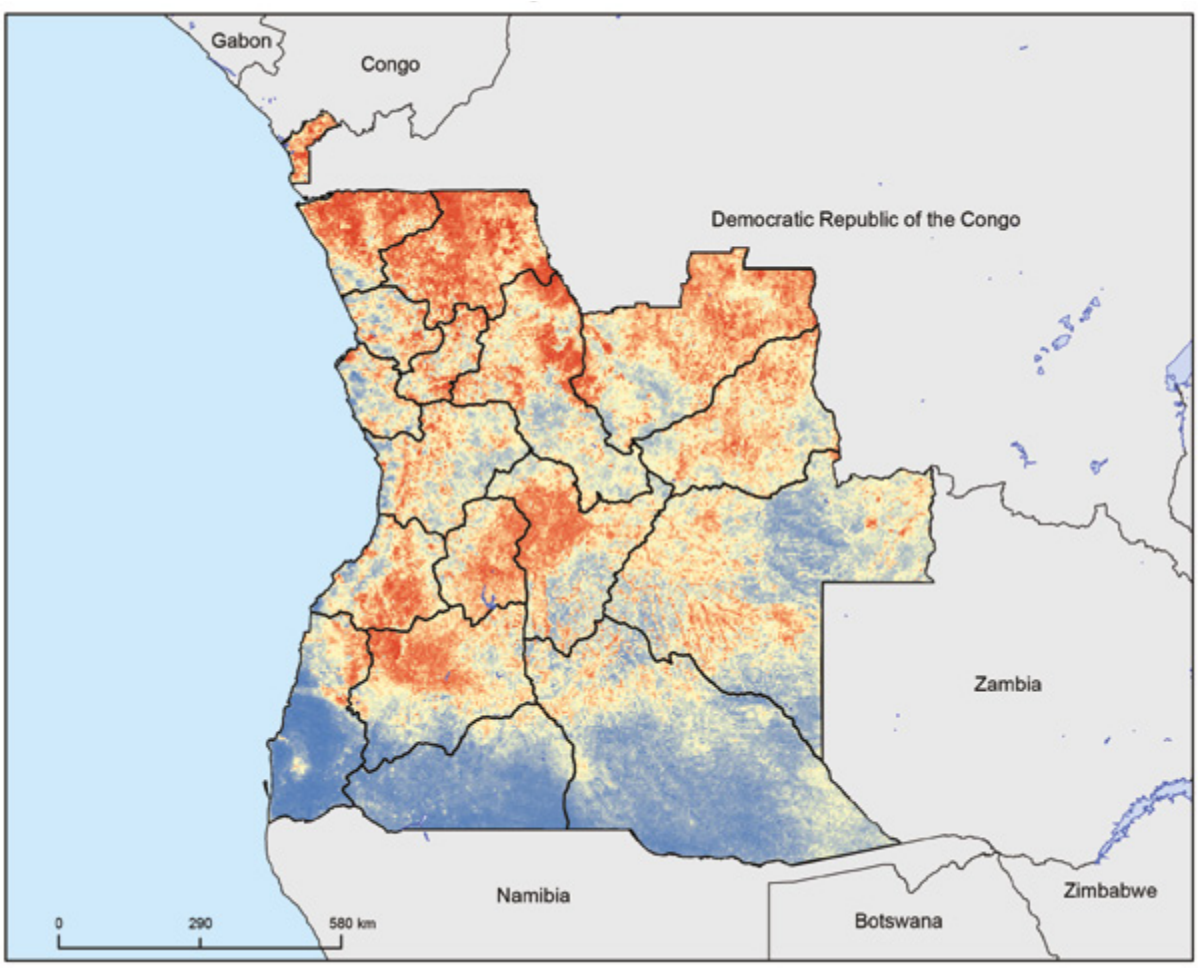
Oncho Endemicity	
0.85 - 1: Very high suitability	0.15 - 0.23: Very low suitability
0.65 - 0.85: High suitability	0 - 0.15: Unsuitable
0.5 - 0.65: Moderate suitability	N/A
0.25 - 0.5: Low suitability	

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Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN
Regional Office for Africa World Health Organization

Fig. A1.4. Predicted environmental suitability for onchocerciasis in Angola





Predicted suitability Upper Bound CI

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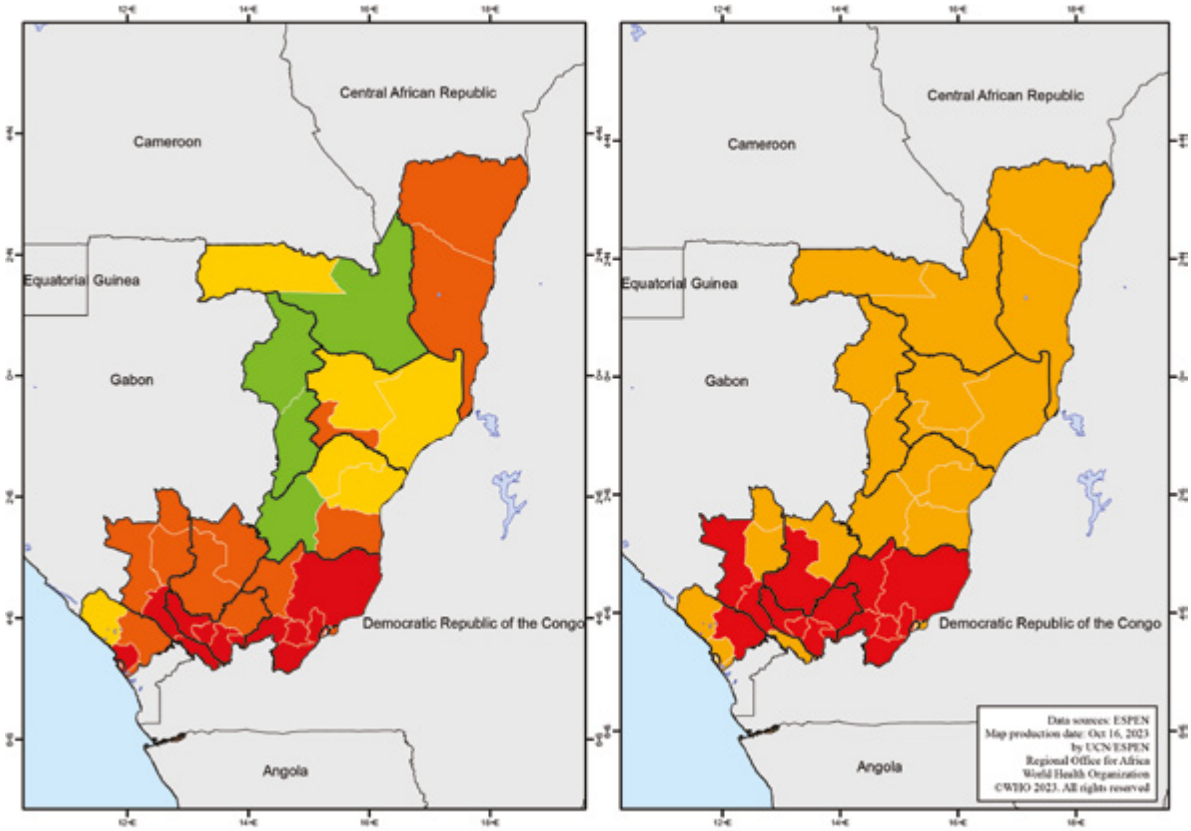
High: 1

Low: 0

Disclaimer: The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN
Regional Office for Africa World Health Organization

Fig. A1.5 Predicted suitability for onchocerciasis and reported endemicity (2020) at implementation unit level in Congo



Predicted Suitability Categorized	
0.85 - 1: Very high suitability	0.15 - 0.23: Very low suitability
0.65 - 0.85: High suitability	0 - 0.15: Unsuitable
0.5 - 0.65: Moderate suitability	N/A
0.25 - 0.5: Low suitability	

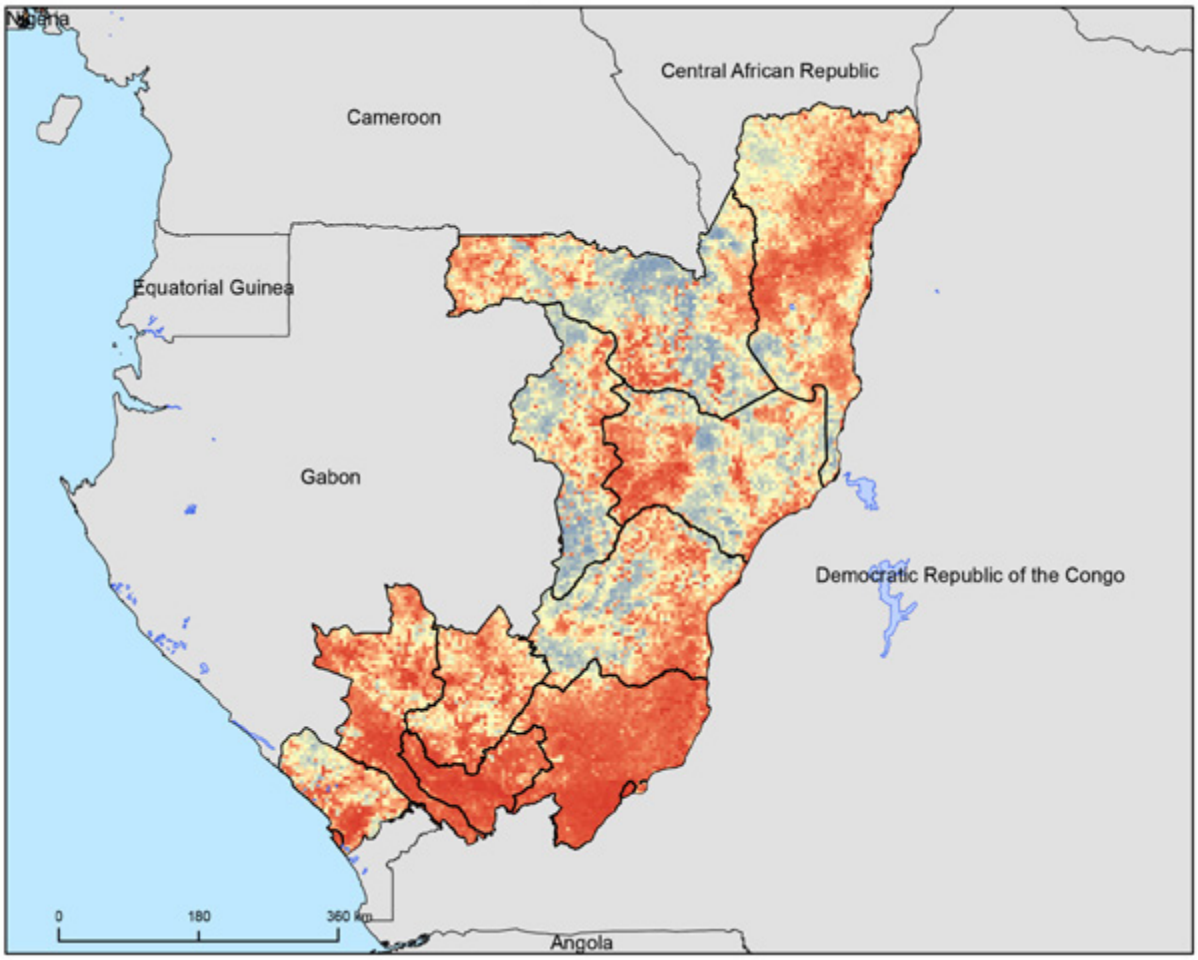
Oncho Endemicity	
0.85 - 1: Very high suitability	0.15 - 0.23: Very low suitability
0.65 - 0.85: High suitability	0 - 0.15: Unsuitable
0.5 - 0.65: Moderate suitability	N/A
0.25 - 0.5: Low suitability	

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Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN
Regional Office for Africa World Health Organization

Fig. A1.6. Predicted environmental suitability for onchocerciasis in Congo





Predicted suitability Upper Bound CI

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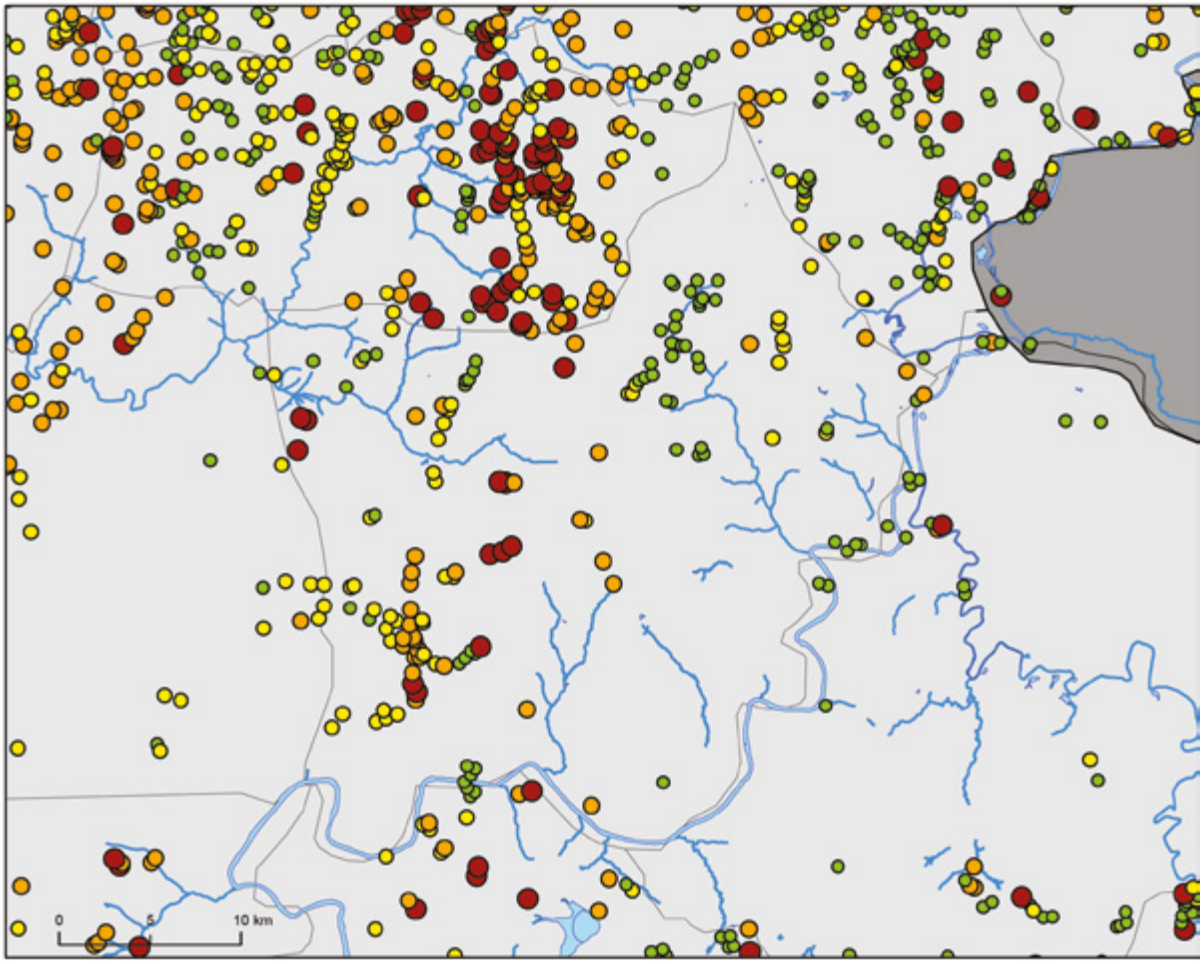
High: 1

Low: 0

Disclaimer: The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN Regional Office for Africa World Health Organization

Fig. A1.7 Distribution of priority communities for OEM in Guinea-Bissau based on environmental suitability modelling





Priority level - OEM GRID3 settlements

0 2,500 5,000 Km

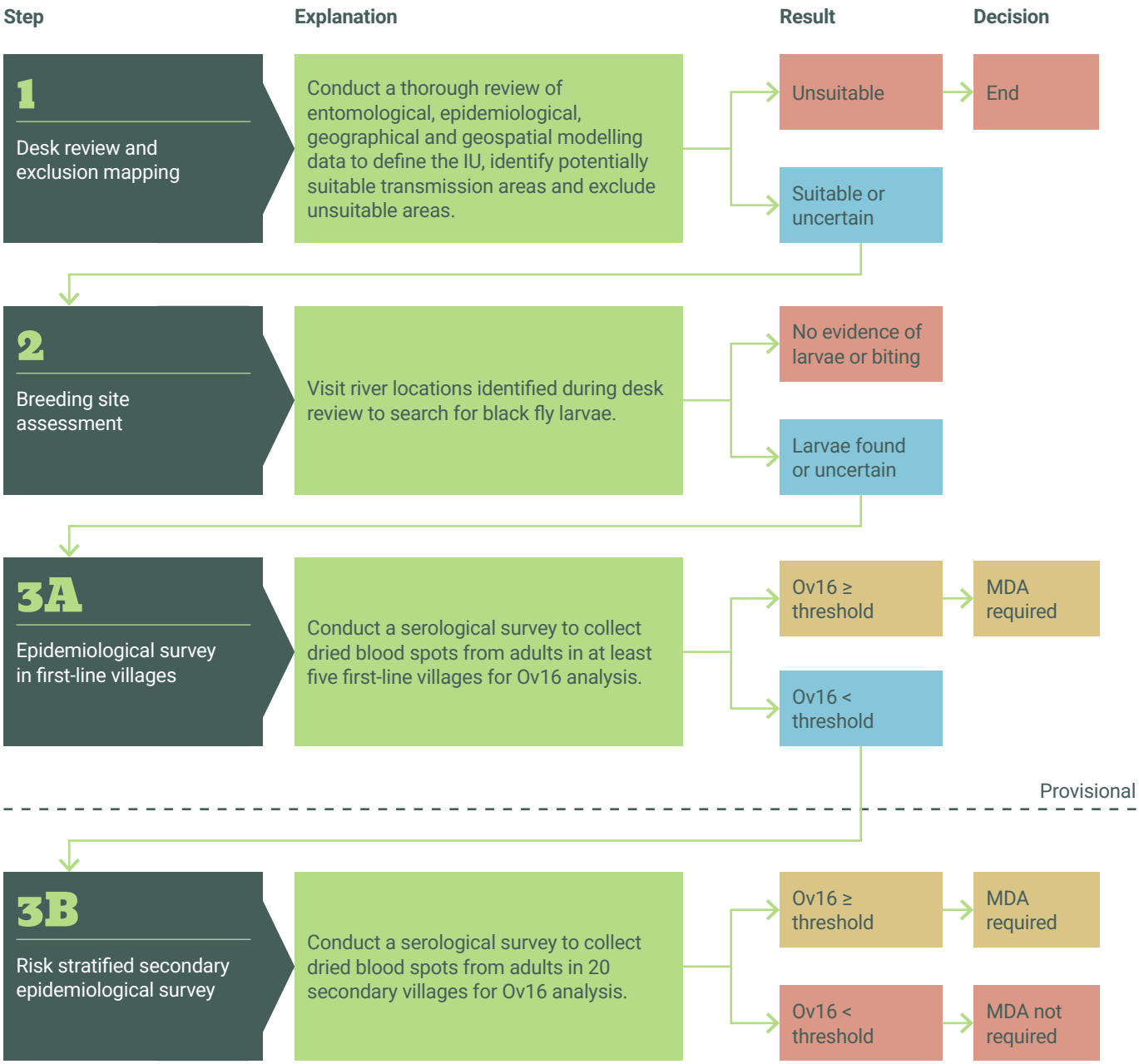
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0	1 - 6	7 - 8	9 - 10	11 - 13	Water bodies	Rivers

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Data sources: ESPEN
Map production date: Oct 16, 2023 by UCN/ESPEN
Regional Office for Africa World Health Organization

Annex 2.
Overview of steps in the OEM process



Annex 3.

Standard operating procedure for collecting dried blood spots

Purpose: This standard operating procedure describes the standard procedure for collecting dried blood spots (DBS)

Audience: Trainers, supervisors and technicians

Responsibilities: Trainers, supervisors and technicians should adhere to this procedure in order to maintain a safe and standardized approach to collecting DBS.

Specific procedure: Blood should only be taken from participants who have agreed to be enrolled in a survey and have consented to a particular blood test. This procedure outlines the basic steps to follow when collecting DBS.

Supplies

- Sterile lancet
- Alcohol swab
- Permanent marker pen for labelling
- Nitrile or other latex-free gloves
- Sharps disposal container
- Waste disposal container or bag
- Dry cotton wool
- Plasters
- Soap and water
- QR code labels
- Desiccant packs
- DBS filter-paper
- Pencil
- Pencil sharpener
- Packaging storage bag or sealable small plastic bags
- Styrofoam sections
- Absorbent bench pads

Preparing to take a DBS

1. Gather the materials listed above in the testing area.
2. Prepare DBS drying pencils by sharpening a normal pencil at both ends and sticking it into a polystyrene foam block.
3. Explain to the participant what the test is for and which procedure you will be following.
4. Wash your hands thoroughly with soap and water, and allow your hands to dry.
5. Put gloves on.
6. Affix the unique QR code label to the DBS filter-paper.
7. Ask the participant to warm his or her hands by rubbing them together to increase blood flow.
8. Clean the participant's fourth finger (ring finger) with the alcohol swab, and allow the finger to dry for 10 seconds.

Performing a DBS

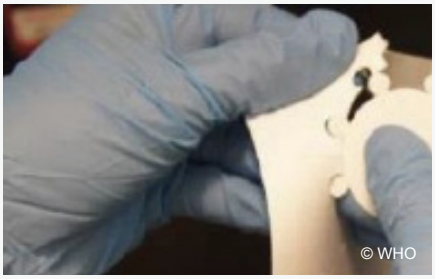
1. Remove the cap from the lancet, then pierce the side of the participant's finger, which has been cleaned with an alcohol swab.
2. Dispose of the lancet in the sharps disposal container.
3. Apply gentle pressure to the finger to express the first drop of blood, then wipe it away with a dry piece of cotton wool.
4. Apply gentle pressure to the finger until a new droplet of blood forms on the finger, then gently squeeze and hold the DBS paper to catch the blood.
5. Allow a large drop of blood to collect, then, holding the filter paper by the edges, touch and hold one of the ears against the drop of blood, allowing enough blood to soak the circle.
6. Repeat this process on each ear of the filter paper until all available ears are full. Do not apply blood to the central portion of the filter paper.
7. Once the DBS filter-paper is full, allow it to dry on a drying pencil. Do not touch the blood on the filter-paper.

Further instructions

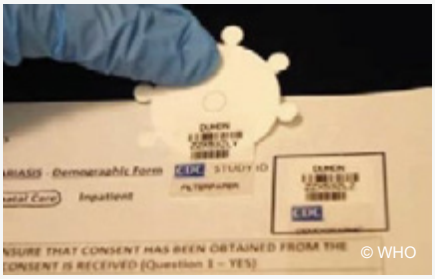
- Always wash your hands before taking a blood sample.
- Always wear gloves when handling filter-paper. Filter-paper is sensitive to contamination.
- Always use universal safety precautions and blood-borne pathogen precautions when handling blood.
- This outline is not intended to replace your standard operating procedure for taking DBS.



1. Collect test supplies:
   - Gloves
   - Filter-paper
   - Lancet
   - QR Code label
   - Alcohol swab
   - Sharps container.



2. **First**, wash hands thoroughly using effective handwashing techniques. Put on gloves. **Then**, gently remove one filter-paper disc from the filter sheet along the perforated lines.



3. Put the QR code sticker labelled "filter-paper" on the disc. Ensure the ID number matches the number on this individual's demographic form.



4. Clean the fingertip with an alcohol swab and allow the fingertip to dry completely.



7. Touch the **edge of one of the filter's extensions** to the blood until the extension is completely filled with blood.



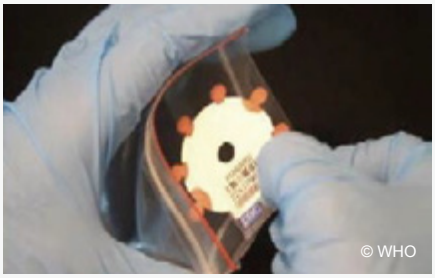
10. Dry the filter-paper discs on a pencil by gently placing them on the pencils so that the extensions do not touch. Place the foam pads into a cardboard box or a cool box in order to stabilize and secure samples.



5. Open the lancet package and pierce the finger on the side of the finger pad. Dispose of the lancet in a sharps container.



8. Continue filling the extensions with blood. **Be sure the extensions are completely filled on both sides of the disc (no white showing)**.



11. When the filter paper disc is completely dry, place the filter-paper in a small sealable plastic bag (one filter-paper disc per bag).



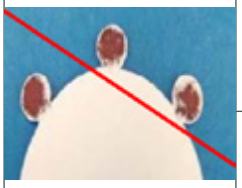
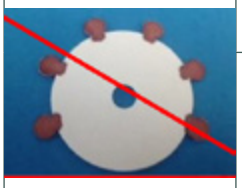
6. Wipe away the first drop of blood with sterile gauze or cotton. Then allow a dro plet of blood to form on the finger.



9. Allow the filter-paper discs to **dry completely (4 hours or overnight)** on a clothesline or pencil. Make sure that the extensions do not touch.



12. Place multiple bags (100–150) into a larger plastic bag. Also place 3–5 desiccant packs in the bag and close tightly. Store at −20 °C.

Spot check for DBS on filter-paper		
VALID		
	**Qualities of a valid DBS collection:**	A sufficient quantity of blood has filled each extension. All extensions are filled with blood. The disc is appropriately labelled with the barcode sticker.
INVALID		
	**Problem:**	Extensions are not completely filled with blood.
	Correction:	Wait until extensions are completely filled with blood before moving to the next extension. It is better to have four correctly filled extensions than six partially filled extensions.
	**Problem:**	One side of the extension has white spots. This is usually because blood was also applied besides, rather than at the edge, of the filter-paper.
	Correction:	Collect blood by touching the edge of the extension to the drop of blood to draw the blood into the filter-paper.
	**Problem:**	The dried blood spot appears supersaturated, clotted or layered. This can happen when blood has been applied to both sides of the paper.
	Correction:	Do not collect blood on both sides of the paper. Instead, collect blood by touching the edge of the extension to the blood.
	**Problem:**	The filter-paper disc is not labelled.
	Correction:	Label the filter-paper disc with a barcode label. Ensure that the ID matches the ID on the demographic form for this participant.
	**Reason:**	The specimen shows serum rings on the central disc area.
	Correction:	Allow the alcohol to dry completely before lancing the finger. Do not allow paper contact with alcohol, lotion or other substances: keep filter-paper dry. Do not apply blood to paper with a capillary tube.

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Annex 4.

Tools for generating maps for desk review and action maps based on results of onchocerciasis eliminating mapping

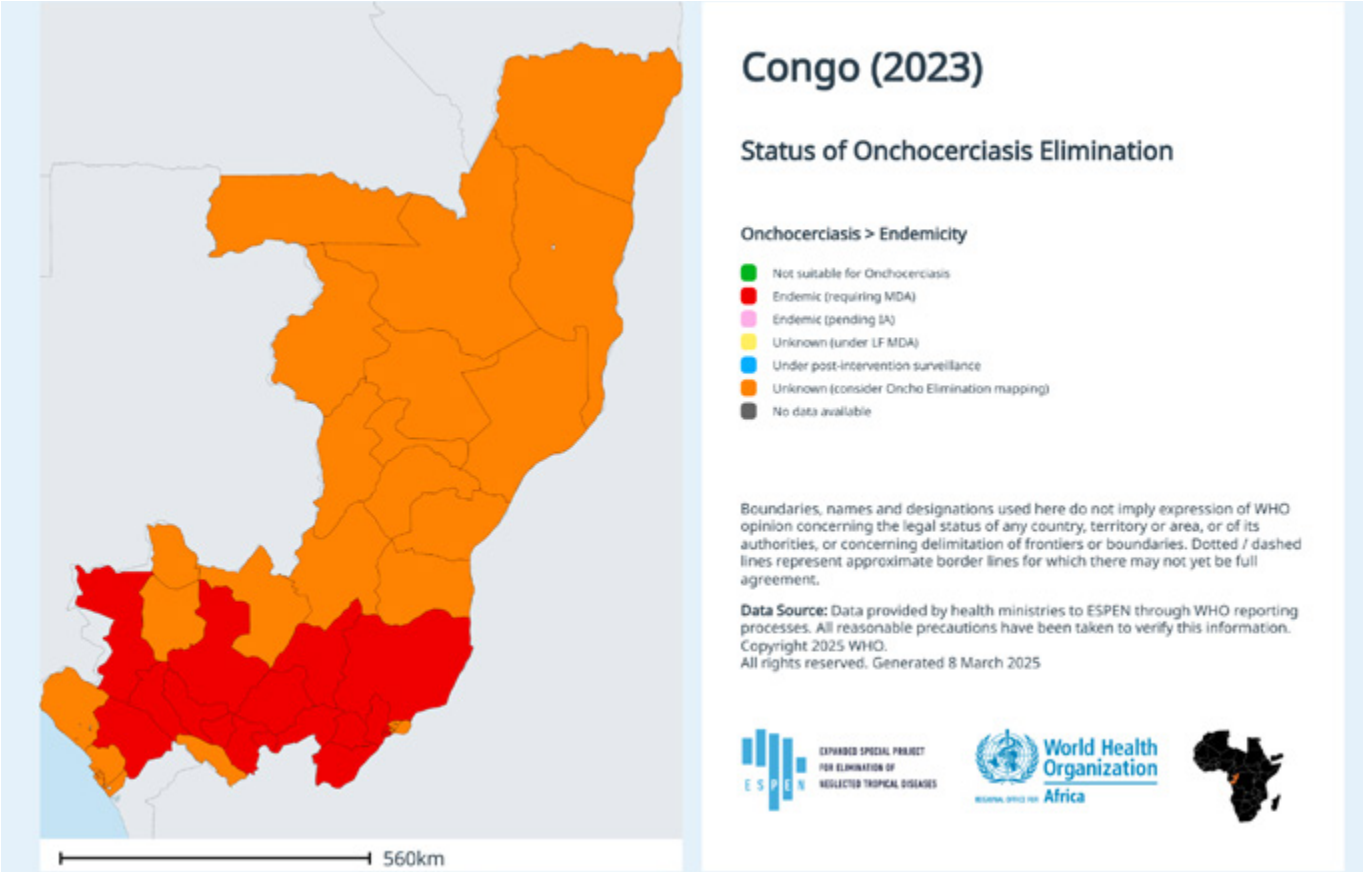
Different options are available to generate bespoke maps with information on onchocerciasis endemicity at the implementation unit (IU) level. These depend on whether or not the programme has staff (data managers or data analysts) with experience in GIS (geographical information system mapping) software and conducting simple spatial analysis.

i) No GIS skills within the programme team.

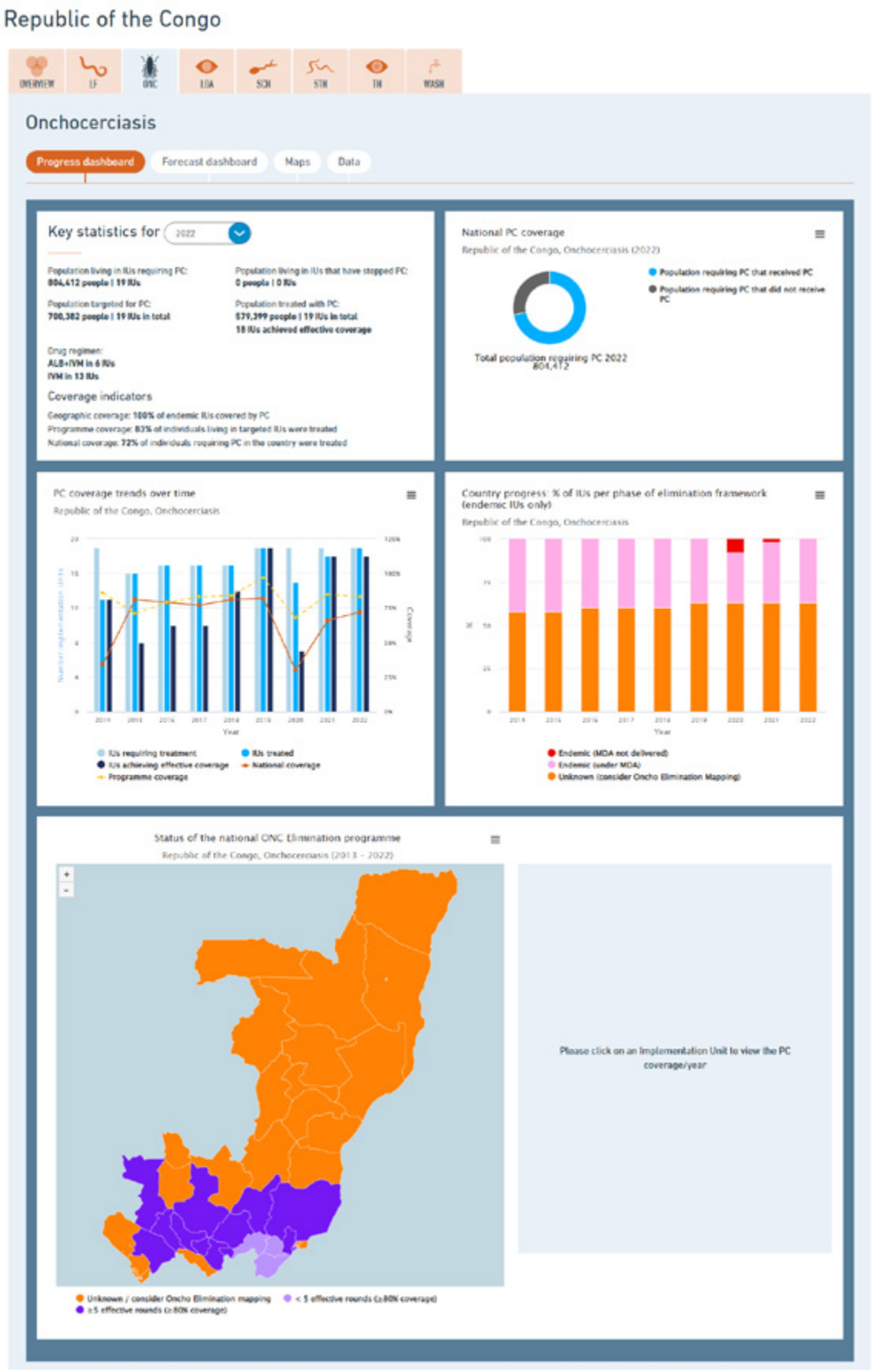
The ESPEN Data Portal (https://espen.afro.who.int/) provides a wealth of GIS-related resources that can be easily used by neglected tropical disease (NTD) programmes implementing preventive chemotherapy.

The ESPEN NTD Portal has made publicly available endemicity and treatment status maps and related data at IU and community levels. These datasets are submitted regularly by PC-NTD programmes through the Joint Application Package (JAP) forms.

Programmes can download static (PDF or PNG format) maps displaying the IU level endemicity and treatments (number of rounds of mass drug administration [MDA]) delivered; they can also interactively explore this information using the analytical dashboards that have been embedded in the ESPEN Data Portal. Below are some examples from Congo.



Source: https://espen.afro.who.int/countries/republic-of-the-congo



Source: https://espen.afro.who.int/countries/republic-of-the-congo

ii) GIS skills within the programme team.

Programmes may decide to tailor their own maps displaying all the spatial data that they may consider relevant, particularly for the desk review stage. Managing and processing data in a GIS environment allow programmes to interact with more variables and produce more precise maps of the areas eligible for onchocerciasis elimination mapping (OEM).

The ESPEN Data Portal (https://espen.afro.who.int/) is also capable of sharing spatial layers of the IU-level division by year since 2013. They can be downloaded from this webpage: https://espen.afro.who.int/tools-resources/cartography-database.

Cartography database

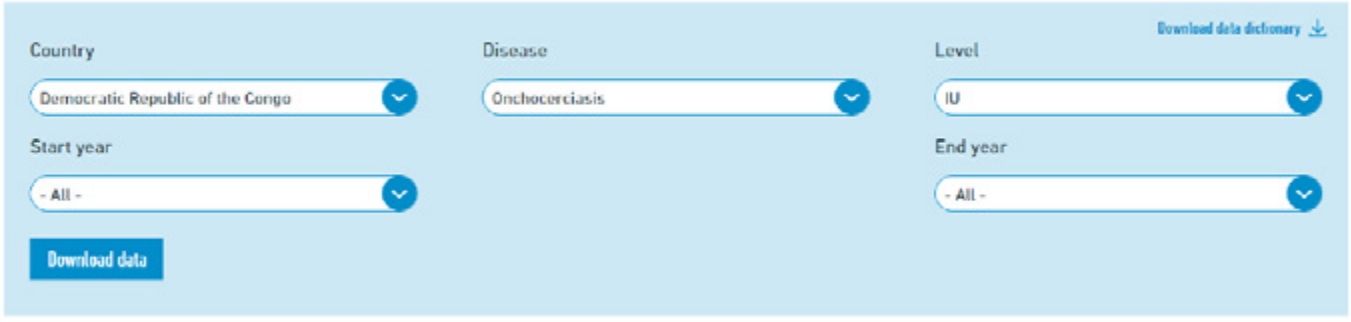
Download

Year	Admin 0	Admin 1	Admin IU
2020	ESPEN_ADM0_2020.zip | 3.64 MB	ESPEN_ADM1_2020.zip | 8.16 MB	ESPEN_IU_2020.zip | 28.08 MB
2019	ESPEN_ADM0_2019.zip | 3.57 MB	ESPEN_ADM1_2019.zip | 8.13 MB	ESPEN_IU_2019.zip | 27.16 MB
2018	ESPEN_ADM0_2018.zip | 3.55 MB	ESPEN_ADM1_2018.zip | 8.04 MB	ESPEN_IU_2018.zip | 26.84 MB
2017	ESPEN_ADM0_2017.zip | 3.55 MB	ESPEN_ADM1_2017.zip | 8.04 MB	ESPEN_IU_2017.zip | 26.84 MB
2016	ESPEN_ADM0_2016.zip | 3.55 MB	ESPEN_ADM1_2016.zip | 8.04 MB	ESPEN_IU_2016.zip | 26.46 MB
2015	ESPEN_ADM0_2015.zip | 4.28 MB	ESPEN_ADM1_2015.zip | 10.14 MB	ESPEN_IU_2015.zip | 35.48 MB
2014	ESPEN_ADM0_2014.zip | 4.28 MB	ESPEN_ADM1_2014.zip | 10.14 MB	ESPEN_IU_2014.zip | 35.48 MB
2013	ESPEN_ADM0_2013.zip | 4.28 MB	ESPEN_ADM1_2013.zip | 10.14 MB	ESPEN_IU_2013.zip | 35.46 MB

Programmes can also download IU-level data on endemicity and treatments delivered as well as survey data submitted by countries through the EPIRF (epidemiological data reporting) form. These datasets are downloadable from the country-specific webpages and also by using the query data tool.

https://espen.afro.who.int/tools-resources/download-data

Download data



Download data dictionary

Other sources of spatial data, mentioned in 3.1.2, may be useful for programmes to delineate the areas of interest during the desk review process.

The OpenStreetMap project is one of the richest repositories of worldwide geographical data. This project generates and updates spatial data on locations, administrative boundaries, river networks and water bodies for every region and country in the world. The Geofabrik website (http://www.geofabrik.de/index.html) provides an easy-to-use platform to access and download OpenStreetMap data in different formats, including the popular shapefile format widely used in GIS analytics. This server has data extracts from the OpenStreetMap project, which are normally updated every day (https://download.geofabrik.de/).



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- ~~africa-internal.osh.pbf~~ The history file contains personal data and is available on the internal server only. See notice above for further information.
- .poly file that describes the extent of this region.
- .osc.gz files that contain all changes in this region, suitable e.g. for Osmosis updates
- raw directory index allowing you to see and download older files

Sub Regions

Click on the region name to see the overview page for that region, or select one of the file extension links for quick access.

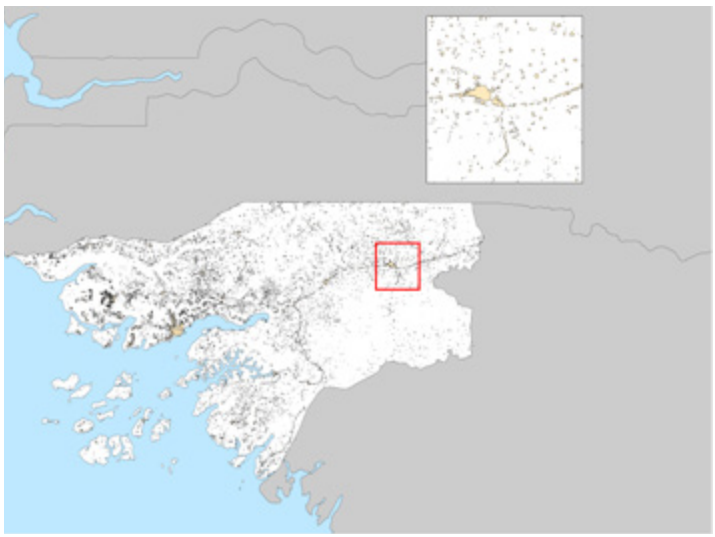
Sub Region	Quick Links		
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Algeria	[.osm.pbf] (109 MB)	[.shp.zip]	[.osm.bz2]
Angola	[.osm.pbf] (57 MB)	[.shp.zip]	[.osm.bz2]
Benin	[.osm.pbf] (39.0 MB)	[.shp.zip]	[.osm.bz2]
Botswana	[.osm.pbf] (80 MB)	[.shp.zip]	[.osm.bz2]
Burkina Faso	[.osm.pbf] (61 MB)	[.shp.zip]	[.osm.bz2]
Burundi	[.osm.pbf] (38.0 MB)	[.shp.zip]	[.osm.bz2]
Cameroon	[.osm.pbf] (178 MB)	[.shp.zip]	[.osm.bz2]
Canary Islands	[.osm.pbf] (45.5 MB)	[.shp.zip]	[.osm.bz2]
Cape Verde	[.osm.pbf] (9.2 MB)	[.shp.zip]	[.osm.bz2]
Central African Republic	[.osm.pbf] (70 MB)	[.shp.zip]	[.osm.bz2]
Chad	[.osm.pbf] (70 MB)	[.shp.zip]	[.osm.bz2]
Comores	[.osm.pbf] (3.3 MB)	[.shp.zip]	[.osm.bz2]
Congo (Republic/Brazzaville)	[.osm.pbf] (24.4 MB)	[.shp.zip]	[.osm.bz2]
Congo (Democratic Republic/Kinshasa)	[.osm.pbf] (330 MB)	[.shp.zip]	[.osm.bz2]
Djibouti	[.osm.pbf] (8.1 MB)	[.shp.zip]	[.osm.bz2]
Egypt	[.osm.pbf] (148 MB)	[.shp.zip]	[.osm.bz2]
Equatorial Guinea	[.osm.pbf] (2.7 MB)	[.shp.zip]	[.osm.bz2]
Eritrea	[.osm.pbf] (9.9 MB)	[.shp.zip]	[.osm.bz2]
Ethiopia	[.osm.pbf] (72 MB)	[.shp.zip]	[.osm.bz2]
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Ghana	[.osm.pbf] (65 MB)	[.shp.zip]	[.osm.bz2]
Guinea	[.osm.pbf] (99 MB)	[.shp.zip]	[.osm.bz2]
Guinea-Bissau	[.osm.pbf] (8.6 MB)	[.shp.zip]	[.osm.bz2]

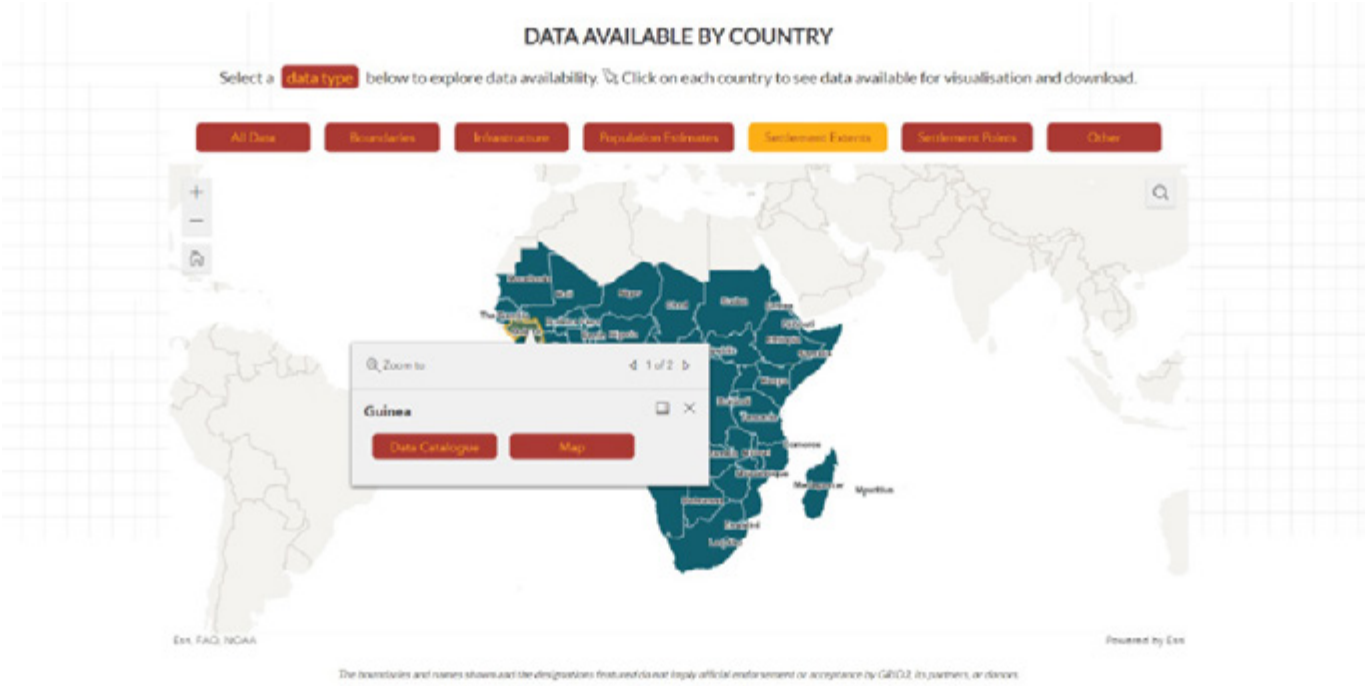
Not what you were looking for? Geofabrik is a consulting and software development firm based in Karlsruhe, Germany specializing in OpenStreetMap services. We're happy to help you with data preparation, processing, server setup and the like. Check out our web site and contact us if we can be of service.

Nicht das Richtige dabei? Die Geofabrik ist ein auf OpenStreetMap spezialisiertes Beratungs- und Softwareentwicklungsunternehmen in Karlsruhe. Gern helfen wir Ihnen bei der Datenaufbereitung, Datenkonvertierung, Serverinstallation und ähnlichen Aufgaben. Besuchen Sie unsere Webseite und sprechen

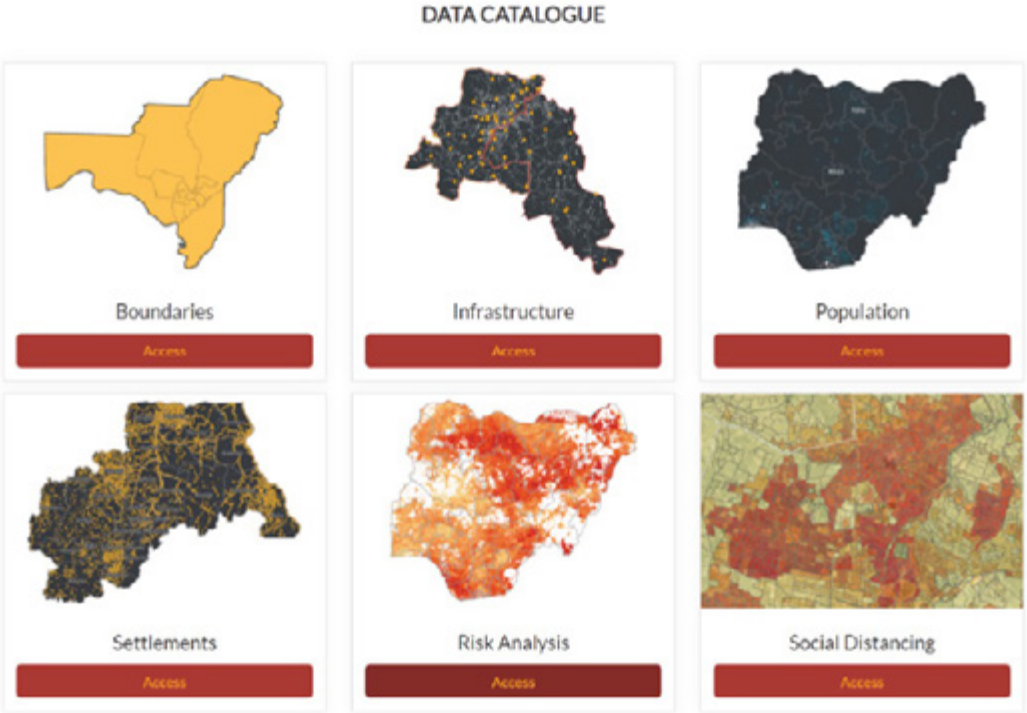
Source: https://download.geofabrik.de/africa.html

Finally, obtaining georeferenced data on settlements without going to the field and collecting the data in situ, which might be unfeasible for large areas, can be a challenge for onchocerciasis programmes. Currently, there are some useful resources that are generating this information based on high-resolution remote-sensed data. The GRID3 project has become one of the most popular resources for precise demographics and spatial data on settlements (https://grid3.org/).





The boundaries and names shown and the designations there used do not imply official endorsement or acceptance by GRID3, its partners, or donors.



Annex 5.
River prospection form

No	Question	Explanation
1.	Recorder ID	Enter the two-digit recorder code provided to each recorder (that is, the person doing the data entry).
2.	Select province	Select the province you are working in from the drop-down list.
3.	Select district	Select the appropriate district from the drop-down list.
4.	Site	Specify the name of the nearest community (text entry).
5.	Enter today's date	Enter using the format DD/MM/YYYY
6.	GPS	Collect site coordinates: must be done outside
7.	River	Provide the name of the river, stream or tributary
8.	River basin	Provide the name of the river basin
9.	Inspection time	At the time of the inspection, did the site appear suitable for blackfly breeding? Yes | No.
10.	Larvae evidence	At the time of the inspection, did the site provide evidence of larvae? Yes | No
11.	Abundance	How many larvae were found? 1–10 11–50 More than 50 (select one)
12.	Notes	Optional additional notes

Annex 6.
Entomology community questionnaire

No	Question	Explanation
1.	Recorder ID	Enter the two-digit recorder code provided to each recorder (that is, the person doing the data entry).
2.	Select district	Select the appropriate district.
3.	Site	Give the name of the community you are in (text entry).
4.	Does this person consent to take part in the study?	Enter Yes or No. in select 1: No will end the survey, and give a warning before exiting the questionnaire
5.	What gender is this person?	Select either Male or Female
6.	What age is this person?	Enter the age in years (number entry).
7.	What is their occupation?	Specify the occupation (text entry).
8.	How long has this person lived in this community?	Give in years (number entry).
9.	Blackfly problem	Are blackfly bites a problem in the community? Yes | No | Don't know
10.	Bites number	If yes, how many bites do you receive each day? How larvae were found? 1–10 11–50 More than 50 (select one)
11.	Blackfly month	Select the month or months when the respondent reports that blackfly biting is most frequent
12.	Blackfly time of day	Morning, midday, afternoon, evening
13.	Notes	Optional additional notes

Annex 7.
First-line village recording form

No	Question	Explanation
1	Recorder ID	Enter the two-digit recorder code provided to each recorder (that is, the person doing the data entry).
2	Select district	Select the appropriate district.
3	River basin	Enter the name of the river basin you are in.
4	Village name	Enter the name of the community you are in.
5	GPS	Collect the site coordinates: must be done outside.
6	Total population of the village	Enter the number (can be obtained from community drug distributors' registers on mass drug administration or from local community guides).
7	Notes	Add optional additional notes.

Annex 8.

Crab collection form for *Simulium neavei* immature stages

District/LGA/State: _____ County: _____

Focus/Transmission zone: _____ Village: _____

River: _____ Site: _____

Coordinates of site: _____ Ph _____

Conductivity: _____ Temp: _____

Name of Field team:											
1.						3.					
2.						4.					
Date:						Date:					
No	Crab size (mm)		*S. neavei* immature stages			No	Crab size (mm)		*S. neavei* immature stages		
	♀	♂	L	P	PCs		♀	♂	L	P	PCs
♀ = female; ♂ = male ; L = larva; P = pupa; PCs = pupal cases.											

Comments: _____

_____

_____

Annex 9.
Standard operating procedure for packaging and shipping dried blood spots

Purpose: To define the standard operating procedure for shipping dried blood spots (DBS) and for ensuring proper packing and preparations, as well as compliance with export regulations at source and import regulations at destination (if exporting outside of country).

Audience: In-country programme managers, supervisors and technicians

Responsibilities: Trainers, supervisors, technicians and programme managers

Specific procedure: If a global courier is used, then all the DBS samples can be sent at one time. However, if another courier is used, then it is recommended for safety that two batches are sent separately in case the samples are lost.

Supplies

- Packing tape
- TropBio filters spotted with dried blood in individual specimen bags
- 1-gallon sealable plastic bags (1 bag per 50 samples)
- Biohazard stickers
- Indicating desiccant (i.e. colour changes if moisture is absorbed)
- Cold packs
- Outer cardboard box for shipping
- Shipping permits, labels and importation documentation
- Commercial invoices
- Absorbent pads

DBS should be completely dry before packaging

To ship DBS specimens you must use the basic **triple-packaging system**:

1. The primary specimen bag contains the individual specimens of absorbed and dried blood.
2. The secondary bag contains the desiccant and the primary specimen bags.
3. The third bag is a large sturdy bag for carrying a number of secondary bags.

Preparing the outer shipping container

1. The outer shipping container must have a complete postal address and delivery address.
2. No content markings are required on the outer shipping container.
3. You must affix the international biohazard symbol on the secondary bags.

Procedure

1. Once the DBS are completely dry, place individual specimens into **completely** sealed individual bags. **Note:** Do not put desiccant in these bags.

Example 1: Sealed specimen bag

Example 2: Slightly larger specimen bags are acceptable

Example 3: Do not add desiccant directly into the bag



2. In a large, sealable plastic bag, add about 15 g of desiccant (Note: 15 g is about 15 mL if measured with a graduated cylinder or tube). Alternatively use moisture-indicating desiccant sachets, which change colour when moisture has been absorbed. Ensure that there are no holes in the bag.



3. Place an orange biohazard sticker on the outside of each large bag.

4. Place approximately 50–75 of the small specimen bags containing the individual filter discs into one of the large, labelled, sealable plastic bags with desiccant. Ensure that ALL the filter discs are facing up with the QR codes visible.



5. When feasible, filter paper bloodspots should be organized which bags by location and date of collection. This information should be indicated on a large plastic bag and on a data sheet placed inside the bag.



Left: note that the outside of the bag is labeled with permanent marker and a data sheet is placed in the bacg that indicates the location and date of collection of the samples

Top view



Side view



6. Place the large biohazard labeled sealable plastic bags into a cardboard box along with enough newspaper or bubble wrap to prevent excessive movement of the bags during transit. Do **NOT** put orange biohazard labels on the outside of the box.

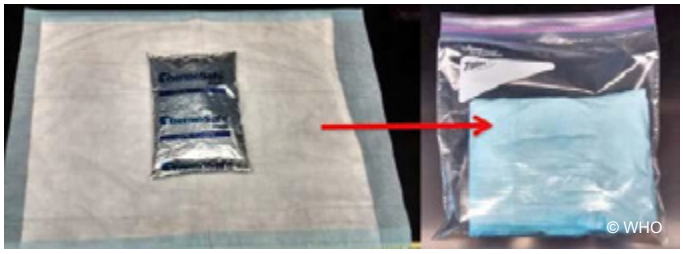




7. Reinforce the corners and edges of the box with multiple layers of packing tape. Seal the box securely with multiple layers of packing tape.

DBS shipping

Ensure that all specimen bags are properly sealed, but not airtight. This is because the lack of air exchange in the inner environment of a sealed bag causes heat build-up and moisture accumulation. These steps will help to prevent moisture from coming in contact with the dried blood spots and causing spoilage. DBS can be shipped at room temperature if arrival is expected within 3 days. If cold packs are used, wrap them in an absorbent material (e.g. paper towels), and place them inside a sealable plastic bag (see diagram below).



Information should be typed up, printed out in triplicate, and attached to the top and two different sites of the shipment container. For the consignee postal address, make sure adequate detail is included to make sure the samples reach their intended destination. Details such as building name, building number, P.O. box, zip code, postcode, etc. should be included in full.

• Sender name:	• Consignee name:
• Sender company or organization:	• Consignee company or organization:
• Sender phone number:	• Consignee phone number:
• Sender email address:	• Consignee email address:
• Sender postal address:	• Consignee postal address:

Annex 10.

Onchocerciasis elimination mapping survey sample builder

Access the Survey Sample Builder at:

https://www.cor-ntd.org/resources/onchocerciasis-elimination-mapping

Annex 11.
Ov16 RDT with dried blood spots standard operating procedures

A11.1 Purpose

This annex describes the procedures for preparing, storing and using dried blood spot (DBS) elution buffer for use with the Standard Diagnostics (SD) Bioline Monoplex Onchocerciasis IgG4 Rapid Test (Abbot [SD]; Republic of Korea; Cat. No. 61FK10) and the SD Bioline Biplex Lymphatic Filariasis/ Onchocerciasis IgG4 Rapid Test (Abbot [SD]; Republic of Korea; Cat. No. 61FK20).

A11.2 Equipment

- Analytical balance, capable of measuring mg (FisherSci Cat#: 01-913-920, or equivalent)
- Weighing spatula (FisherSci Cat#: 14-371C, or equivalent)
- Stir plate (FisherSci Cat#: 90-691-23, or equivalent)
- Magnetic stir bar (FisherSci Cat#: 14-513-60SIX, or equivalent)
- Graduated cylinder appropriate to batch volume
- Beaker appropriate to batch volume
- pH meter (FisherSci Cat#: 13-636-AE150, or equivalent)
- Vacuum for filtering

A11.3 A12.3 Supplies

- Syringe (FisherSci Cat#: 14-823-16D, or equivalent)
- Blunt pipetting needle 16 g x 4” (Cadence Science Cat#: 9872)
- Weigh paper or weigh boats (FisherSci Cat#: 02-202-101, or equivalent)
- 0.45 µm vacuum filter with attached 500 mL storage bottle – FisherSci, Corning™ Disposable Vacuum Filter/Storage Systems Cat#: 09-761-142, ESSENTIAL
- 50 mL conical tubes (FisherSci Cat#: 14-432-22, or equivalent)
- Aluminium foil

A11.4 A12.4 Reagents/media

- Sodium borate, decahydrate, reagent grade (Sigma Aldrich, Cat#: SX0355)
- Sodium chloride, reagent grade (Sigma Aldrich, Cat#: S7653)
- Tween® 20, reagent grade (Sigma Aldrich, Cat#: P1379)
- Sodium caseinate, reagent grade (Sigma Aldrich, Cat#: C8654)
- Water, high purity (ISO grade II or better)

A11.5 A12.5 Procedure for preparing elution buffer

11.5.1 Determine the batch volume of buffer to be prepared and record it in the Bill of Materials (see 11A below).

11.5.2 Using the Bill of Materials, calculate the "Quantity Required" of each reagent.

11.5.3 Add the required amount of water to the beaker.

11.5.4 Add a clean stir bar to the beaker and place it on a stir plate; begin stirring at room temperature.

11.5.5 Measure out the required amount of sodium borate; record the measured amount in the Bill of Materials; add to the beaker.

11.5.6 Allow all of the sodium borate crystals to dissolve (Note: this may take up to 3 hours).

11.5.7 After the sodium borate dissolves, measure out the required amount of sodium chloride; record the measured amount in the Bill of Materials; add to the beaker; and stir until completely dissolved.

11.5.8 Measure the pH (should be 9.1–9.3); adjust as necessary with HCl or NaOH and record the final pH on the Bill of Materials.

11.5.9 Measure out the required amount of sodium caseinate; record the measured amount in the Bill of Materials; add to the beaker.

11.5.10 Stir until the sodium caseinate is completely dissolved (Note: this will take several hours; covering with foil and stirring overnight may be necessary).

11.5.11 Following the steps in 12.5.11.1–12.5.11.4, measure the required amount of Tween; record the measured amount in the Bill of Materials; add to the beaker.

11.5.11.1 Attach a blunt 16 g x 4" pipetting needle to a syringe and fill the

11.5.11.2 syringe to the 5 mL mark with Tween solution. As the Tween is viscous, pull the plunger slowly and give the syringe time to fill completely.

11.5.11.3 Remove the needle and press the plunger to dispense the Tween back into the stock bottle until you reach your required volume + 1 mL (i.e. if the required volume of Tween is 1 mL for 500 mL total buffer, dispense to the 2 mL mark).

11.5.11.4 Slowly expel the required amount of Tween into the solution until the plunger is at the 1 mL mark.

11.5.11.5 Discard the remaining Tween and the syringe. Rinse the needle with plenty of water and allow it to dry.

11.5.12 Adjust the volume to the planned batch volume with high-purity water.

11.5.13 Stir for 15 minutes to homogenize.

11.5.14 Set up the filtration apparatus (0.45 µm) and filter the batch into a clean storage bottle (Note: add solution slowly to the filter apparatus to avoid clogging).

11.5.15 Confirm the final batch volume and assign a unique lot number. Using the preparation date is suggested (i.e. DD/MM/YYYY). Record this lot number on the Bill of Materials and keep this document for laboratory records.

11.5.16 Aliquot buffer into 50 mL conical tubes at 40 mL per tube, label each tube with a description, lot number and expiration date (see 11B below).

11.5.17 Store at 2–8 °C when not in use; DO NOT FREEZE.

11.5.18 Shelf-life: 1 month.

Appendix 11A. Label

DBS elution buffer for LF/Ov16 Biplex RDT			
Lot number		Amount	
Store at		Expiry date dd/mm/yyyy	
Prepared by initials		Date dd/mm/yyyy	

Appendix 11B. Bill of materials

Buffer lot number: _____		Date: _____		Final pH: _____	
Description	**Lot number**	**Quantity per litre**	**Quantity per 300 mL**	**Quantity required[a]**	**Quantity used**
Sodium borate, decahydrate, reagent grade		19.07 g	5.72 g		
Sodium chloride, reagent grade		1.75 g	0.525 g		
Water, high purity		0.8 L	0.24 L		
Sodium caseinate, reagent grade		2 g	0.6 g		
Tween® 20, reagent grade		2 mL	0.6 ml		
Water, high purity		qs to 1 L	qs to 300ml		

[a] This table can be used to make 300 mL of buffer or to calculate required amounts for other volumes as needed.

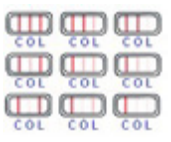
Appendix 11C. Assay of DBS using the SD Bioline Onchocerciasis/Lymphatic filariasis Biplex rapid test

DBS elution – DAY 1

1. Cut the DBS spot samples (TropBio paper type) and place in a 96-well elution plate (Corning 3788 or similar).
2. Record the DBS sample ID in the RDT plate map (see Appendix D).
3. Add **60 µL of RDT buffer** to each well containing a DBS spot. Push the spot down with the pipet tip to completely submerge it in buffer, with no air bubbles surrounding the spot. (**NOTE:** use **100 uL RDT buffer** for 6 mm Whatman paper.
4. Cover the plate and incubate in the refrigerator overnight (12–24 hours) at 2–8 °C.

Running DBS on the RDT – DAYS 2 and 3

1. Prepare your workbench: bring the protocol, the elution plate map, clock/watch, pen, marker, pipettes, tips, RDT buffer, RDTs and the eluted samples.
2. Write the RDT LOT number and the expiration date on your plate map. Record the laboratory temperature.
3. Open the RDT pouch and check the desiccant indicator beads. Only use the RDT if the indicator beads are YELLOW.
4. Write on the RDT cassette the DBS ID.
5. Run the eluted DBS according to the following method:

1. Add 10 µL of RDT buffer to the sample well (round). Wait for the buffer to be absorbed.		
2. Carefully mix the eluted spot with pipetting until uniform in colour. Add 10 µL sample of DBS eluate to the same sample well (round). Wait for the sample to be absorbed.		
3. Snap off the dropper tip of the RDT assay diluent to use the pipette. Add 70 µL of assay diluent to the assay diluent well (square).		
4. Record the time in the lower right-hand corner of the RDT cassette	TIME	
5. After 1 hour, confirm the RDT has a control line, indicated with a "C" on the cassette. If a control line is not visible, repeat the sample on a new cassette.	Onchocerciasis reactive	
	Lymphatic filariasis reactive	
6. Read the test result after 24 hours for a final result and record them in the plate map. Any visible line is considered a positive result. If desired, the Intensity Scale can be used to define weak and strong results (see Appendix E). 	Onchocerciasis and lymphatic filariasis reactive	
	Nonreactive	
	Invalid	

Appendix D. RDT plate map

Plate #: _____ Date: _____

Teh Name: _____ Room Temp: _____

Sample Type: Serum ☐ DBS ☐

RDT Kit Info Expiration Date: _____

Kit Lot #: _____

Test: Biplex ☐ Monoplex ☐

	1	2	3	4	5	6	7	8	9	10	11	12
A	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
B	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
C	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
D	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
E	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
F	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
G	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L
H	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L	O L

Comments: _____

_____

_____

Appendix E. Intensity scale



Annex 12.

Onchocerciasis Elimination Program for the Americas: standard operating procedures for enzyme-linked immunosorbent assays

A. Reagents

1. **PBS 10x (store at room temperature)**

- NaCl 80 g
- KCl 2.0 g
- KH_2PO_4 2.4 g
- Na_2HPO_4 14.4 g

Dissolve components in 800 mL of distilled water. Check that the pH is between 7.0 and 7.4. Bring the solution to a final volume of 1 litre with distilled water.

2. **PBST (store at room temperature)**

- PBS 1X
- Tween 20 0.05% (V/V)

Mix 100 mL of 10x PBS with 800 mL of distilled water. Add 0.5 mL of Tween 20 and mix thoroughly. Tween 20 is quite viscous and heating it at 42 °C in a water bath for 30 minutes or so will thin it out and make it easier to measure. When the Tween is mixed in, adjust the volume to 1 litre by adding distilled water.

3. **PBST-BSA (store at 4 °C)**

- 1x PBST, bovine serum albumin 5% (W/V)

Dissolve 5 g of bovine serum albumin in 100 mL of PBST.

4. **Coating buffer (store at 4 °C)**

- 0.1 M $NaHCO_3$ (pH 9.6)

Dissolve 8.4g of sodium bicarbonate in 900 mL of distilled or deionized water. Check the pH of the solution with a pH meter: it should be 9.5–9.7. Add water to reach a final volume of 1 litre. Store at 4 °C and keep for no longer than a week.

5. 1 mg/mL p-Nitrophenyl phosphate (pNPP) solution (Sigma N-9389)
6. Anti-human IgG4 antibody (mouse monoclonal) conjugated to biotin (Sigma B3648)
7. Streptavidin-AP conjugate (Invitrogen 19542018)
8. Ov16 GST antigen (supplied by Unnasch laboratory)
9. GST antigen (supplied by Unnasch laboratory).
10. Positive control sera: humanized Ov16 monoclonal antibody or McAb (BioRad)
11. Immunolon 2HB flat-bottomed plates (Thermo Scientific)

B. Procedure

1. **Create a map: Use the map below. The samples are run in duplicate as indicated.**

	1	2	3	4	5	6	7	8	9	10	11	12
A	STD 640	STD 640	Blank	Blank	S7	S7	S15	S15	S23	S23	S31	S31
B	STD 320	STD 320	Blank	Blank	S8	S8	S16	S16	S24	S24	S32	S32
C	STD 160	STD 160	S1	S3	S9	S9	S17	S17	S25	S25	S33	S33
D	STD 80	STD 80	S1	S3	S10	S10	S18	S18	S26	S26	S34	S34
E	STD 40	STD 40	S2	S4	S11	S11	S19	S19	S27	S27	S35	S35
F	STD 20	STD 20	S2	S4	S12	S12	S20	S20	S28	S28	S36	S36
G	STD 10	STD 10	S5	S5	S13	S13	S21	S21	S29	S29	S37	S37
H	STD 5	STD 5	S6	S6	S14	S14	S22	S22	S30	S30	S38	S38

S = sample wells; STD = positive control dilutions (STD 640 = 50 uL of 250 ng/mL)

2. **Prepare elution from filter-paper samples:** Punch out duplicate spots from the dried blood samples collected on the filter-papers using a standard 6-mm paper punch. Using your map, place the duplicate punches of the blood spots into the sample wells. Add 200 µL of PBST-BSA to each sample. Push the punches to the bottom of the well and then mix 10 times by pipetting. Cover the plates with a plate sealer and incubate them at 4 °C overnight. Store the eluted serum samples at −20 °C.

3. **Coat the plate with antigen:** Dilute Ov16-GST antigen to 2.0 µg/mL in coating buffer. Add 100 µL to each well. Place the plate in a sealable plastic bag and incubate it overnight at 4 °C.

4. **Wash the plates:** Wash the plates four times with PBST, using a wash bottle. Do not dry between washes. Blot dry the plate after the last wash.

5. **Block the plates:** Add 100 µL of PBST-BSA, place the plate in a sealable plastic bag and incubate it at 4 °C for 1 hour.

6. **Prepare a standard curve and blank:** During the incubation of step 5, prepare 250 uL of a 250 ng/mL of the McAb in PBST-5% BSA. This is equivalent to 640 units (per 50 ul) of the old, pooled serum-positive control. Prepare a series of seven two-fold dilutions of the McAb. Start with 125 uL of the 250 ng/mL stock, and add it to 125 uL of antibody dilution buffer. This will be a solution of 125 ng/mL or 320 units/50 uL. Continue making the additional dilutions until you have a total of 8 dilutions in the series, which will range from 640 units/50 uL to 5 units/50 uL. Set aside 250 uL of PBST-5% BSA to use as blanks.

7. **Empty the plate:** After the incubation period of step 5, empty the PBST-BSA in the sink and dry the plate. Do not wash.

8. **Add samples:** Using your map, add 50 µL of each serum sample (positive control dilution, blanks and eluted samples) to the corresponding wells on the plate map. Place the plate in a sealable plastic bag and incubate it at room temperature for 2 hours.

9. **Wash the plate:** Wash the plate four times with PBST, using a wash bottle. Dry the plate after the first wash, then carry out the remaining three washes without drying the plate between the wash steps. Dry the plate after the fourth wash.

10. **Add conjugate:** 10 minutes before the time is up for the last incubation, prepare the conjugate. Dilute the anti-human IgG4 antibody conjugated to biotin 1:10 000 in PBST. Add 50 µL of the diluted conjugate to all wells. Place the plate in a sealable plastic bag and incubate it at room temperature for 1 hour.

11. **Wash the plates:** Wash the plates four times with PBST, using a wash bottle. Do not dry between washes. But dry the plates after the last wash.

12. **Add Streptavidin-AP:** 10 minutes before the time is up for the last incubation, prepare the streptavidin-AP. Dilute streptavidin-AP 1:2000 in PBST. Add 50 µL to all wells, cover the plate and incubate it at room temperature for 1 hour.

13. **Wash the plates:** Wash the plates four times with PBST, using a wash bottle. Do not dry between washes. But dry the plates after the last wash.

14. **Substrate:** 15 minutes before the incubation period is over, switch on the ELISA plate reader and set it to read at 404 nm. Prepare PNPP solution with the substrate buffer by dissolving 1 tablet in 5 mL of the buffer provided. Add 50 µL to each well. If your reader reads the whole plate at once, then read the plate until the 640 standard is around OD 1.5. If your reader reads well by well, then read only the two wells of the 640 standard every minute, and shake the plate between readings. When they are around 0.9 OD, start reading more frequently. When you reach OD 1.1, stop the reaction by adding 25 µL of 3M NaOH to each well. Shake the plates to stop the reaction. Incubate the plate for 5 minutes at room temperature and read it at 405 nm.

C. Quality assurance

All the ELISAs should be analysed using the following criteria; if the ELISA run does not meet these criteria it should be repeated.

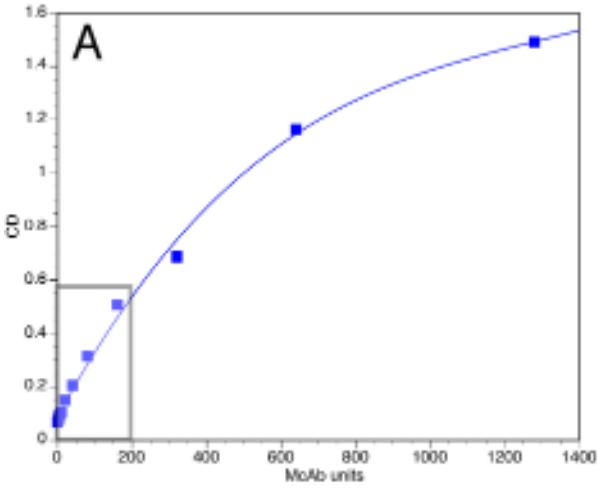
The standard curve dilutions are assigned the following arbitrary units:

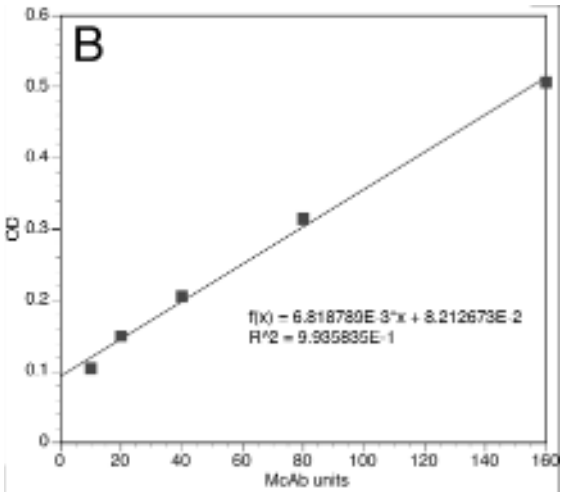
Dilution	Units
250 ng/mL	640
125 ng/mL	320
62.5 ng/mL	160
31 ng/mL	80
15.5 ng/mL	40
8 ng/mL	20
4 ng/mL	10
2 ng/mL	5

A plot of OD versus units should be linear in the range of 10–160 units, with an r2 value for the line of > 0.95. The dynamic range for the linear part of the curve should be at least 5 fold, i.e. the OD value for the 160 unit points should be at least four times greater than the 10 unit points. The 40 unit points should be at least twice the background ODs. Duplicate wells in the standard curve should be within 10% of one another.

Fig. A13.1. An example of a standard curve that has passed QC standards. Panel A: Plot of all OD values versus McAb units.

Note the OD for the 640 unit point is approximately 1.1, indicating the assay was terminated at an appropriate time. The boxed area indicates the linear area of the curve. Panel B: A re-plot of the linear area of the curve. Note the r2 value is 0.993, which is greater than 0.95. The OD value of the 160 unit point is 0.51, which is more than four times the value of the 10 unit point (0.10).





D. Analysis of results

The cut-off is set at 40 arbitrary units. Any sample for which both duplicate samples give a reading at or above this point is considered putatively positive. If the sample is putatively positive, repeat the assay. Samples giving OD values in both assays that are at least 40 units should be then tested in the same way on wells coated with GST. In running the GST plate, the first column should be coated with Ov16 and the standard curve run as usual. Coat the remaining wells to be tested with the serum samples with GST alone. Samples giving an OD value greater than 40 units in both of the assays when tested against Ov16 and which give an OD corresponding to less than 20 units when tested against GST alone are confirmed positives.

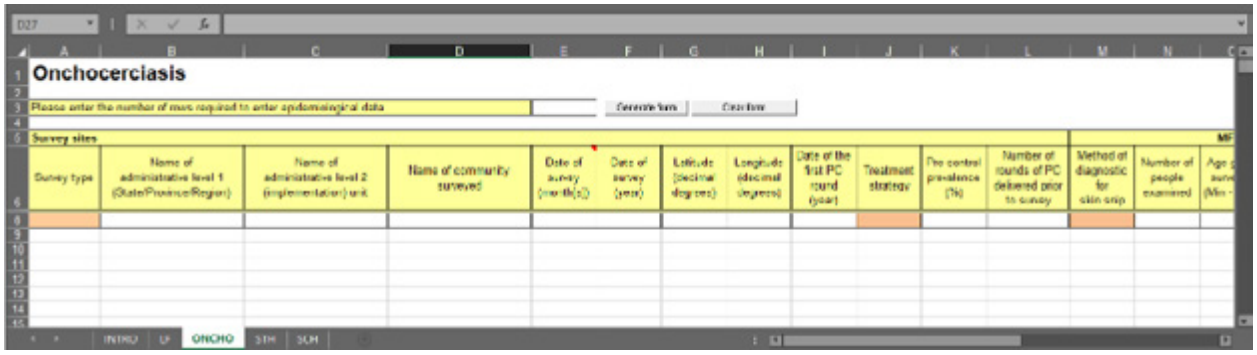
Annex 13.

Reporting onchocerciasis elimination mapping results using the epidemiological reporting form

Results formatted to the epidemiological reporting form

Following the conclusion of the survey field-based data collection activities and post-submission data cleaning, survey results are formatted and applied to the epidemiological reporting form (EPIRF), which is submitted annually as part of the Joint Application Package (JAP). The ESPEN Collect team will process the survey data into the format needed for reporting on the EPIRF, use these data to fill in the EPIRF template, then email the completed EPIRF template to national programmes.

Image 1. The EPIRF

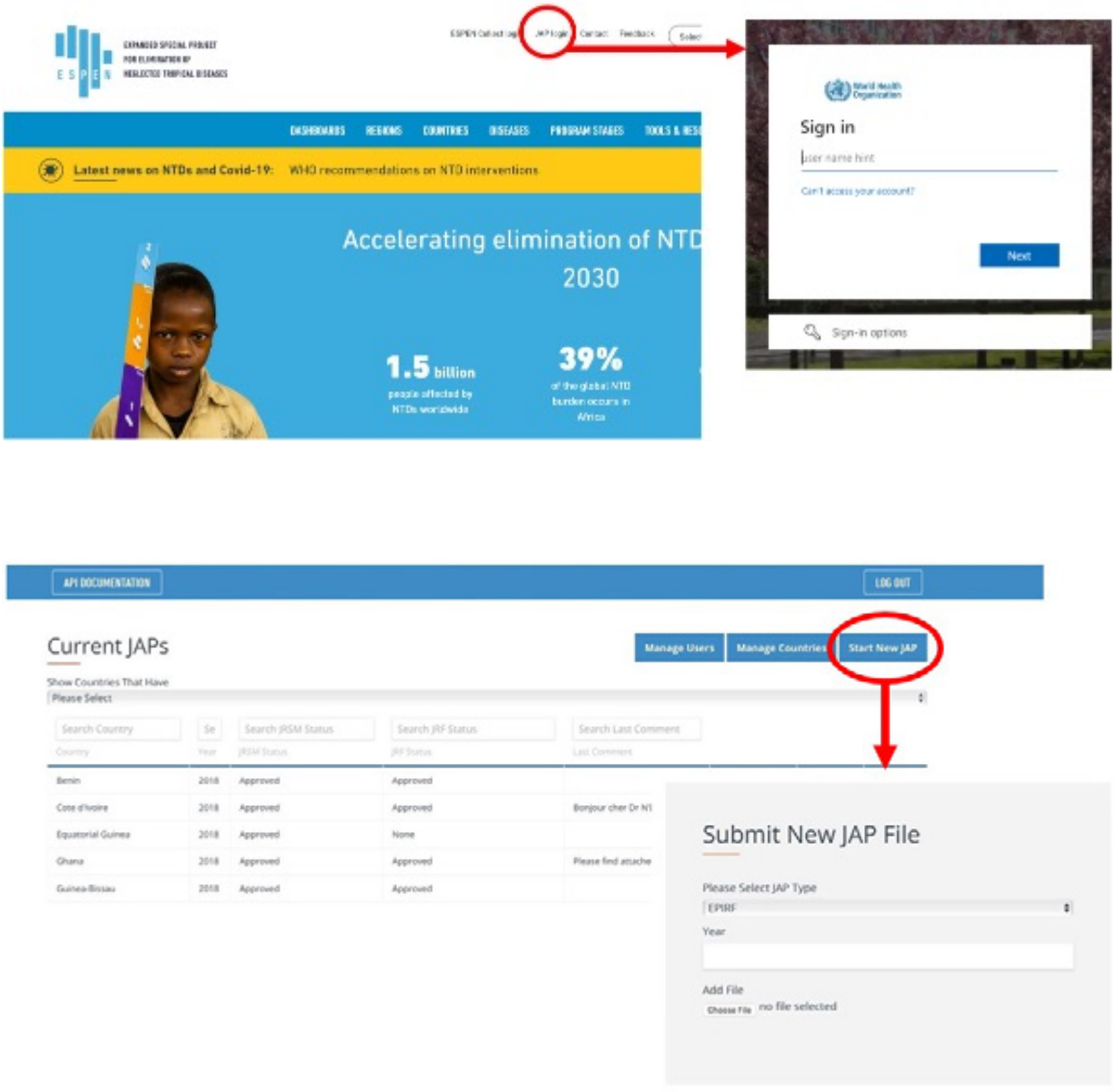


National NTD Programme Managers will review these survey results, usually within one week following the completion of field-based data collection activities. National NTD Programme Managers can submit these data once they have been reviewed and approved or combine these data with any additional surveys conducted using tools other than ESPEN Collect.

ESPEN Portal: JAP upload tool

Once the national managers of neglected tropical disease (NTD) programmes are ready to submit their EPIRF, they can go to the ESPEN Portal to access the JAP upload tool. This tool requires a WHO account to access restricted sections of the ESPEN Portal. Once the authorized user has successfully logged into the system, he or she can search for an existing JAP file or submit a new JAP file. New versions of existing files can be submitted. The newest version will be the active version. However, all previous versions will be stored and are available on the ESPEN Portal JAP upload tool.

Image 2. Access restricted sections of the ESPEN Portal



Annex 14.
Evaluation forms for trainers

TRAINERS' DAILY DEBRIEFING GUIDE

Day of training:

Date of training (DD/MM/YYYY):

Location of training:

Trainers:

At the end of each day, trainers are recommended to sit down, discuss and make a note of the following:

1. What worked well today?

2. What did not work well? Do you know why?

3. What modules/components/concepts did participants seem to understand well?

4. Were there any clinical concepts that participants appeared to not understand?

5. What (if anything) could we do differently next time in presenting today's training content?

6. Do you have any other comments or suggestions?

TRAINERS' END-OF-TRAINING EVALUATION FOCUS GROUP: TOPIC GUIDE

It will be helpful to have an assigned focus group facilitator to ask the questions and keep the group engaged, and a scribe to take note of or write answers on a flip chart so everyone can see them.

1. What worked well during the training?

2. What did not work well during the training?

3. What were the strengths of the training?

4. What were the weaknesses of the training?

5. What recommendations were made for future training?

6. What topic or parts did trainees find most difficult in your opinion?

7. Do you have any other comments or suggestions?

Annex 15.
Evaluation forms for trainees

DAILY EVALUATION FORM FOR TRAINEES

Day of training:

Date of training (DD/MM/YYYY):

Location of training:

Trainers:

1. What did you enjoy most about today's training?

2. What did you least enjoy about today's training?

3. Was there anything you did not understand during today's sessions? Please provide specific examples.

4. What is the most valuable thing you learnt today (knowledge or skills)?

5. Do you have any recommendations on how to improve this training day?

6. Do you have any other comments or suggestions?

END OF TRAINING EVALUATION FORM

Title of training:

Date of training (DD/MM/YYYY):

Location of training:

Names of trainers:

Instructions: Please tick your level of agreement with the statements listed below	Strongly Agree	Agree	Disagree	Strongly disagree	Not relevant to this event
1. The objectives of the training were clearly defined					
2. The training objectives were met					
3. The trainers were well prepared					
4. The presentation materials were relevant to me					
5. The content of the course was organized and easy to follow					
6. The trainers were well prepared and able to answer any questions					
7. The course length was appropriate					
8. The exercises/role plays were helpful and relevant					
9. What was most useful during this training?					
10. What was least useful during this training?					
11. What recommendations can you provide us with to improve the training?					
12. Do you have any other comments or suggestions?					

Thank you very much for attending the OEM training course, and thank you for completing this evaluation form. We will use your feedback to help improve the course contents and its delivery.

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